

Peace or Push? Understanding Global Patterns of Refugee Return

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Abstract

Return is the main way refugee crises end, yet there have been few cross-national studies of when it actually occurs. A central expectation of the refugee regime is that refugees remain abroad while home is unsafe and return as security improves. Yet using a novel global dataset, I show that most returns occur during ongoing conflict, while many refugees remain abroad long after wars end. To explain these patterns, I argue that asylum countries and international organizations play major and underrecognized roles in shaping return, largely overlooked by existing micro-level research focused on refugees' concerns about security at home. Return is substantially higher when refugees face exclusion policies or conflict in asylum countries, and when donor-funded UNHCR assistance lowers the costs of return. I introduce the Global Refugee Return Dataset, covering 293 origin–asylum dyads from 1980 to 2023, which documents returns systematically undercounted by UNHCR and provides new global data on *de facto* refugee conditions and UNHCR financing.

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1 Introduction

When do refugees return? Both the international refugee regime and existing research generally expect return to follow improvements in conditions at home. The regime is designed to protect refugees in asylum for as long as they would face persecution or serious harm upon return, with the principle of *non-refoulement* as its legal cornerstone.¹ Once those conditions improve, UNHCR treats voluntary repatriation as its preferred “durable solution” for ending refugee crises.² Existing research points in the same direction: studies of return intentions generally find that improved origin-country security is the strongest predictor of willingness to return (e.g., Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025; Al Husein and Wagner, 2023).

Global historical patterns of refugee return, however, do not align with these expectations. The Global Refugee Return (GRR) dataset introduced below, which covers 293 origin–asylum dyads from 1980 to 2023, reveals two striking patterns. First, four-fifths of recorded returns occurred while conflict was ongoing in the country of origin. Second, peace often does not lead to mass return: in only one-third of post-war cases did even half of refugees return within five years. Refugees therefore frequently return when prevailing accounts predict that they should remain abroad, while many remain displaced when those accounts predict that they should return.

Why do refugees return during war, yet often remain in asylum after peace? To answer these questions, I argue that we have to look beyond refugees’ individual or household-level preferences to the population-level conditions that determine whether those preferences can be acted upon. Most quantitative research examines return intentions within particular contexts, showing that willingness to return depends especially on security at home relative to conditions in asylum (e.g., Alrababah et al., 2023; Al Husein and Wagner, 2023; Adema et al., 2025). This helps explain where refugees would prefer to live, but not necessarily when large refugee populations actually move.

I show that realized return is concentrated around two population-level shifts that map onto constraints on refugees’ choices: whether they are able to remain in asylum and whether they have the resources required to return. First, a large share of return occurs amid asylum-side push, particularly exclusion policies designed to force refugees out and conflict in the country of asylum. Both exclusion policies and conflict can make staying in asylum untenable even when conditions at home remain dangerous. Almost half of recorded return occurs in years with either asylum shock, compared with fifteen per cent in the three years after war ends. These pressures are common because

¹The principle is codified in Article 33(1) of the 1951 Convention Relating to the Status of Refugees and its 1967 Protocol. It is also recognized as a rule of customary international law and therefore binds states regardless of whether they are parties to the Convention.

²The other two durable solutions are local integration and third-country resettlement. UNHCR has treated voluntary repatriation as its preferred durable solution since the mid-1980s (Barnett and Finnemore, 2004; Loescher, 2001).

most refugees live in low- and middle-income countries, often in neighboring states where they may face new conflicts or governments that restrict their ability to remain (UNHCR, 2025).

Second, peace may fail to produce return not only because refugees prefer to remain in asylum, as existing research emphasizes, but also because those who want to return may lack the resources to do so. After years of restricted employment or camp residence, refugees may be unable to finance transportation and rebuild their lives at home. This makes UNHCR a critical but largely neglected actor in explaining when return occurs. Its repatriation programs can relax feasibility constraints by providing transportation and reintegration assistance, but whether such assistance is available depends not only on conditions of displacement, but also on donor funding and decisions by international bureaucrats.

To evaluate this argument, I use within-dyad models, stacked event studies, observable implications, and historical cases. Within dyads, severe exclusion policies are associated with a threefold increase in return, low-intensity conflict in asylum with a 2.2-fold increase, and high UNHCR repatriation funding with a 1.8-fold increase, compared with 2.7-fold following war endings. Event studies show similar increases after the onset of asylum-side push and major repatriation programs, with little evidence of pre-trends. Return is also highly concentrated, varies across asylum countries for refugees from the same origin, and remains associated with asylum-side push even during ongoing conflict at home. Historical cases illustrate these mechanisms, with Rwandan refugees returning at different times because of war and coercion in neighboring asylum countries, and Eritrean refugees remaining in Sudan after independence until a UNHCR repatriation program began.

These findings do not imply that conditions at home are unimportant. War endings are associated with substantially higher return, at magnitudes comparable to exclusion policies, and refugees are less likely to return to more repressive origin countries. Other origin-country conditions, including income and conflict intensity, are less consistently associated with return. The broader point is that improvements at home alone do not explain when most realized return occurs globally. Asylum-side push and repatriation assistance are at least as important empirically, yet have received far less attention in existing research.

This research makes three contributions. First, despite the centrality of repatriation to the international refugee regime, macro-level patterns of return remain understudied. Existing quantitative work has taught us a great deal about where specific refugee groups would prefer to live, but much less about when displacement actually ends. I provide the first global analysis of realized refugee return using improved historical data and examine coercive repatriation and the role of UNHCR in facilitating return, two factors long discussed in historical and legal scholarship but rarely examined systematically, in part because cross-national data have not previously been available.

Second, the paper makes a major data contribution. The principal existing UNHCR data systematically undercount coerced returns and therefore mischaracterize when and why refugees return. With 36 research assistants, I constructed the GRR dataset by digitizing and coding 7,526 dyad-years of archival country reports based on field visits, interviews, and media reporting. GRR provides a more accurate measure of realized return alongside new explanatory measures of exclusion policies and UNHCR repatriation financing. When I re-estimate the analysis using UNHCR data, the resulting patterns are difficult to reconcile with asylum push factors or existing explanations, consistent with UNHCR’s systematic undercounting of forced return. GRR also records broader *de facto* asylum conditions, complementing *de jure* datasets such as the Dataset of World Refugee and Asylum Policies (DWRAP) (Blair, Grossman and Weinstein, 2022).

Third, the paper connects research on refugees and international security with IPE scholarship on international regimes. Relative to regimes governing trade, finance, the environment, and human rights, the international refugee system has received comparatively little systematic attention in quantitative IPE research.³ The data and results here enable an assessment of compliance with its core legal principle of *non-refoulement*, revealing systematic violations in some cases and continued protection in others. This variation raises a question central to research on international regimes (e.g., Chayes and Chayes, 1993; Simmons, 2000; Mitchell, 1994; Stone, 2011) but rarely examined systematically in the refugee context: why are core rules upheld in some cases but violated at scale in others, and when do international organizations constrain rather than accommodate those violations?

This article proceeds as follows. I first review existing research and document the prevalence of return during conflict. I then develop the argument, introduce the GRR dataset and its advantages over UNHCR data, and test the theory using within-dyad models, event studies, and observable implications, before tracing the proposed mechanisms through Rwandan and Eritrean cases. I conclude by discussing the implications of the findings and directions for future research.

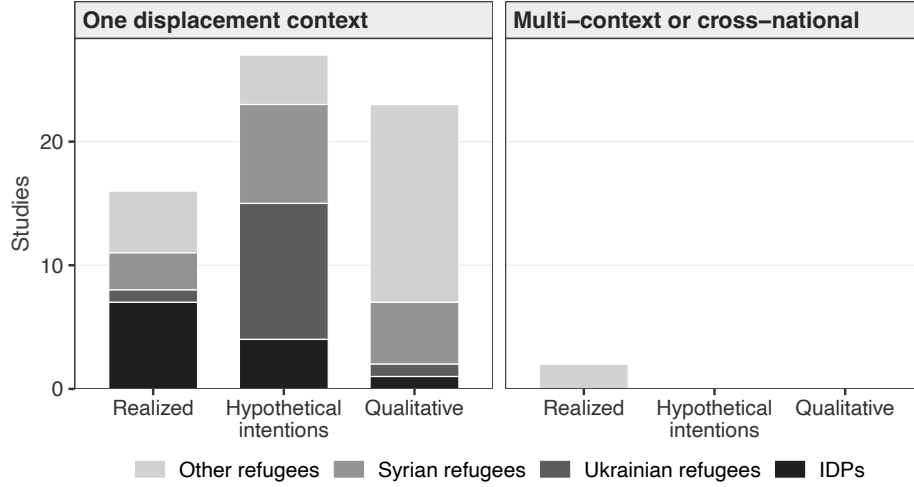
2 Puzzling Patterns of Refugee Return

Figure 1 shows that existing research on refugee return has focused primarily on stated return intentions within particular displacement contexts, especially among Syrian and Ukrainian refugees.⁴ Drawing on household models of migration choice (Borjas, 1987),

³Important policy, IR, historical, and legal work on the refugee regime includes Loescher (2001), Barnett (2001*a*), Chimni (2000), Skran (1995), Betts (2009), Betts (2010), Abdelaaty (2021), Arar (2017), and Davutoglu (2025).

⁴The main cross-national exception is Zakirova and Buzurukov (2021), which examines realized return but emphasizes many of the same political and economic factors used to explain return intentions. It also relies on UNHCR return data, whose limitations are discussed in Section 4.1.

Figure 1: Literature review of empirical research on refugee return



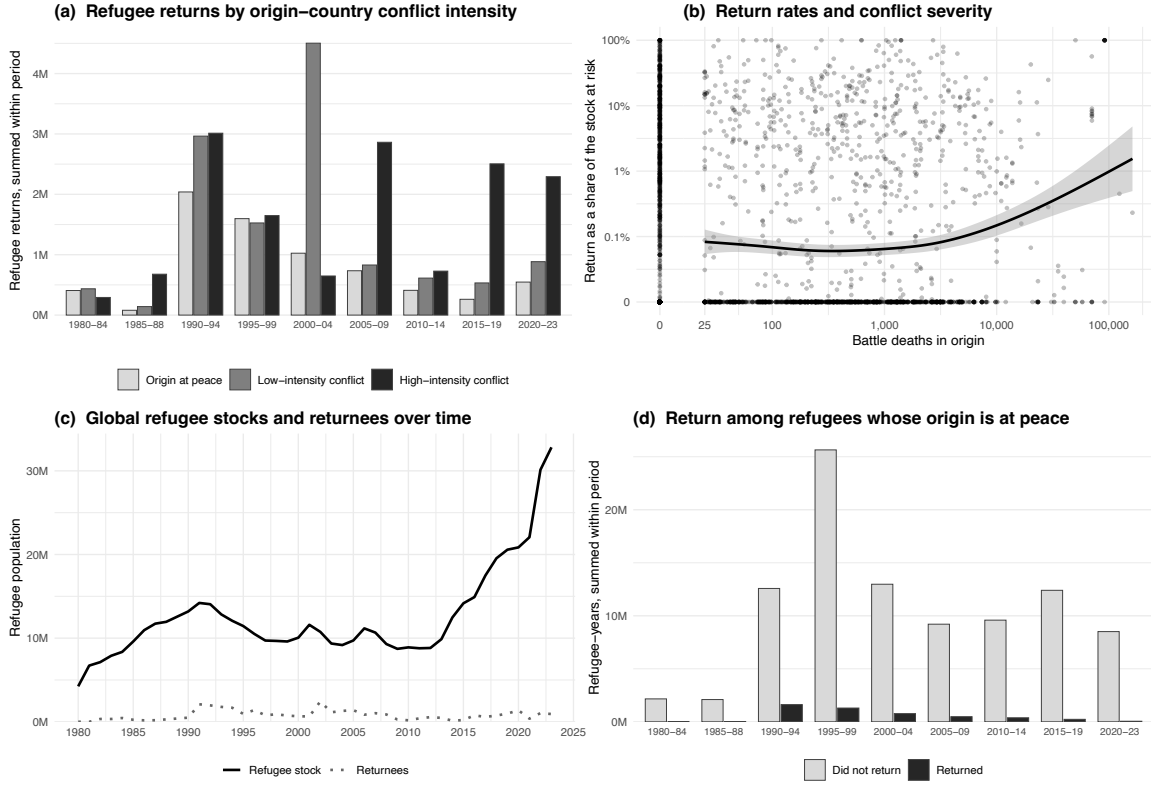
Notes: Search: 19 journals, 2006–2026; citation tracing; 2024 World Bank–UNHCR Joint Data Center review; targeted searches. Displacement contexts may span multiple origins within one host or one origin across multiple hosts; cross-national studies span multiple contexts. See Section 4 of the appendix for the study list.

this work generally asks whether refugees want to return once it is safe to do so, reflecting security concerns and attachment to their country of origin, or whether economic opportunities and living conditions in asylum make remaining abroad more attractive.

The consistent finding of this research is that origin security conditions are the primary determinant of return intentions (Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025). The “threshold model” developed by Alrababah et al. (2023), for example, argues that refugees consider asylum-country economic factors only once a minimum level of safety at home has been reached: Syrian refugees in Lebanon were unlikely to consider returning while Syria remained unsafe, even when they faced poor living conditions in asylum. Political conditions and attachment to home also shape return intentions (Kaya and Orchard, 2020; Ghosn et al., 2021). By contrast, evidence on asylum-country conditions is mixed: some studies find they shape return intentions, others find little relationship, and Beaman, Onder and Onder (2022) find that better conditions in exile are associated with greater realized return (Kayaoglu, Şahin-Mencütek and Erdoğan, 2022; Müller-Funk and Fransen, 2023; Alrababah et al., 2023).

This literature yields two expectations for realized return. First, return should be more common when the origin country is at peace. Second, large-scale return should be rare while intense conflict continues. To examine whether realized return follows these expectations globally, I use the Global Refugee Return (GRR) Dataset, which records annual refugee stocks and return flows for 293 origin–asylum dyads from 1980 to 2023 and is described in Section 4.2.

Figure 2: Return during conflict and non-return during peace



Notes: Panel (a): return by origin-conflict intensity (UCDP categories). Panel (b): return rate by log battle deaths. LOESS smoothers; 1989 omitted. Panel (c): global refugee stock and recorded return. Panel (d): return among war-displaced refugees during peace. Sources: GRR; UCDP/PRIO.

Figure 2 shows a striking pattern: most refugees return to war, not peace. Panel (a) shows that between 1980 and 2023, four-fifths of all recorded returns took place while conflict was ongoing in the country of origin.⁵ Panel (b), which scales returns by the refugee stock at risk, reinforces this point, showing that returns do not decline as battle deaths increase.

Panels (c) and (d) show the other side of the puzzle: peace often does not lead to return. Panel (c) plots the total refugee stock against returns over time and shows that returns have generally been rare. Across dyad-years with more than 1,000 refugees, the median return rate is zero and the mean is 2.99%; in 89% of dyad-years, fewer

⁵Some of this pattern reflects exposure, since the largest refugee populations disproportionately come from countries experiencing protracted conflict. Dyad-years with conflict in the origin constitute 34.1% of observations but contain 72.5% of refugee stock-years at risk and account for 79.0% of recorded returns. Thus, conflict's share of return exceeds its share of the population at risk by only 6.5 percentage points. But exposure does not resolve the puzzle: it explains why most refugees at risk come from countries at war, not why they return while war continues.

than 5% of refugees return. Panel (d) shows that return remains limited even when origin countries are at peace. Among origin–asylum dyads with no ongoing conflict at home, the average annual return rate is only 3.03%.⁶ Only 8 of 24 wars that ended and remained at peace for at least five years saw half of the refugee population return within that period.

3 Theory

Why do refugees often return during ongoing conflict, yet remain in asylum even after peace? To understand these patterns, I argue that we need to look beyond refugees’ hypothetical preferences over where they would ideally live to the constraints that determine whether those preferences can be acted upon. Refugees who prefer to remain abroad may nevertheless return when conditions in asylum make staying very difficult, while those who want to return may lack the resources needed to move. These constraints are difficult to study through within-context surveys of individual intentions, which have limited leverage on the population-wide shocks that drive major return episodes.

3.1 Constraints on Remaining in Asylum

Two types of shocks can make remaining in asylum very difficult: host-government coercion and armed conflict in the country of asylum. First, host governments may pressure refugees to leave before conditions at home have improved. Despite the prohibition on forced return under the 1951 Refugee Convention, states may circumvent or ignore *non-refoulement* (Schwartz, 2025; Chimni, 2004; Harrell-Bond, 1989; Long, 2013; Stein and Cuny, 1994). Deportation is the most severe and obvious violation. But governments may also revoke residence permits, restrict access to work and services, withdraw assistance, close camps, or threaten detention and expulsion. These measures can make remaining increasingly difficult while the resulting return is still recorded as “voluntary.”⁷ Consider the treatment of Afghan refugees in Iran in 2005:

“UNHCR [complained] that Iran was pressuring [Afghan refugees] to return by suspending education and medical services and revoking their residence permits. UNHCR also accused Iran of levying a punitive tax on refugees to force them to repatriate. Afghan media reports accused Iranian police of beating refugees and demanding bribes to extend residence permits.”

⁶This calculation is restricted to origin–asylum dyads in which refugees generally fled war rather than broader non-conflict persecution.

⁷UNHCR may facilitate repatriation under these conditions, both to reduce the harm refugees would face if expelled without assistance and because it depends on host-government cooperation to maintain access to refugee populations (Barnett, 2001*a,b*; Barnett and Finnemore, 2004; Whitaker, 2003). Its participation does not necessarily establish that refugees would have returned in the absence of government pressure.

Second, armed conflict in the country of asylum can make remaining there more dangerous than returning home. Most refugees remain near their countries of origin, often in neighboring states that share similar structural risk factors for conflict and are vulnerable to the cross-border spread of war (UNHCR, 2025; Fearon and Laitin, 2003; Collier and Hoeffler, 2004; Gleditsch, 2007; Lischer, 2005). Refugees may therefore escape one conflict only to encounter another in asylum. Violence can directly harm refugees or disrupt the humanitarian assistance on which they depend. Under these conditions, refugees may return even while their country of origin remains at war. The 1996 return of Rwandan refugees from the DRC illustrates this mechanism:⁸

“In October, an outbreak of civil war in Zaire [DRC], which engulfed Rwandan refugee camps there, produced what UNHCR described as the ‘largest and swiftest’ repatriation in memory. About a half-million Rwandan refugees returned home during a four-day period in mid-November. Most returnees repatriated on foot, jamming the main Rwandan highway from the border. The sea of humanity stretched nearly 100 miles during the height of the return.”

1997, Zaire [DRC]

When refugees are forced to leave asylum, onward movement to a third country is possible in principle but often infeasible. Onward migration may seem like a realistic alternative in high-income asylum countries, where refugees have often already traveled considerable distances. But most refugees live in low- and middle-income countries, often near the border they crossed and without the resources or legal permission to travel elsewhere. Many remain in peripheral regions because of community ties or government efforts to contain them there (Camarena, 2024); 79% of UNHCR camps in Africa lie within 50 kilometers of an international border (Fein and Van Den Hoek, 2022). Reaching a third country may therefore require a long, costly, and illegal journey that authorities can block, particularly when host governments are seeking to return refugees home (Whitaker, 2003). When conflict drives refugees from asylum, traveling farther into the host country may also be dangerous. For many refugees unable to remain in asylum, their country of origin is therefore the only destination they can realistically reach.

3.2 Constraints on Returning Home

Returning home can require far more resources than an ordinary international move. Refugees are often returning to places devastated by war, where homes and farmland have been destroyed, local economies have contracted, and there may be few public

⁸The country was officially known as Zaire in 1996 and was renamed the Democratic Republic of the Congo in May 1997. For consistency with current usage, I refer to it as the DRC in the text; quotations and source labels retain the contemporaneous name.

services. Return therefore requires not only paying for transportation across a border, but resources to rebuild livelihoods and to sustain families while finding new shelter and income sources in a post-conflict economy.

These costs are especially difficult to bear because displacement often limits refugees’ ability to accumulate resources. Many spend years in camps or other settings where access to employment is restricted.⁹ At the same time, remaining in asylum may offer a degree of material security, whether through humanitarian assistance or through informal work, businesses, and social networks established over years or even decades. Returning means giving up these relatively predictable sources of support for a much riskier environment. Refugees may therefore want to return while lacking the resources needed to rebuild their lives at home.

UNHCR repatriation programs can make such returns feasible. These are often large logistical operations that provide transportation, information, cash, food, shelter materials, legal assistance, and support for restoring housing and livelihoods. The repatriation of Sierra Leonean refugees from Guinea in the early 2000s illustrates the scale of this support:

“About half of the returning refugees received assistance from UNHCR and other agencies in the form of transportation, cooked meals, and medical care during the journey home. . . Returnees received blankets, cooking utensils, shelter materials, and a two-month food supply. UNHCR appealed for a nine-fold increase in its budget, to \$18 million, to support reintegration programs including schools, health clinics, and livelihood programs.”

2002–2004, Guinea & Sierra Leone

These operations require substantial donor financing, and funding availability can determine how many refugees UNHCR is able to assist and the resources it can provide per person. In 2023, for example, more than 24,000 Burundian refugees had expressed a desire to return, but UNHCR reported a funding shortfall of more than \$13 million for transportation and assistance restarting their lives. At UNHCR’s annual pledging conference, High Commissioner Filippo Grandi said “repatriation is possible, is demanded by refugees, but we can do much less than we would like because there is lack of resources” (Grandi, 2023).

3.3 Formalization

For clarity, I formalize the argument using a standard random utility model of migration choice (Borjas, 1987). Such models allow individual preferences to remain partly unobserved while deriving how changes in common conditions affect the probability

⁹Fifty-five percent of refugees live in countries that significantly restrict their right to work in practice (Ginn et al., 2022).

of return. This is particularly useful here because major shocks need not induce every refugee to leave. Instead, they shift the relative attractiveness of return for many refugees at once, causing a larger share of the population to cross individual return thresholds.

I extend the standard framework in two ways. First, I incorporate asylum-country push factors, allowing refugees to return even when conditions at home remain unsafe. Second, I incorporate the transition costs of return, allowing refugees who would prefer living at home to remain in asylum because undertaking the move is costly.

Refugee i compares the expected utility of remaining in the asylum country a with that of returning to the origin country o . Utility from remaining in asylum is

$$U_{ai} = v_a - \phi_a - \kappa_a + \varepsilon_{ai},$$

where v_a captures the average value of living in the asylum country based on general economic, political, security, and social conditions. The terms $\phi_a \geq 0$ and $\kappa_a \geq 0$ capture population-level shocks that enter separately from these baseline conditions: ϕ_a captures host-government exclusionary policies directed at refugees, while κ_a captures insecurity generated by armed conflict in the asylum country. The term ε_{ai} captures individual-specific factors affecting the value of remaining in asylum, in terms of individual deviations from the average values.

Utility from returning to the origin country is

$$U_{oi} = v_o - \tau + r + \varepsilon_{oi},$$

where v_o captures the value of living in the origin country based on general economic, political, security, and social conditions, including psychological attachment to a refugee's home country. The term $\tau \geq 0$ captures the transition cost of return, while $r \geq 0$ captures resources available to support return, including refugees' private resources and UNHCR repatriation assistance. The term ε_{oi} captures individual-specific factors affecting the value of return.

I say that a refugee has a baseline *preference for return* when

$$v_o + \varepsilon_{oi} > v_a + \varepsilon_{ai}.$$

Whether refugee i actually returns, however, depends on the full utility comparison:

$$U_{oi} > U_{ai} \iff v_o - \tau + r + \varepsilon_{oi} > v_a - \phi_a - \kappa_a + \varepsilon_{ai}.$$

Rearranging,

$$v_o - v_a + \phi_a + \kappa_a + r - \tau > \varepsilon_{ai} - \varepsilon_{oi}. \quad (1)$$

The idiosyncratic utility terms imply that refugees differ in their thresholds for return, so changes in common conditions shift the share who undertake return. Improved conditions at home raise U_{oi} , while exclusionary pressure or conflict in asylum reduce U_{ai} and can induce return even without improvements at home. Repatriation resources instead offset the transition cost τ , increasing the likelihood that a preference for return translates into realized movement. Thus, refugees may prefer to return but remain abroad because moving is costly, or may return despite poor conditions at home because remaining in asylum has become still less attractive.

If the distribution of $\varepsilon_{ai} - \varepsilon_{oi}$ is centered at zero and approximately bell shaped, Equation 1 also clarifies why large return movements should generally require large changes in common conditions. When few refugees return, the common determinants of return are sufficiently unfavorable that only individuals with unusually strong idiosyncratic reasons for returning cross the return threshold. A large increase in return therefore requires a substantial shift in common conditions, large enough to move many more refugees across their individual thresholds. Such shifts may come from a large improvement in v_o , as when conflict in the origin country ends; a large increase in κ_a , as when conflict begins or intensifies in the asylum country; or a substantial increase in $r - \tau$, most plausibly through an increase in the resources r available to support return.

3.4 Empirical Implications

The model generates two main hypotheses. First, exclusionary pressure and conflict in asylum reduce the utility of remaining there, without requiring any improvement in origin-country conditions.

H1 (Asylum-side push). *Refugee return increases following severe exclusion policies and armed conflict in the asylum country, including when conflict continues in the country of origin.*

Second, repatriation assistance offsets the transition costs of return and therefore increases the likelihood that refugees undertake the move.

H2 (Repatriation assistance). *Refugee return is higher when a major UNHCR repatriation program is operating and increases with the resources available for repatriation assistance.*

Beyond these average associations, the model generates four observable implications. First, return should be concentrated in particular dyads and years rather than occur as a steady flow. Common shocks can move many refugees at once, whereas household-level changes are more dispersed across people and time. Second, if conditions in asylum

independently shape return, refugees from the same origin should return at different rates across asylum countries depending on push factors, holding origin-country conditions constant. Third, because asylum-side push does not require an improvement at home, its association with return should persist while conflict continues in the country of origin. Fourth, if resources constrain return, improvements in origin-country security should produce more return when repatriation assistance is available than when it is not.

These predictions complement rather than replace conventional explanations. More favorable conditions in the origin country should continue to raise v_o relative to v_a and therefore increase return. The model instead highlights additional factors related to asylum conditions and international organization financing that can shape whether those preferences translate into realized movement.

4 Data

4.1 Limitations of UNHCR Data

Testing the theory requires cross-national data on realized return. The principal existing source, UNHCR’s population statistics, has important limitations for this purpose. These data are designed primarily to track voluntary repatriation and can systematically undercount return when refugees are coerced or forcibly removed.

Undercounting can occur for several reasons. First, refugees may intentionally be removed from UNHCR’s statistics before they are forcibly returned. Host governments sometimes withdraw refugee status precisely to facilitate deportation; once individuals are no longer classified as refugees, their subsequent movement would not be recorded as a violation of *non-refoulement*. Forced returns may also occur where UNHCR has limited access, reducing its ability to observe and record them. Governments hostile to refugee protection may restrict its activities, while conflict in the country of asylum may force personnel to evacuate. For example, UNHCR records no returns from South Africa to Mozambique despite reports of removals:

“With no UNHCR presence in South Africa ... [n]either country recognizes them as refugees ... During 1985, South Africa was returning some 1,500 a month to the danger they had fled.”

1986, South Africa

Refugee population figures can also become politically contested. Host governments may understate return to preserve higher refugee counts and associated aid. In Somalia, for example:

“Somalia refused to accept any reduced estimate ... UNHCR reluctantly contin-

ued to provide assistance for the supposed 840,000 refugees . . . [before] publicly refut[ing] that figure, saying that the actual number is less than half that.”

1989, Somalia

These measurement problems are not random. Returns are especially likely to go unrecorded when refugees lose formal status, move outside assisted channels, or return from settings where humanitarian access is limited. Analyses based only on UNHCR data may therefore understate forced return precisely in the cases central to this paper.

4.2 The Global Refugee Return Dataset

To study realized return, I created the Global Refugee Return (GRR) Dataset. GRR includes origin populations with at least 25,000 refugees in exile in any year, consistent with UNHCR’s definition of a protracted refugee situation, and adopts a broad definition of refugees that includes formally recognized refugees, asylum seekers, and unregistered refugees. For each origin population, I include the principal asylum countries that collectively host at least 75% of that population.¹⁰

To address the measurement problems described above, GRR draws primarily on contemporaneous country reports that combine locally collected information with official and nonofficial sources. These reports often capture refugee dynamics missing from administrative data, including unregistered populations, spontaneous and forced returns, and disputes over official figures. Their coverage is also less dependent on the presence of any single international organization and, because the reporting organizations did not administer refugee assistance, less directly exposed to the political pressures surrounding aid-linked population estimates.

The principal source is the *World Refugee Survey* (WRS), published annually by the U.S. Committee for Refugees and Immigrants from 1980 through 2008. USCRI describes its reporting process as follows:

“[A] rigorous method including, when possible, site visits to the countries on which we report. In most countries, we have Research Partners—local human rights NGOs that interview officials, refugees, and representatives of local civil society and international agencies based on our standard questionnaire . . . We consult with and seek the input of representatives of each government . . . We receive extensive input from the field offices of UNHCR and UNRWA. Finally, we crosscheck all of this with secondary sources in the media, on the internet, and in major governmental and nongovernmental human rights reports.”

USCRI, 2008

¹⁰The 75% threshold is calculated using UNHCR data, supplemented with data on Palestinian refugees under UNRWA and Mozambican refugees in South Africa, whom the UNHCR data list under an unknown country of asylum.

The *core* dataset covers Global South asylum countries from 1980 through 2008, where WRS reporting is most consistent.¹¹ The *extended* series adds high-income asylum countries and continues through 2023 using the U.S. Department of State’s annual country reports on human rights practices. These reports contain country-level reporting on refugees and migrants, including forced return, exclusion policies, and unregistered populations (U.S. Department of State, 2024). Because the *State* reports do not always provide refugee stock or voluntary return estimates, the extended series uses UNHCR figures as a baseline and supplements them with additional evidence on forced returns and unregistered populations. The extended series therefore provides broader geographic and temporal coverage, while the core series offers more consistent measurement.

Constructing GRR required digitizing WRS reports, many available only as scans or physical volumes. I hand-coded the 25 largest origin populations, covering 83 dyads, and worked with 36 trained research assistants to code the remainder using a common protocol. I then reviewed and adjudicated every observation, returning to the original reports, adjacent years, and additional desk research when GRR and UNHCR figures differed substantially. To my knowledge, this is the first project to digitize and systematically code the full WRS corpus, producing 7,526 dyad-year observations.

A comparison with UNHCR data, reported in Section 1.3 of the appendix, lends support to the measurement concerns raised above. Across the GRR core sample, UNHCR records 19.1 million returns compared with 19.5 million voluntary returns in GRR, indicating close agreement on voluntary repatriation. However, UNHCR data appear to underreport coerced returns. GRR records an additional 3.6 million forced returns that UNHCR does not capture, meaning that UNHCR records only 83% of total realized return. Nearly one-third of forced-return episodes in GRR are entirely absent from UNHCR data, and almost half of forced-return volume occurs in episodes that are either absent or only minimally recorded.

5 Empirical Analysis

The empirical analysis first estimates within-dyad associations using PPML and stacked event studies, then examines additional observable implications of the theory and briefly assesses the aggregate importance of the principal predictors.

5.1 Within-Dyad PPML Estimates

I estimate Poisson pseudo-maximum likelihood (PPML) models with dyad and year fixed effects. The unit of analysis is the directed origin–asylum dyad-year, and the

¹¹It contains 105 dyads across 58 origin populations and 61 asylum countries. The WRS was not published in 1990, so GRR has no observations for 1989.

Table 1: Explanatory variables

Variable	Measure	Source	Period
Asylum-country push			
Exclusion Policy	Host-government pressure to induce return: 0 = none, 1 = moderate, 2 = severe. Severe policies are implemented measures intended to affect at least 10% of the refugee stock; coding is based on policy, not subsequent return. <i>Enters as separate moderate and severe indicators.</i>	WRS; U.S. State Department	1980–2023
Conflict in COA	Armed conflict in asylum: none, low (25–999 battle deaths), or high ($\geq 1,000$); based on conflict location. <i>Enters as separate low and high indicators.</i>	UCDP	1980–2023
UNHCR repatriation assistance			
Repatriation Assistance	UNHCR repatriation funding per refugee in the origin country’s global refugee population, measured at the origin-year level. <i>Enters as high and low indicators, split at the median of positive values (\$19.88 per refugee per year).</i>	UNHCR ExCom; Yearbooks	1980–1999; 2011–2020
Existing explanations			
Conflict in COO	Armed conflict in origin: none, low (25–999 battle deaths), or high ($\geq 1,000$); matched to the conflict producing displacement. <i>Enters as separate low and high indicators.</i>	UCDP; author matching	1980–2023
War Ending	First three peaceful years after the displacement-producing war ends, provided conflict does not resume. <i>Binary.</i>	Fearon (2017); author extension	1980–2023
Civil Liberties Restrictions (COO)	Freedom House civil liberties index: 1 = greatest protection, 7 = least protection. <i>Standardized.</i>	Freedom House	1980–2023
GDP (COO)	GDP per capita in the country of origin. <i>Standardized log.</i>	World Bank WDI	1980–2023
GDP (COA)	GDP per capita in the country of asylum. <i>Standardized log.</i>	World Bank WDI	1980–2023
Refugee Rights	De jure refugee and asylum policy; higher values indicate more liberal policies. <i>Standardized.</i>	DWRAP (Blair, Grossman and Weinstein, 2022)	1980–2022
Conflict Duration	Age of the displacement-producing conflict spell, used as a proxy for time in asylum; one quiet year is tolerated and the counter resets after two. <i>Standardized log, $\log(\text{years} + 0.01)$.</i>	Derived from UCDP	1980–2023

dependent variable is the number of refugees who return during the year. Table 1 summarizes the explanatory variables, their measurement, sources, and temporal coverage.

I employ a PPML estimator because it accommodates a non-negative, highly skewed outcome with many zeros without transforming the outcome or dropping zero-return observations. In the model, dyad fixed effects absorb time-invariant differences across refugee populations and origin–asylum relationships, so the estimates are identified from changes within the same dyad over time. Year fixed effects absorb shocks common to all dyads. Because the analysis is observational, I interpret the estimates as conditional associations rather than causal effects.

I include the log refugee population at risk as an offset, so coefficients describe propor-

tional differences in the return rate rather than differences driven by population size. The population at risk is the previous year’s refugee stock.¹² I restrict the analysis to dyad-years with more than 1,000 refugees at risk. The baseline model is

$$E[Y_{it} \mid \mathbf{X}_{it}] = \exp(\beta' \mathbf{X}_{it} + \alpha_i + \gamma_t + \ln S_{it-1}), \quad (2)$$

where Y_{it} is the number of returnees in dyad i and year t , $\mathbf{X}_{it}$ is the vector of time-varying covariates, α_i and γ_t are dyad and year fixed effects, and $\ln S_{it-1}$ is the offset. Standard errors are clustered by dyad, with alternative clustering strategies reported in appendix Section 2.4.1.

Table 2 presents the main estimates, and panel (a) of Figure 3 plots the corresponding return-rate ratios. Model 1 is the primary specification. Model 2 adds UNHCR repatriation assistance and uses the smaller subset of origin-years for which funding data are available. The mean return rate in the Model 1 sample is 6.6%.¹³

H1 predicts that asylum-side push increases return even when conditions at home have not improved. The strongest evidence is for exclusion policies. Relative to years without an exclusion policy, a severe policy is associated with 2.97 times the return rate (coefficient 1.09, SE = 0.19), the largest association in panel (a) of Figure 3. A moderate policy is associated with a 62% higher return rate (coefficient 0.48, SE = 0.22; return-rate ratio = 1.62).

Conflict in the country of asylum is also associated with higher return. Low-intensity conflict is associated with 2.24 times the return rate (coefficient 0.81, SE = 0.21), while high-intensity conflict is associated with 1.80 times the rate (coefficient 0.59, SE = 0.31). The low-intensity estimate is statistically distinguishable from zero, whereas the high-intensity estimate is less precise, and a Wald test does not reject equality of the two coefficients. The results therefore support an association between conflict in asylum and return, but not a clear difference by conflict intensity. This finding is less robust than that for exclusion policies, and leave-one-region-out checks in Section 2.5.3 of the appendix suggest that the association is concentrated in Sub-Saharan Africa.

H2 predicts that refugees who want to return may nevertheless remain abroad if they lack the resources to move, so return should increase with the resources available for

¹²In 21 dyad-years, a large inflow arrives and returns within the same year, causing return to exceed the lagged stock; for these observations, I use the estimated within-year peak population.

¹³This is higher than in the broader estimation frame because fixed-effects PPML excludes dyads that never record return. The estimation sample contains 2,327 dyad-years across 110 dyads, from an estimation frame of 5,131 dyad-years in 285 dyads. Of the 175 excluded dyads, 148 record no return in any year, 25 are lost to covariate missingness, and 2 are single-year dyads absorbed by the fixed effects. The fitted sample contains 77.7% of all return recorded in GRR and 58.7% of refugee stock-years in the estimation frame. Section 3.3 of the appendix reports a cross-national model that retains dyads without within-dyad variation in return.

Table 2: Poisson pseudo-maximum likelihood estimates of refugee return

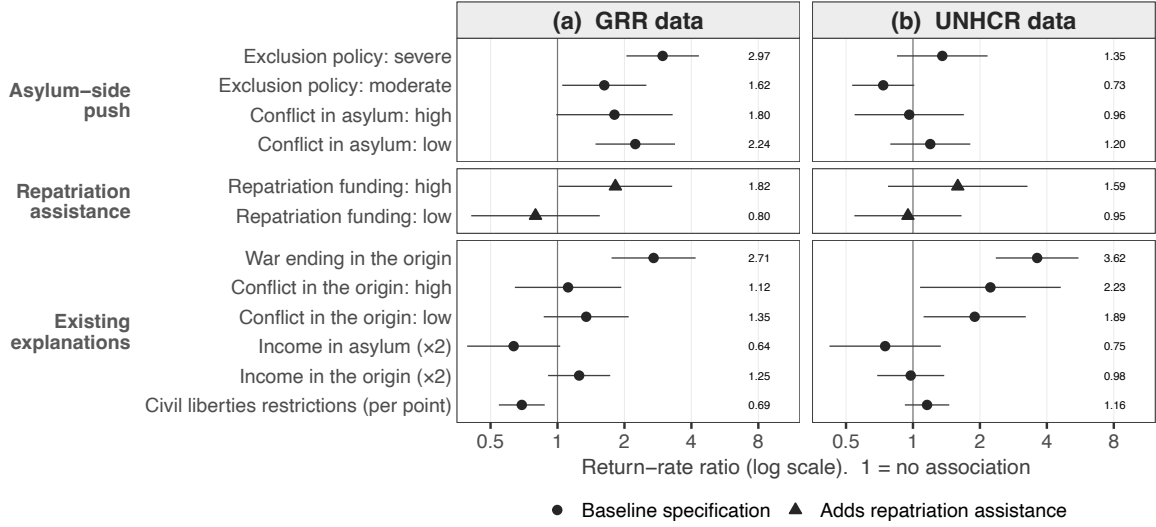
Term	(1) Main	(2) With assistance
Asylum-side push		
Exclusion policy: severe	1.09*** (0.19)	1.31*** (0.35)
Exclusion policy: moderate	0.48* (0.22)	0.16 (0.27)
Conflict in COA: high	0.59 [†] (0.31)	0.88* (0.44)
Conflict in COA: low	0.81*** (0.21)	0.95** (0.35)
Repatriation assistance		
Repatriation funding: high	—	0.60* (0.30)
Repatriation funding: low	—	-0.23 (0.34)
Existing explanations		
War Ending	1.00*** (0.22)	0.97** (0.36)
Conflict in COO: high	0.11 (0.28)	0.35 (0.34)
Conflict in COO: low	0.30 (0.22)	0.35 (0.25)
GDP (COO)	0.39 (0.28)	0.67 (0.47)
GDP (COA)	-1.09 [†] (0.59)	-1.23 (0.82)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.09 (0.24)
Refugee Rights	-0.02 (0.17)	-0.10 (0.22)
Conflict Duration	-0.03 (0.08)	-0.11 (0.13)
Specification and fit		
Dyad and year fixed effects	Yes	Yes
Sample	Extended GRR	Funding coded
Years	1980–2022	1980–1999, 2011–2020
Dyad-years	2,327	1,378
Dyads	110	100
Pseudo R ²	0.799	0.783

Notes: PPML with the log refugee stock at risk as an offset and standard errors clustered by dyad. Column 2 includes repatriation assistance and is estimated on a smaller sample.

repatriation. Consistent with this expectation, high funding is associated with 1.82 times the return rate relative to none (coefficient 0.60, SE = 0.30), and moving from low to high UNHCR repatriation funding is associated with approximately 2.3 times the return rate. Section 2.2.3 of the appendix shows similar patterns using two alternative measures of repatriation assistance.

Among existing explanations, the clearest predictor is the end of war. The post-war window is associated with 2.71 times the return rate (coefficient 1.00, SE = 0.22). By contrast, neither low- nor high-intensity conflict in the origin generally is significantly associated with return relative to peace. Greater civil-liberties restrictions in the origin are associated with less return, while higher asylum-country income is also negatively associated with return, though imprecisely estimated. Origin-country income, refugee rights in asylum, and conflict duration are not distinguishable from zero.

Figure 3: Predictors of refugee return in GRR and UNHCR data



Notes: PPML; dyad and year FE; log refugee-stock offset; dyad-clustered SEs; 95% CIs. Panel (a): GRR. Circles: Model 1 of Table 2, N=2,327 dyad-years. Triangles: Model 2 of Table 2, N=1,378. Panel (b): UNHCR (Section 1.3.4 of the appendix). N=2,126. Triangles: N=1,207.

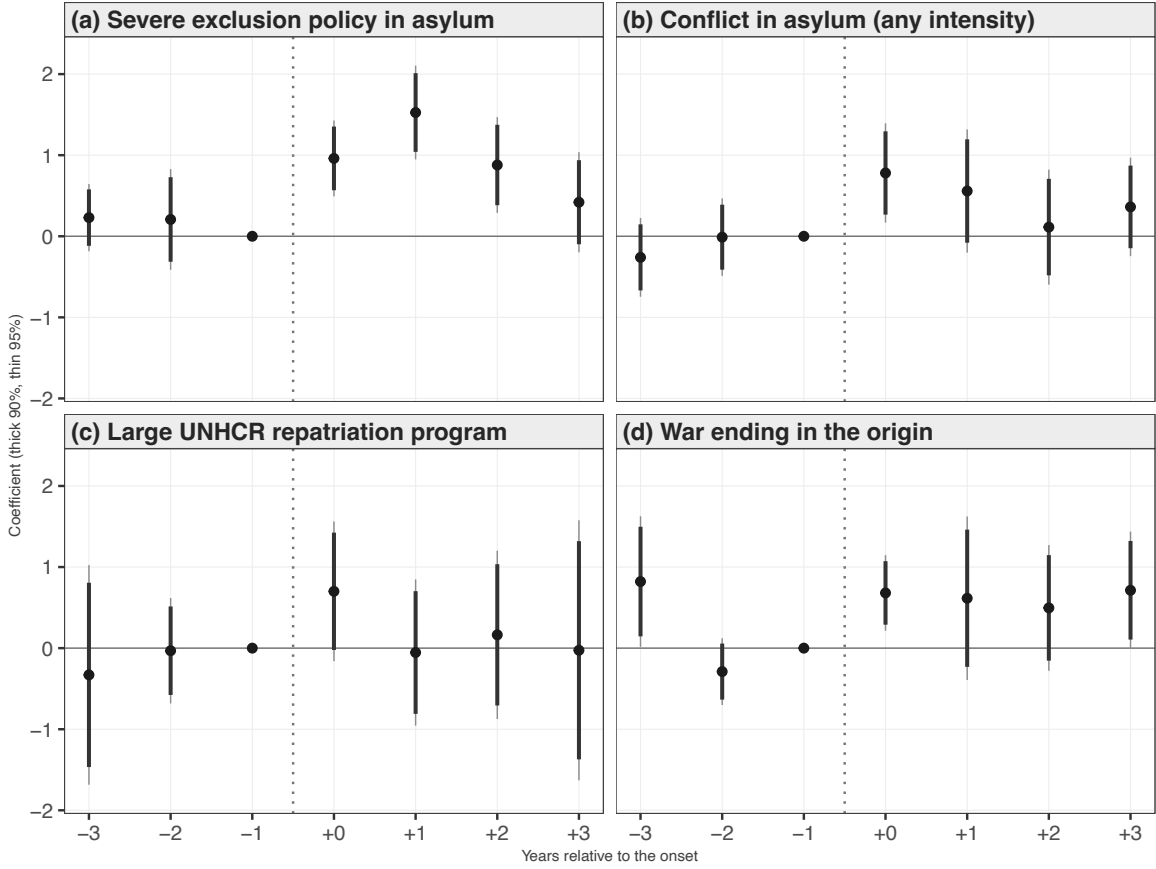
5.2 Event-Study Evidence

I next examine the timing of these relationships using stacked event studies. This provides a complementary use of the within-dyad variation underlying Table 2 and, importantly, tests whether return was already changing before each condition occurred. Because these conditions are not absorbing, I stack a separate comparison around each onset. Treated dyads are observed from three years before to three years after onset and compared with dyads that remain unexposed to the corresponding condition throughout the window; the year before onset is the reference period.¹⁴

Figure 4 broadly reproduces the main PPML findings while providing assurance that the asylum-side relationships do not appear to reflect return already underway. Following the onset of a severe exclusion policy, return rises to approximately 2.6 times its pre-onset level in the onset year and 4.6 times the following year. Conflict in the asylum country is likewise followed by approximately 2.2 times as much return in the onset year. In both cases, the pre-onset coefficients are jointly indistinguishable from zero ($p = 0.55$ and $p = 0.45$, respectively). For comparison, war endings are likewise followed by higher return, but exhibit a statistically significant pre-trend ($p = 0.011$). Return was therefore already increasing before the war formally ended.

¹⁴A cohort enters only if the condition is observed and absent throughout the three preceding years. Conditions may persist or lapse after onset. Section 2.6 of the appendix reports the specification and additional details.

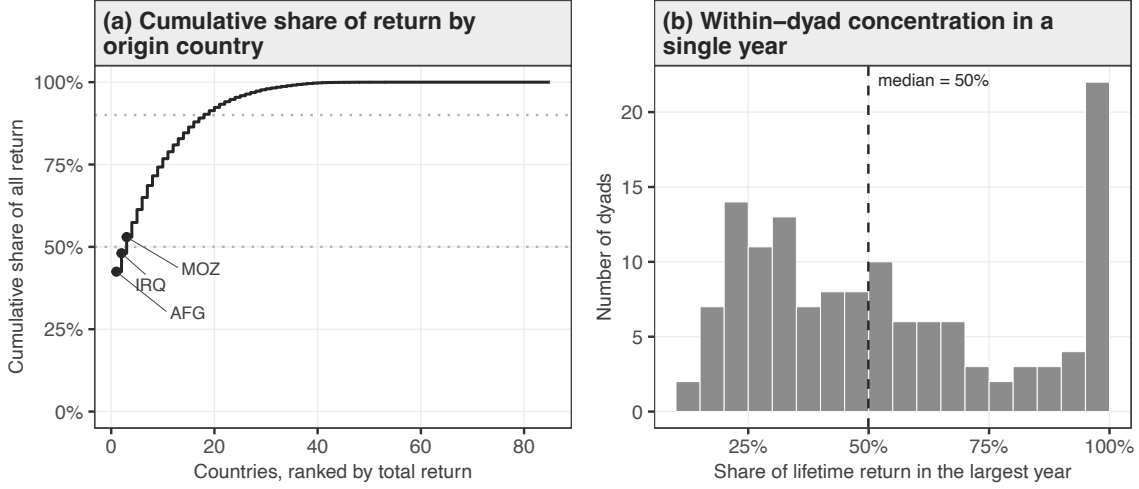
Figure 4: Event-study estimates of refugee return



Notes: Stacked PPML; ± 3 yrs; cohort $\times$ dyad and cohort $\times$ year FE; log stock-at-risk offset; dyad-clustered SEs; $t-1$ omitted. Cohorts require the condition absent 3 prior yrs; controls unexposed in-window. Bars: 90%/95%. Cohorts and treated dyads, with the p -value from a joint test that all pre-onset coefficients are zero: (a) 23 cohorts, 17 treated dyads, $p = 0.55$; (b) 45 and 36, $p = 0.45$; (c) 29 and 24, $p = 0.85$; (d) 49 and 44, $p = 0.01$.

The pattern for repatriation assistance is similar but less precisely estimated, potentially reflecting the smaller sample given the shorter period for which assistance data are available. Return approximately doubles when a major UNHCR repatriation program begins, while there is no evidence that return was already changing beforehand ($p = 0.85$ for the joint test of the leads). This timing is reassuring against the concern that UNHCR initiates repatriation programs only after return is already underway, although it cannot rule out broader selection into where and when assistance is provided, including the possibility that UNHCR anticipates conditions favorable to future return.

Figure 5: Concentration of recorded refugee return



Notes: Panel (a): extended GRR sample, 7,498 dyad-years, 292 dyads, 85 origins; dotted lines: 50th/90th percentiles of cumulative return. Panel (b): 135 dyads observed for at least 5 yrs with recorded return.

5.3 Observable Implications

The theory generates several observable implications beyond the average associations examined above. First, if population-level shocks are important for realized movement, return should be concentrated in particular places and years rather than occur as a steady annual flow. Figure 5 shows precisely this pattern. The top 5% of dyad-years account for 93.8% of return, and the top 1% for 63.5%. Among dyads that record any return, 49.6% have at least half their lifetime return in one year, while 19.3% experience at least 90% in one year. The timing of these episodes also suggests that war endings alone do not account for this concentration: asylum-side shocks coincide with 47.4% of recorded return, compared with 15.4% during war-ending windows.

Second, if conditions in asylum independently shape realized return, refugees from the same origin should return at different rates across asylum countries. I test this using origin-by-year fixed effects, which absorb all origin-specific conditions in a given year and identify asylum-side associations from differences across host countries. Section 2.4 of the appendix shows that the main patterns persist: refugees are more likely to return from asylum countries experiencing exclusionary policies or conflict than are refugees from the same origin elsewhere. Complementary asylum-by-year fixed effects likewise show that, within the same host country and year, groups targeted by exclusionary policies return at higher rates than untargeted groups.

Third, asylum-side push should increase return even when conditions at home remain

unsafe. Interaction models reported in Section 3.2 of the appendix show that the associations between severe exclusion policies, asylum-country conflict, and return are similar whether conflict continues in the country of origin or has ended.

Fourth, if resources constrain return, improvements in origin-country security should translate into greater realized movement when repatriation assistance is available. The descriptive pattern is consistent with this implication. In dyad-years in the post-war period without a UNHCR repatriation program, the return rate is 4.4% compared to 31.3% when a UNHCR program is operating. The within-dyad interaction points in the same direction: a war ending is associated with 1.8 times the return rate without substantial repatriation funding, compared with 3.4 times the rate when substantial funding is present. The interaction is imprecisely estimated (0.66, SE = 0.56), so I interpret this evidence as consistent with, rather than a decisive test of, the resource mechanism.

Beyond these observable implications, I assess the aggregate importance of the principal predictors. I use Model 2 to compare fitted return under the observed data with fitted return when each factor is set to its reference value, holding other covariates fixed. These simulations are illustrative rather than causal and ask how much lower fitted return would be in the absence of each condition. Fitted aggregate return is 22.1% lower without severe exclusion policies, 15.1% lower without large repatriation assistance, and 12.1% lower without high-intensity conflict in asylum, compared with 9.9% lower without the post-war window.

6 Alternative Explanations and Robustness

I briefly discuss three broad concerns about the results here and report the full set of robustness analyses in Section 2 of the appendix. First, I examine alternative explanations for the main results. One possibility is that the relationship between conflict and return reflects anticipatory return before wars end rather than asylum-country push factors. There is some evidence of this: return is modestly elevated before war endings, as shown in the event-study evidence in Section 5.2. However, anticipation cannot account for the main result. Only 11 of 129 severe-policy dyad-years fall within three years before a war ending, and excluding all dyad-years within three years on either side of an ending leaves the severe-policy estimate essentially unchanged.

Another possibility is that higher return during conflict reflects failed asylum claims: people flee, host governments reject their claims, and they return to conflict, with this movement counted as repatriation. This is unlikely to explain the results. Stock estimates exclude people turned away at the border, while the return data and archival reports show that large return episodes generally involve recognized refugees or people who had already been living in the asylum country without formal status. Large asylum-seeker populations are also concentrated in the Global North, because many

Global South governments automatically grant *prima facie* refugee status to people fleeing particular conflicts (UNHCR, 2026). Finally, severe exclusion policies tend to target large, long-established refugee populations (see Section 3.5 of the appendix).

Section 3.3 of the appendix also considers a broader set of explanations, including differences across dyads. Changes in the political standing of refugees' principal ethnic groups in the origin country, or of co-ethnics in asylum, do not predict return. Across dyads, return is lower between countries sharing a widely spoken language and higher where religious composition is more similar, while I find no relationship with transnational ethnic kin.

A second concern is selection into repatriation assistance: UNHCR may direct resources toward cases where it anticipates substantial return. Such selection is likely, but does not imply that assistance plays no causal role. The event study shows little evidence that return was already increasing before major programs began, and assistance often arrives after peace: 24% of post-war programs begin at least two years after war ends and 10% at least five years later. In the fourteen cases where assistance begins at least three years after war's end, return averages 0.8% in the preceding two years and 9.9% over the following three. Among funded cases, greater funding per refugee is also associated with higher return (coefficient 0.82, SE = 0.21), consistent with a resource mechanism in which assistance relaxes constraints on return. These patterns do not eliminate selection, but suggest that resources help determine whether favorable conditions translate into realized movement. The Eritrean case below illustrates this mechanism.

The third set of checks concerns the validity of the exclusion policy measure. Because coding archival reports requires judgment, I make each classification auditable by providing the supporting text and coding rules in the GRR data file. Policies are coded based on their intent and implementation, not on whether forced return actually occurs. Consistent with this, 36% of moderate-policy and 8% of severe-policy dyad-years record no return. Severe policy remains statistically significant when the outcome is restricted to returns coded as voluntary. As a consistency check, exclusion policies are less common where DWRAP records stronger protections against refoulement.¹⁵

I examine the robustness of the findings across alternative measures of return, conflict, and war endings; different estimators and clustering strategies; lagged covariates and additional fixed effects; alternative samples, including core GRR; and the exclusion of refugee-warrior cases, particular regions, and the largest origins, asylum countries, dyads, and return episodes. The severe-policy association is the most robust, remaining positive and significant across nearly all of these checks. The asylum-country conflict

¹⁵To further assess the measure's validity using an independent source, I am constructing a parallel measure from Amnesty International reports using the same coding rules. The data collection and analysis is underway.

and repatriation funding results are less robust but generally persist. Among the existing explanations, restrictions on civil liberties at home are relatively robust, while the post-war association is more sensitive.

I also assess the value of the new data by examining whether the findings replicate with UNHCR’s official statistics. Panel (b) of Figure 3 re-estimates the same specifications using UNHCR return counts and shows markedly different patterns. The associations with asylum push factors largely disappear, a moderate exclusion policy is associated with *less* recorded return, UNHCR-recorded return is *higher* during conflict in the country of origin, and there is no clear association between repatriation funding and return. These patterns are difficult to reconcile either with the asylum-side mechanisms documented above or with the conventional expectation that return should increase as conditions at home improve. They are instead consistent with the measurement evidence that UNHCR omits many coerced returns captured by GRR. Without a ground-truth measure, however, this comparison cannot establish that measurement differences fully explain the discrepancy.

7 Case Studies

To illustrate the mechanisms underlying the cross-national results, I examine two historical cases. The first compares Rwandan refugees across neighboring countries in 1996, showing how asylum conditions produced return at different times despite broadly similar conditions at home. The second examines Eritrean refugees in Sudan, showing that peace did not immediately produce large-scale return, but increased after UNHCR repatriation assistance became available.

7.1 Asylum-Side Push: Rwandan Refugees in Burundi, the DRC, and Tanzania, 1996

After the 1994 genocide and the subsequent victory of the Tutsi-led Rwandan Patriotic Front, hundreds of thousands of Hutu fled to neighboring countries, including members of the former government and armed forces implicated in the genocide. Although the new Rwandan government encouraged return and UNHCR intensified repatriation efforts in 1995, most refugees remained in Burundi, the DRC, and Tanzania. Many feared violence or arrest in Rwanda, while former-regime leaders intimidated refugees and sometimes prevented them from leaving the camps. Efforts to promote voluntary repatriation therefore initially produced relatively little return.

Mass return instead followed distinct shocks in each country of asylum. In Burundi, the government increasingly pressured Rwandan refugees to leave during 1996. Officials told refugees they were no longer welcome, soldiers and local militias attacked and burned camps, and authorities forcibly transported refugees across the border. In

July, soldiers expelled 15,000 refugees and temporarily blocked UNHCR access to two camps. By year's end, "about 98 percent of Rwandan refugees had left Burundi, due to fear for their safety and coercive policies by Burundian authorities" (1997, Burundi).

In the DRC, repatriation efforts likewise produced little movement until civil war reached the refugee camps in October 1996. Rebel attacks dispersed some former-regime leaders who had obstructed departure, while the fighting itself forced refugees from the camps. More than half a million returned within four days in what was described as the "largest and swiftest repatriation in memory" (1997, Zaire). At least 200,000 instead moved deeper into the DRC, showing that many did not prefer to return.

Return from Tanzania followed yet another asylum-side shock. Only 2% of Rwandan refugees returned during the first eleven months of 1996. On December 5, following the return from the DRC, the government proclaimed that "all Rwandese refugees can now return to their country in safety" and imposed a year-end deadline for departure (1997, Tanzania). Hundreds of thousands initially moved deeper into Tanzania, away from Rwanda, but security forces blocked their movement, closed the camps, and directed them toward the border. Return subsequently reached as many as 15,000 people per hour, while approximately 50,000 avoided repatriation by dispersing elsewhere in Tanzania or presenting themselves as Burundian.

The staggered timing across these three countries is difficult to explain through changes in Rwanda alone. Refugees faced the same Rwandan government and broadly similar conditions at home, yet mass return followed coercion in Burundi, conflict reaching the camps in the DRC, and an expulsion order in Tanzania. Their efforts to move deeper into the DRC and Tanzania further show that many did not prefer to return. The episode therefore illustrates how asylum-side shocks can produce mass return without corresponding improvements at home.

7.2 Post-War Return and Repatriation Assistance: Eritrean Refugees in Sudan, 1991–2005

Hundreds of thousands of Tigrayans and Eritreans fled Ethiopia's civil war to seek refuge in neighboring Sudan. In May 1991, the Mengistu regime fell and the EPLF took control of Eritrea, ending the conflict. Yet only about 10,000 refugees returned in the year the war ended, despite the improvement in security at home.

One obstacle was the absence of a large repatriation operation. A dispute between the Eritrean government and UNHCR over the scale and financing of reconstruction assistance prevented the immediate launch of a return program. Eritrea sought more rehabilitation assistance than donors were willing to provide. A UNHCR repatriation program eventually began in 1994 and approximately 81,000 refugees returned over

the following two years, but roughly 320,000 remained in Sudan. Return subsequently stalled amid “financial disputes over the scope of the repatriation program, and deteriorating relations between the governments of Sudan and Eritrea” (WRS, Sudan, 1997), alongside insecurity near the border.

A second repatriation effort around 2000 provided transportation, reintegration grants, and housing assistance. Approximately 130,000 refugees returned over the following five years. The timing is consistent with the mechanism in H2: after the war had already ended, return increased when institutional and financial support reduced the costs of moving and rebuilding a life in Eritrea.

Many refugees continued to remain in Sudan even once assistance became available. By this point, some had lived there for decades and had established livelihoods and social ties. In the early 1990s, half lived in camps with access to food aid, health and education programs, and technical training, often near relatives or co-ethnics and with access to agricultural land, while the remainder lived in urban areas. When encouraged to return in 2001, “many would-be returnees chose to remain longer in Sudan to harvest their crops” (WRS, Sudan, 2002). A survey reported that roughly 85 percent of refugee households considered assistance levels in Eritrea an important determinant of whether they would return.

Even the formal termination of refugee status did not produce universal repatriation. In 2002, UNHCR invoked the cessation clause, requiring Eritreans to “repatriate, file individualized asylum claims to remain in Sudan as refugees, or take steps to become permanent legal residents of Sudan” (WRS, Sudan, 2003). Only 25,000 repatriated in 2002 and 9,500 in 2003, while many remained in Sudan.

Together, the cases illustrate why origin-country security alone does not determine realized return. Rwandan refugees returned when coercion and conflict made remaining in asylum difficult, while Eritrean return increased only after organized repatriation assistance became available. At the same time, many Eritreans chose to remain when given that option. Whether refugees actually move therefore depends not only on conditions at home, but also on the constraints they face in asylum and the resources available for return.

8 Conclusion

This paper revisits a central expectation of the international refugee regime: that refugees receive protection while conditions at home remain unsafe and return once those conditions improve. Using the Global Refugee Return dataset, which covers 293 origin–asylum dyads from 1980 to 2023, I show that realized return often follows a different pattern. Most recorded returns occur while conflict is still ongoing in the country of origin, while many refugees remain abroad long after war has ended.

I argue that realized return depends not only on refugees' individual or household-level preferences, but also on population-level conditions that determine whether remaining abroad and undertaking return are feasible. Host-government coercion and conflict in asylum can make staying very difficult, while repatriation assistance can make return possible for refugees who otherwise lack the resources to move.

The empirical results support both parts of the argument. Within the same refugee populations over time, severe exclusion policies and conflict in asylum are associated with substantially higher return, including while conflict continues at home, while greater UNHCR repatriation assistance is also associated with more return. Stacked event studies show return rising after the onset of each, with little evidence that these patterns were already underway beforehand. The examination of broader observable implications supports these results: return is highly concentrated in particular dyads and years, refugees from the same origin return at different rates across asylum countries, and post-war return is substantially higher when repatriation assistance is available.

This analysis helps to reconcile existing research on return intentions with global patterns of realized movement. Individual preferences shape whether refugees want to return and may depend primarily on safety at home, but conditions in asylum and the resources available for return help determine whether they actually do. Refugees may return without improvements at home when remaining in asylum becomes very difficult, while those who want to return may remain abroad after peace because they lack the resources to move.

These findings complicate how observed repatriation should be interpreted within the refugee regime. Voluntary return after peace can provide a durable solution when refugees want to return and can do so safely, but return itself is not necessarily evidence that these conditions have been met. Some of the largest movements instead occur under coercive pressure and while conflict continues at home, while other returns depend heavily on international assistance to finance transportation and reintegration, assistance that may be threatened by recent aid cuts. The conditions under which return occurs may shape both returnees' welfare and reintegration (Fransen, Ruiz and Vargas-Silva, 2017) and the consequences of mass return for origin communities and post-conflict politics (Schwartz, 2019; Blair, Van Dijke and Wright, 2026).

The analysis also underscores the difficulty of measuring refugee return, showing that existing UNHCR statistics omit many coerced returns. GRR reconstructs refugee stocks and realized return and adds new data on exclusion policies, but it still relies on expert reporting and remains an imperfect measure of movements that are often difficult to observe precisely. Other poorly measured displacement dynamics remain beyond its scope and important for future research, including efforts to prevent access to asylum through maritime pushbacks and border closures (FitzGerald, 2019; Goodwin-Gill, 2011; Schwartz, 2025), as well as temporary and cyclical movement between origin

and asylum (Masterson and Vidarte, 2026).

Several questions follow. First, when and why do governments impose exclusion policies? Section 3.5 in the appendix suggests that severe coercion is most consistently associated with the size of the refugee population and tends to emerge two to three years after displacement begins. Future research should examine the domestic pressures behind these decisions, the role of international aid and diplomacy, and whether host governments face meaningful costs for violations. It should also examine the role of origin governments, which may encourage return to signal normalization but resist it when they lack capacity to absorb returnees or view particular groups as political or security threats.

More broadly, the paper connects the refugee regime to longstanding questions about the performance of international institutions. *Non-refoulement* is the regime's core legal principle, yet the analysis reveals systematic violations at scale. UNHCR may even facilitate return under coercive conditions when refusing to do so would leave refugees without assistance or jeopardize its access. This variation raises a question familiar from research on other international institutions but rarely examined systematically in the refugee regime: why are core rules upheld in some cases but violated in others, and when do international organizations constrain rather than accommodate those violations?

Taken together, the findings show that repatriation does not automatically follow the restoration of peace. Displacement may end because protection in asylum erodes or foreign aid makes return possible, or it may persist long after the conflict that produced it has ended. Explaining when displacement ends therefore requires understanding the political processes that determine whether states continue to provide asylum or push refugees to leave, when donors finance repatriation, and when UNHCR facilitates return, alongside where refugees actually want to live.

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Peace or Push? Understanding Global Patterns of Refugee Return

Mae MacDonald

Supplementary Materials

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1 Global Refugee Return (GRR) Dataset

The Global Refugee Return (GRR) dataset is an original global dataset on refugee stocks, returns, and the conditions shaping repatriation. It covers origin–asylum dyad-years from 1980 to 2023. The dataset makes two core contributions: (1) improved stock and repatriation data that corrects systematic errors in UNHCR figures and incorporates forced returns; and (2) new indicators of *de facto* refugee conditions in asylum, including coercive return policies.

Unit of analysis. Each observation records conditions for refugees from a specific country of origin residing in a specific country of asylum in a given year.

Geographic coverage. The dataset includes all origin countries with at least 25,000 refugees in exile in any given year, consistent with UNHCR’s threshold for a protracted refugee situation. For each origin country, the dataset includes the principal asylum countries that together account for at least 75% of the origin’s total refugee population, calculated using UNHCR refugee stock data. Dyad-years in which no refugee population is present are excluded.

Temporal coverage. The dataset spans 1980–2023. The year 1989 is missing for all variables coded from the World Refugee Survey (WRS), because the 1990 WRS report was not published.

Primary sources. GRR draws on three primary sources: (1) the World Refugee Survey (WRS), 1980–2008, providing annual country reports on refugee stocks and conditions in asylum; (2) the U.S. Department of State Country Reports on Human Rights Practices (USHR), 2009–2023, with data extracted from Section 2.d on freedom of movement; and (3) the UNHCR population statistics database, which supplies the stock and voluntary-return baseline for the years after 2008 and for the high-income asylum countries that enter only in the extended series. Additional variables are drawn from the UCDP Armed Conflict Dataset, the Ethnic Power Relations (EPR) Dataset, DWRAP (Blair et al., 2022), the World Bank World Development Indicators, and the CEPII GeoDist dataset. The UNHCR financing measures are compiled from UNHCR Executive Committee reports and archival documents for 1980–1999 and from UNHCR statistical yearbooks and financial reports for 2011–2020. Neither source contributes to the refugee stock or return series.

Core and extended versions. There are two versions of GRR. The *core* dataset covers Global South asylum countries from 1980 through 2008 and rests on the World Refugee Survey evidence in which I have the greatest confidence: 1,623 dyad-years across 105 dyads, 58 origin populations and 61 asylum countries. The *extended* series adds high-income asylum countries and continues through 2023 using the U.S. Department of State’s annual country reports on human rights practices, giving 7,526 dyad-years across 293 dyads, 87 origin populations and 93 asylum countries. Because the State Department reports do not always provide stock or voluntary-return estimates, the extended series takes UNHCR figures as a baseline and supplements them with additional evidence on forced returns and unregistered populations. The extended series therefore offers broader geographic and temporal coverage, and the core series more consistent measurement.

Limitations. GRR remains an imperfect measure of refugee stocks and returns. The *World Refugee Survey* was designed to document conditions in asylum rather than to provide a precise administrative series, and its reports sometimes give rounded estimates or ranges. In high-income countries, where refugee populations from many individual origin countries are small, the reports often do not disaggregate information by origin. In these cases, asylum-country policies are coded as applying to all origin-country dyads. This motivates the restriction of the core dataset and the use of UNHCR stock and voluntary-return figures as a baseline in the extended series. Finally, variables drawn from the WRS are missing for 1989 because no 1990 survey was published; I leave this gap missing rather than interpolate across it.

1.1 Summary Statistics

Table 1 reports summary statistics for every variable in the main specification. Statistics are computed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year). Figure 1 shows the distribution of the dependent variable. The left panel displays the full range of total repatriation counts with the y-axis capped to show detail among non-zero values. The right panel plots positive values on a log scale, showing a roughly log-normal distribution among dyad-years with nonzero repatriation. The heavy

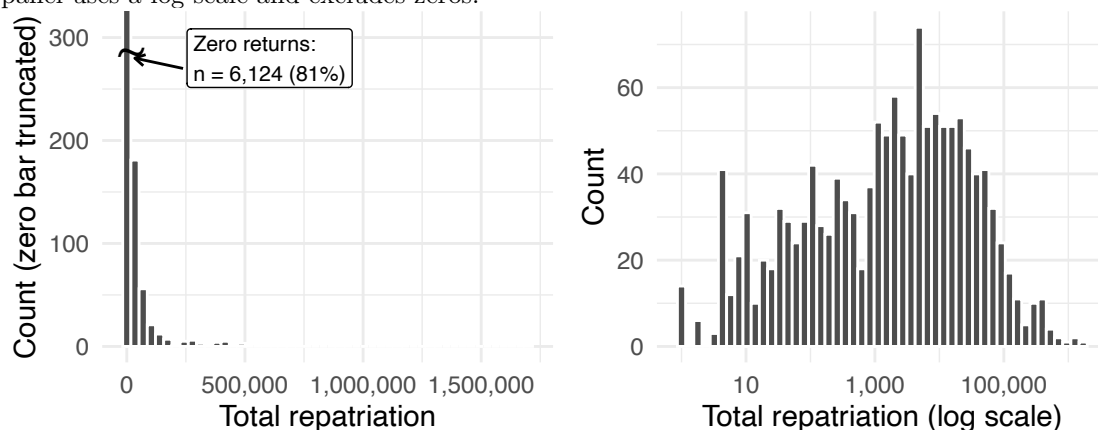
concentration of zeros, extreme right skew, and count nature of the dependent variable motivate the use of Poisson pseudo-maximum likelihood estimation with the log stock at risk as the offset.

Table 1: Summary statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Return and stock					
Total Repatriation	5109	6,513.72	48,844.54	0.00	1,700,000.00
Refugee Stock (Lagged)	5131	129,495.83	400,251.60	1,002.00	6,013,249.00
Return Rate (% of stock at risk)	5109	2.99	11.03	0.00	100.00
Asylum-side push					
Exclusion policy: severe	4415	0.03	0.18	0.00	1.00
Exclusion policy: moderate	4415	0.07	0.25	0.00	1.00
Conflict in COA: high	5131	0.07	0.26	0.00	1.00
Conflict in COA: low	5131	0.18	0.38	0.00	1.00
Repatriation assistance					
Repatriation funding: high	3015	0.14	0.34	0.00	1.00
Repatriation funding: low	3015	0.14	0.34	0.00	1.00
Existing explanations					
War Ending	5131	0.04	0.20	0.00	1.00
Conflict in COO: high	5131	0.12	0.32	0.00	1.00
Conflict in COO: low	5131	0.22	0.42	0.00	1.00
GDP (COO)	4845	-0.00	1.00	-2.98	2.22
GDP (COA)	5095	-0.00	1.00	-2.23	1.45
Civil Liberties Restrictions (COO)	5086	0.00	1.00	-2.82	1.44
Refugee Rights	4911	-0.00	1.00	-2.16	3.79
Conflict Duration	5131	0.00	1.00	-0.78	1.83

Note: Computed on the filtered sample: dyad-years with a lagged refugee stock above 1,000.

Figure 1: Distribution of the dependent variable (total repatriation). Left panel shows the raw distribution; right panel uses a log scale and excludes zeros.



1.2 Limitations of UNHCR Data

The main paper briefly discusses why UNHCR data may undercount forced returns. This section provides additional detail on the four sources of measurement error. The examples below are drawn from the *World Refugee Survey*.

Unregistered refugees. The term “refugee” encompasses several legal and statistical categories. UNHCR’s published definitions cover recognized refugees, asylum-seekers, people in refugee-like situations, and others in need of international protection. These data often miss unregistered refugees, including people who qualify as refugees but never apply for asylum and those whose refugee status is later revoked and who are subsequently treated as irregular migrants.

This gap matters especially for measuring forced return, because withdrawing formal status can itself facilitate removal. Governments may revoke refugees' status before deporting them, so those individuals may no longer be counted as refugees and their return may not appear in official refugee statistics. The Croatian government's treatment of Bosnian refugees in 1995 illustrates the problem:

"Croatia revoked the refugee status of 100,000 Bosnian refugees ... saying it was now safe for them to return. ... The irony here is that the Croatian army, which is occupying many of those same towns, has refused to allow Muslims from Bosnia to come into those same places."

1996, Croatia

Dependence on institutional presence. Accurate return estimates often depend on UNHCR's access and operational presence in the country. UNHCR may lack access because governments restrict its activities, because it has limited resources or no office, or because war and insecurity force personnel to evacuate. This creates a systematic measurement problem, since some of the same conditions that increase forced return – host-government reluctance to cooperate on refugee protection, and conflict in asylum – also make UNHCR less able to observe and record those movements. UNHCR data record no returns from South Africa to Mozambique, for example, despite reports of large-scale removals:

"With no UNHCR presence in South Africa ... [n]either country recognizes them as refugees ... During 1985, South Africa was returning some 1,500 a month to the danger they had fled."

1986, South Africa

The Somali civil war created a similar problem when it reached areas hosting Ethiopian refugees. UNHCR evacuated its personnel, and the resulting returns do not appear in its data:

"Beginning in December 1990, as fighting in the Somali civil war intensified ... UNHCR personnel were evacuated from Somalia ... Large numbers of Ethiopian refugees streamed home to escape the fighting, many on foot. ... More than 34,000 returned in January alone. ... By the end of May, 64,000 returnees had reached Ethiopia."

1992, Ethiopia

Measurement of forced and spontaneous returns. UNHCR's return statistics are designed to measure voluntary repatriation, not forced return; its own data-content documentation describes the solutions series in those terms. Deportations and refoulement therefore generally fall outside the data, while coerced movements may still be recorded as voluntary when they occur through formal repatriation channels. The data can thus both omit forced returns and obscure coercion among the returns they do record. Spontaneous returns are also difficult to observe: refugees may go uncounted if they leave without deregistering or cross at unmonitored border points.

The politics of population figures. UNHCR figures can also become objects of political negotiation. Host governments may understate returns in order to maintain higher refugee counts and attract more foreign aid. UNHCR may accept these figures when it cannot verify them independently, or when challenging them could jeopardize government cooperation or access to refugee populations. Somalia provides one example:

"UNHCR had no firm estimate of the number of refugees in Somalia because the government did not permit a census. ... During 1988, large numbers of Ethiopian refugees returned to Ethiopia ... [but] [t]he Somali government denied that a mass repatriation had occurred. ... The Somali government claimed that about 537,000 refugees remained in the country at year's end, but some voluntary agencies put the total at less than 400,000."

1989, Somalia

These sources of error are systematic rather than random. Forced returns are especially likely to go unrecorded when refugees lose formal status, move outside assisted channels, return after humanitarian access has collapsed, or are classified as deportees rather than returnees. Analyses based only on UNHCR's return series may therefore understate forced return, particularly during ongoing conflict or following coercive policies in the country of asylum. Section 1.3 quantifies how much of GRR's recorded return the UNHCR series misses.

1.3 Comparison of GRR and UNHCR

This section compares GRR with UNHCR data on both refugee returns and stocks. The comparison uses the GRR core dataset and excludes populations not recorded by UNHCR, such as Palestinians under UNRWA; restricting the sample to dyad-years observed in both sources leaves 1,415 dyad-years across 96 dyads.

1.3.1 Comparing Voluntary and Forced Return

Table 2 compares the return data. At the aggregate level the two datasets agree closely on voluntary return. UNHCR records 19.1 million returns across the comparison sample, compared with 19.5 million voluntary returns in GRR.

Table 2: Comparison of GRR and UNHCR return data

	1980s	1990s	2000s	All
Sample				
Dyad-years compared	319	563	533	1,415
Dyads	51	83	79	96
Aggregate calibration				
GRR voluntary return	1,770,932	10,050,798	7,643,847	19,465,577
GRR forced return	259,027	957,565	2,337,900	3,554,492
UNHCR recorded return	1,250,530	9,757,187	8,105,436	19,113,153
UNHCR / GRR voluntary	0.71	0.97	1.06	0.98
UNHCR / GRR total	0.62	0.89	0.81	0.83
Episode capture				
Forced-return episodes	21	61	75	157
invisible in UNHCR	14 (67%)	14 (23%)	21 (28%)	49 (31%)
token in UNHCR	0	2	5	7
Share of forced-return volume				
Forced-return volume in invisible episodes	71%	16%	34%	32%
in invisible or token episodes	71%	26%	53%	47%
where UNHCR $\approx$ GRR voluntary	60%	36%	72%	62%

Note: Comparison of GRR core and UNHCR data. 2000s covers 2000–2008. Episode: dyad-year in which GRR records forced return. *Invisible*: UNHCR records no return in that year or either adjacent year; *token*: UNHCR records a positive number smaller than one-tenth of GRR forced-return. *UNHCR $\approx$ GRR voluntary* denotes forced-return volume in dyad-years where UNHCR’s figure is within 5% of GRR’s voluntary-return estimate.

Once GRR’s 3.6 million forced returns are included, however, UNHCR records only 83% of total realized return. Of the 157 dyad-years in which GRR identifies forced return, 49 (31%) are invisible in UNHCR: it records no return in that dyad-year, the preceding year, or the following year. Another seven appear only as token movements, where UNHCR’s count across those three years is positive but less than one-tenth of GRR’s forced-return estimate for that one dyad-year. Together, these episodes account for 47% of GRR’s forced-return volume.

Moreover, 62% of forced-return volume occurs in dyad-years where UNHCR’s figure closely matches GRR’s voluntary-return estimate, which suggests that the forced component is omitted rather than reclassified. In other cases coercive movements appear in the UNHCR data as voluntary repatriation. GRR classifies Rohingya returns from Bangladesh between 1994 and 1997 as forced, for example, while UNHCR records approximately the same number as voluntary.

As Table 2 shows, the discrepancies are greatest in the 1980s but persist into later decades. In the 1980s UNHCR records 71% of GRR’s voluntary-return total and two-thirds of GRR’s forced-return episodes are invisible; the correspondence improves considerably thereafter, but forced-return episodes remain likely to be omitted through the 2000s and the two series never fully converge. Neither dataset provides an independent ground truth for every movement, but the concentration of disagreement in forced-return episodes is consistent with the coverage problems set out in Section 1.2.

1.3.2 Comparing Refugee Stocks

Table 3 compares the stock data. Logged formal stocks correlate at 0.86 and are exactly equal in 7% of dyad-years. The two series therefore track each other closely on formal refugees.

However, similar to the returns data, there are discrepancies once unregistered refugees, often absent from UNHCR’s formal statistics, are included. For registered refugees, UNHCR records 0.82 of GRR’s total. For the broad stock, the ratio falls to 0.61, a difference of 90.6 million unregistered refugee stock-years. Of the 424 dyad-years in which GRR records an unregistered population, 108, or 25%, have no UNHCR stock at all, while another 62 have an UNHCR figure below one-tenth of GRR’s.

Table 3: Registered and unregistered refugee stocks, GRR against UNHCR

Quantity	1980s	1990s	2000s	All
Sample				
Dyad-years compared	371	638	605	1,614
Dyads	55	90	89	104
Aggregate calibration				
GRR registered stock	78,321,850	104,907,809	83,528,312	266,757,971
GRR unregistered stock	14,177,285	27,405,166	49,059,410	90,641,861
UNHCR recorded stock	73,605,396	86,767,560	58,795,551	219,168,507
UNHCR / GRR registered	0.94	0.83	0.70	0.82
UNHCR / GRR total	0.80	0.66	0.44	0.61
Population capture				
Dyad-years with unregistered population	98	162	164	424
invisible in UNHCR	38 (39%)	40 (25%)	30 (18%)	108 (25%)
token in UNHCR	5	26	31	62
Share of unregistered volume				
Unregistered stock-years in invisible dyad-years	35%	45%	10%	25%

Note: Core dataset only: 1980–2008, asylum outside Europe and North America. Stocks are summed over dyad-years, so the unit is a stock-year. A dyad-year is *invisible* when UNHCR records no stock for it and *token* when UNHCR’s figure is below a tenth of GRR’s broad stock. Built to the same row blocks as the comparison of voluntary and forced return above.

In years after 2008, which use the GRR extended dataset, GRR and UNHCR are more closely aligned. That convergence partly reflects a change in how GRR is constructed. After the *World Refugee Survey* ends, GRR takes UNHCR’s registered stock as its starting point, then adds unregistered refugee populations identified in State Department reports and other archival sources. The two series therefore share the same baseline after 2009, but GRR remains broader where those additional populations can be identified.

During the *World Refugee Survey* period, the two stock series are identical in 55.9% of dyad-years, while GRR exceeds UNHCR in roughly one-quarter. In the State Department period, they are identical in 95.3%, and GRR exceeds UNHCR in fewer than one dyad-year in twenty. Where GRR does exceed UNHCR, the median difference is 76%.

The return data show the same shift. GRR and UNHCR report identical return figures in 9.7% of dyad-years before 2009, compared with 84.0% after 2009. GRR’s independent contribution is therefore concentrated before 2009. In later years, it is better understood as UNHCR’s series supplemented with archival additions and corrections. The comparison between the two sources is therefore most informative for 1980–2008, when they were produced more independently.

1.3.3 Internal Consistency

Table 4: Internal-consistency diagnostics on the stock series

Series	Returns exceed prior stock (%)	Unexplained collapse (%)	V-shaped reversal (%)	Spike then collapse (%)	Unchanged from prior year (%)	Exact multiple of 10,000 (%)	Median abs. log change
GRR stock	0.365	3.28	0.848	0.348	8.37	7.49	0.111
UNHCR stock	0.437	3.54	0.941	0.282	4.15	1.82	0.113

The internal-consistency checks are broadly similar across the two sources. GRR is more likely than UNHCR to carry the previous year’s stock estimate forward and to report values rounded to multiples of ten thousand. Both patterns reflect the coding rules and the underlying archival sources. When a reported range does not change, GRR retains the previous estimate rather than shifting between the bounds, and the source reports themselves often use rounded figures. As a result, the GRR series is somewhat smoother within dyads but also less granular.

1.3.4 Replicating the Main Analysis

Table 5: Alternative return series and offsets

Term	(1) GRR returns GRR stock	(2) UNHCR returns UNHCR stock	(3) UNHCR returns UNHCR stock + assistance
Asylum-side push			
Exclusion policy: severe	1.09*** (0.19)	0.30 (0.24)	0.24 (0.32)
Exclusion policy: moderate	0.48* (0.22)	-0.31 [†] (0.16)	-0.55* (0.22)
Conflict in COA: high	0.59 [†] (0.31)	-0.04 (0.29)	-0.09 (0.42)
Conflict in COA: low	0.81*** (0.21)	0.18 (0.21)	0.28 (0.47)
Existing explanations			
War Ending	1.00*** (0.22)	1.29*** (0.22)	1.25** (0.46)
Conflict in COO: high	0.11 (0.28)	0.80* (0.37)	1.21* (0.47)
Conflict in COO: low	0.30 (0.22)	0.64* (0.27)	0.65 (0.45)
GDP (COO)	0.39 (0.28)	-0.04 (0.31)	-0.22 (0.66)
GDP (COA)	-1.09 [†] (0.59)	-0.69 (0.71)	-0.23 (0.63)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	0.20 (0.16)	1.00*** (0.28)
Refugee Rights	-0.02 (0.17)	0.02 (0.22)	-0.43 [†] (0.23)
Conflict Duration	-0.03 (0.08)	-0.02 (0.12)	-0.25 (0.16)
Repatriation assistance			
Repatriation funding: high	—	—	0.46 (0.37)
Repatriation funding: low	—	—	-0.05 (0.28)
Dyad-years	2,327	2,126	1,207
Dyads	110	107	94

Note: Every column is the paper’s specification with the outcome and offset swapped: dyad and yr FE, log stock at risk as offset, dyad-clustered SEs. Col. 1 reproduces the baseline. Col. 3 is the UNHCR analogue of main table col. 2, with funding as none / low (to \$19.88/refugee/yr) / high, none omitted. Samples differ: each column drops dyad-years with a missing or zero offset, and col. 3 also drops uncoded funding yrs. Significance: [†] 0.10, * 0.05, ** 0.01, *** 0.001.

This section replaces GRR’s return and stock measures with UNHCR data and examines how the results change. In Table 5, Column 1 reproduces the main specification, Column 2 replaces both the outcome and refugee stock with UNHCR measures, and Column 3 adds repatriation assistance to provide the UNHCR analogue of Model 2 in the main paper.

The results change substantially when UNHCR measures are used. Most importantly, the positive associations between return and both severe exclusion policies and conflict in the asylum country disappear. War ending remains strongly associated with return, while conflict in the origin becomes more positively associated with return. The repatriation-assistance coefficients retain the same ordering as in the main analysis, with a positive coefficient for high assistance and little association with low assistance, although both estimates are imprecise. The coefficient on civil-liberties restrictions in the origin also becomes positive when using the UNHCR data. Overall, substituting UNHCR measures substantially alters the estimated relationship between return and the asylum-country conditions at the center of the paper’s argument. The resulting pattern is consistent with the measurement concerns discussed in Section 1.2: UNHCR may record return more completely in difficult origin-country contexts where it is more actively involved, while recording less return in asylum countries where governments are hostile to refugees and UNHCR has more limited access.

1.3.5 Dyadic Return Patterns

Figures 2 and 3 plot refugee stocks and total returns over time for the two largest asylum destinations for each of the twenty largest origin countries by displaced population. GRR is shown in black and UNHCR in red. Across most dyads, UNHCR records lower stocks and returns than GRR.

Figure 2: Refugee Stocks and Returns by Dyad (1 of 2)

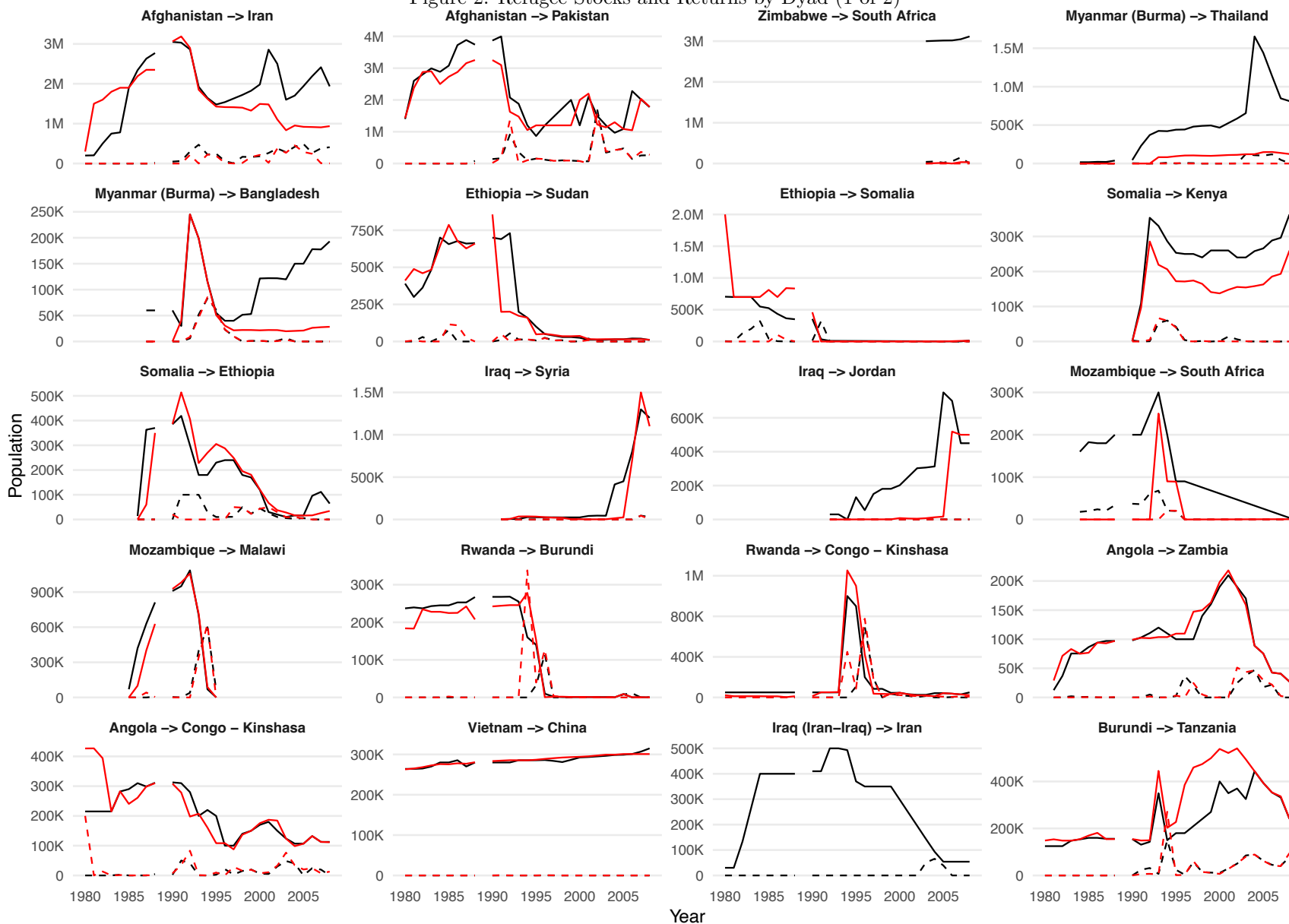
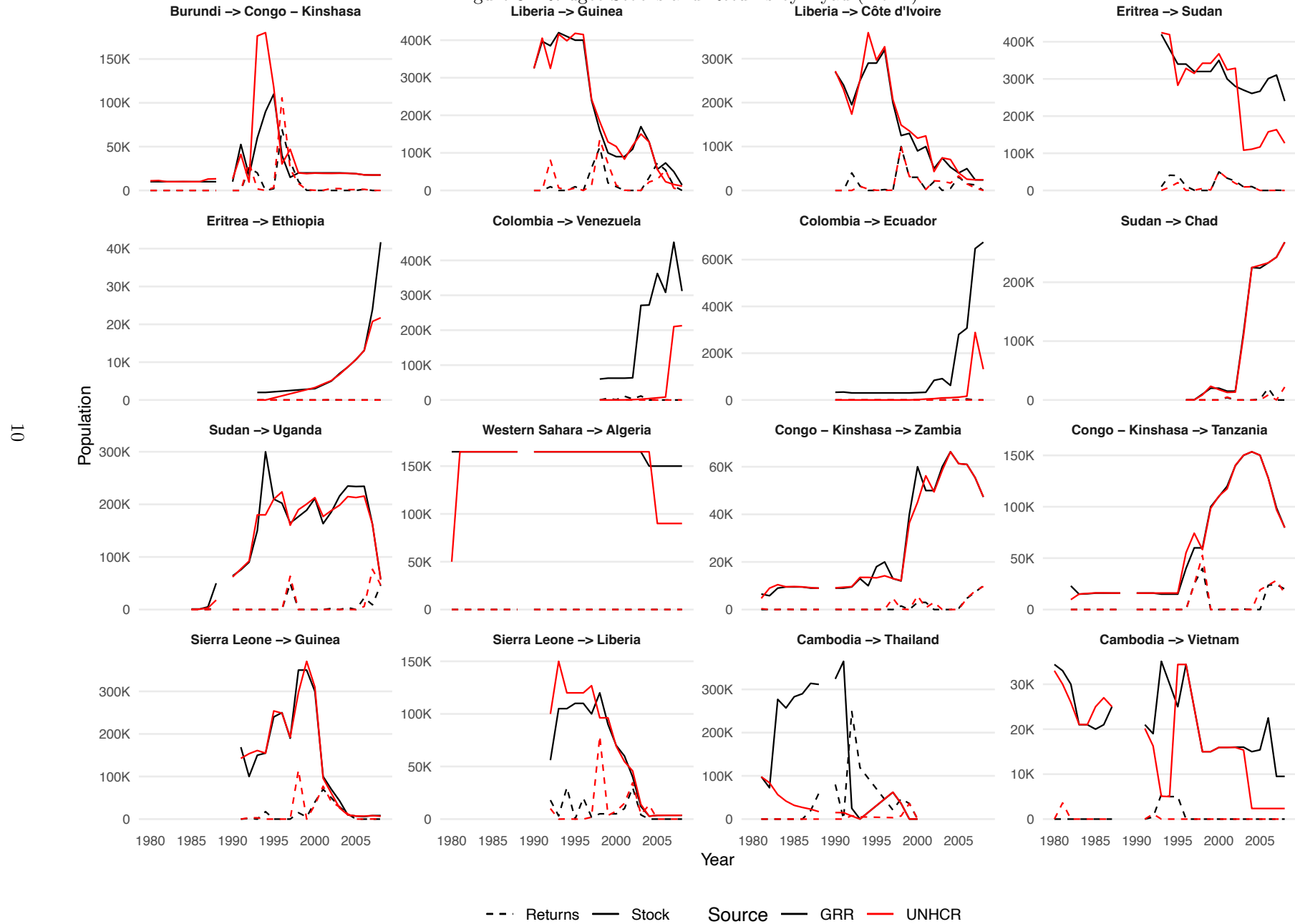


Figure 3: Refugee Stocks and Returns by Dyad (2 of 2)



2 Robustness Checks

Unless otherwise noted, each robustness model changes one feature of column 1 in the main table while holding the remainder of the specification fixed. I use Model 1 for these checks because it has a larger sample than Model 2, which excludes years for which repatriation assistance is not measured. A parallel set of robustness checks for the repatriation-assistance results is reported in Section 2.2.3.2. I discuss individual specifications only when the core results change. By core results, I mean the positive and statistically significant coefficients on severe exclusion policy, asylum-country conflict, and war ending.

2.1 Dependent Variable

The main analysis examines within-dyad changes in the number of refugees returning over time. One concern is that year-to-year changes in the recorded return count may partly reflect measurement error rather than real changes in return. I address this by varying the dependent variable in several ways. Table 6 reports results using the baseline return count; voluntary return alone, excluding forced returns; and a centered three-year moving average. I also construct three measures of substantial return episodes, which focus on large changes in return that should be less sensitive to noise in the annual count: return exceeding 10% of the population at risk, a return rate more than one within-dyad standard deviation above the dyad mean, and the onset of a 10% return episode. These measures deliberately discard small year-to-year differences in recorded return. The main results are robust across these specifications, with two qualifications. The association with asylum-country conflict is less consistent when using the discrete return-episode measures, and it disappears when the outcome is restricted to voluntary return, suggesting that this relationship is driven primarily by forced return.

Table 6: Alternative Measures of the Dependent Variable

Model:	Total Returns (1) Poisson	Voluntary Returns (2) Poisson	3-Year MA (3) Poisson	Return above 10% (4) Logit	Mean + 1 SD (5) Logit	Episode Onset (6) Logit
Asylum-side push						
Exclusion policy: severe	1.09*** (0.19)	0.52** (0.17)	0.95*** (0.21)	2.75*** (0.33)	2.17*** (0.38)	2.71*** (0.53)
Exclusion policy: moderate	0.48* (0.22)	-0.18 (0.15)	0.49* (0.20)	0.48† (0.28)	0.75* (0.31)	0.86** (0.33)
Conflict in COA: high	0.59† (0.31)	0.38 (0.31)	0.39† (0.23)	-0.03 (0.43)	0.25 (0.43)	0.50 (0.56)
Conflict in COA: low	0.81*** (0.21)	0.22 (0.15)	0.67*** (0.16)	-0.17 (0.30)	-0.15 (0.32)	0.06 (0.36)
Existing explanations						
War Ending	1.00*** (0.22)	1.20*** (0.20)	0.59*** (0.18)	1.26** (0.39)	1.15*** (0.28)	1.25* (0.53)
Conflict in COO: high	0.11 (0.28)	0.16 (0.29)	0.27 (0.31)	-0.42 (0.38)	-0.10 (0.35)	0.69 (0.57)
Conflict in COO: low	0.30 (0.22)	0.51* (0.23)	0.27 (0.20)	-0.14 (0.28)	-0.30 (0.28)	0.20 (0.44)
GDP (COO)	0.39 (0.28)	0.27 (0.28)	0.16 (0.26)	0.18 (0.32)	0.90* (0.40)	0.44 (0.41)
GDP (COA)	-1.09† (0.59)	-0.26 (0.57)	-0.97 (0.59)	-0.59 (0.66)	-0.43 (0.64)	-1.18† (0.69)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.44** (0.17)	-0.47*** (0.14)	-0.74** (0.27)	-0.46* (0.20)	-0.43 (0.29)
Refugee Rights	-0.02 (0.17)	0.06 (0.17)	-0.03 (0.16)	0.38* (0.15)	0.10 (0.15)	0.44** (0.17)
Conflict Duration	-0.03 (0.08)	-0.05 (0.09)	-0.10 (0.09)	0.37* (0.17)	0.41** (0.16)	0.17 (0.26)
Estimator	PPML	PPML	PPML	Logit	Logit	Logit
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,281	2,207	1,834	2,300	1,266

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Dyad and yr FE; dyad-clustered SEs throughout. M1–M3: PPML, log stock-at-risk offset. M4: episode is return above 10% of the stock at risk. M5: episode is a return rate more than one within-dyad SD above its mean. M6: onset of a 10% episode, on dyad-years at risk of beginning one. Conditional-logit coefficients are log odds ratios and are not comparable in magnitude with the PPML columns.

2.2 Explanatory Variables

2.2.1 Exclusion Policy

2.2.1.1 Measurement

The *Exclusion Policy* variable is hand-coded from the qualitative country narratives in the *World Refugee Survey* and, after the Survey ceased publication, the U.S. State Department’s *Country Reports on Human Rights Practices*. Non-zero dyad-years are assigned to one of two categories, *moderate* or *severe*, using two criteria: (i) whether forced return is implemented or only threatened, and (ii) the share of the refugee stock the policy is intended or expected to affect.

A dyad-year is coded *severe* when the asylum government implements an exclusion policy intended or expected to affect at least 10% of the refugee stock. Severe policies include large-scale arrests and deportations, non-renewal of residence cards or refugee status, closures of camp schools and marketplaces, and cuts to food aid expressly intended to induce return. A dyad-year is coded *moderate* when the asylum government either (a) threatens or encourages forced return without policy implementation (e.g., political rhetoric, parliamentary debate, public ultimatums), or (b) implements an exclusion policy targeting fewer than 10% of the refugee stock. The 10% threshold separates broad coercive policies, which I expect to shift return rates, from narrower actions targeting specific individuals or groups but not intended to bring about widespread return.

Coding depends only on intentions to encourage return and the policy itself, not on the return outcome. A policy that cuts food aid across all camps to induce return is coded as severe regardless of whether refugees subsequently repatriate. This rules out the most direct form of measurement endogeneity: the variable captures the policy environment refugees face rather than the return rate it is intended to predict. Table 7 provides illustrative examples of moderate and severe coding decisions. The GRR codebook records a verbatim source quote for every non-zero observation, allowing external verification of all coding decisions.

Table 7: Examples of Exclusion Policies

Moderate exclusion policies	Severe exclusion policies
“In a move protested by UNHCR, authorities in Congo-Brazzaville forced 19 asylum seekers back to Congo-Kinshasa in April.” (COD-COG, 2002)	“South Africa enacted new legislation in 1991, the Aliens Control Act, giving immigration authorities even more power to seek out and deport Mozambicans... South Africa has forcibly repatriated thousands of Mozambican refugees to a land of deprivation and death in recent years. This practice violates international standards and must cease,” USCR stated. (MOZ-ZAF, WRS 1993)
“The Tibetan refugee leadership encourages most of those on pilgrimages and many who come to study to return to Tibet, both to counteract the overwhelming Chinese presence in Tibet and because the two dozen Tibetan refugee settlements in India cannot absorb so many people.” (CHN-IND, 1999)	“In April, the Government announced that it would expel all Rwandans whose cases it rejected and the number of applicants declined substantially... Deterred by the low acceptance rates, some 2,000 never registered with the Government and more than 10,500 of those who registered dropped their claims and returned.” (RWA-BDI, WRS 2006)
“There were scattered reports that the government intended to expel Palestinians who had not arrived in 1948, but who had come to Lebanon from other Arab countries, such as Egypt and Syria. If implemented, this could affect up to 50,000.” (PAL-LEB, 1993)	“The events in Zaire produced strains in Congo that resulted in the Congolese authorities forcibly deporting some 10,000 Zaireans in November.” (COD-COG, WRS 1992)
“UNHCR expressed concern that some refugees were being manipulated to repatriate prematurely to Congo-Kinshasa based on misinformation about conditions there. Some refugees reportedly repatriated and had to flee again to Rwanda because conditions in Congo-Kinshasa remained dangerous.” (COD-RWA, 2000)	“Yemeni authorities forcibly returned 438 Somalis, at least 200 of whom were UNHCR-recognized refugees. They had been part of a group of 1,200 persons who had been rounded up in Aden as part of a sweep of the city for illegal immigrants.” (SOM-YEM, 1996)
“In western Sudan, 45,000 Chadian refugees remain in UNHCR-assisted reception centers despite government and UNHCR attempts to persuade them to return to Chad.”	“UNHCR protested when the government’s roundup of 1,000 undocumented immigrants in January resulted in the detention of several refugees who possessed UNHCR permits. Persons judged to be undocumented were deported without UNHCR screening to determine claims to refugee status.” (COD-ZMB, 1993)

2.2.1.2 Validation

In core GRR, exclusion policies and forced-return counts are coded from the same archival sources, raising the possibility of a mechanical association. I address this in three ways: by asking whether policies are coded in years with no return, whether the result persists when forced returns are removed from the outcome, and whether the measure agrees with multiple independently coded measures.

Table 8: Return in the years a policy is coded

Policy	Dyad-years	Zero return	Return below 1% of stock	Median return rate
None	3,988	76.3%	84.8%	0.00%
Moderate	290	35.9%	58.3%	0.35%
Severe	150	8.0%	22.0%	10.64%

Note: If the measure were read off the return, the zero-return column would be near empty.

Table 8 verifies that return counts are coded independently of the exclusion policy. Moderate policy is coded in 290 dyad-years, and 36% record no return at all. Severe policy is coded in 150 dyad-years, of which 8% record no return. This is possible because the policy measure captures threats, ultimatums, camp closures, deadlines, and other government actions, rather than return itself. Moreover, most returns are not coded from forced-return passages: 81% of all returns in GRR are voluntary, and forced returns occur in only 4.1% of dyad-years.

Table 6 replicates the main analysis using only voluntary returns as the dependent variable. The severe-policy coefficient falls from 1.09 to 0.52 when forced returns are excluded, or to 47 per cent of the baseline estimate, but remains statistically significant. This is reassuring because, if the baseline association were driven mainly by mechanical overlap between the coding of exclusion policy and forced-return counts, it should largely disappear once forced returns are removed from the outcome. Its persistence for voluntary return instead suggests that coercive policies may also increase returns classified as voluntary but undertaken under substantial pressure. The smaller coefficient is therefore consistent with severe policies producing both directly forced return and nominally voluntary return under coercive conditions.

2.2.1.3 Correlation with DWRAP *Non-Refoulement* Legislation

To increase confidence in the exclusion policy measure, I examine its correlation with independently-coded measures that capture related measures. First, I examine whether cases of exclusion policies are negatively correlated with governments that provide protection against *non-refoulement*. If the measure is accurate, assuming that there is at least some correlation between law and practice, we should expect that places with more legal protections are less likely to have exclusion policies. To measure this, I employ DWRAP's statutory measure of non-refoulement. DWRAP covers 90 host countries and 95% of the panel.

Table 9: Exclusion policy by statutory position on refoulement

Statutory position	Dyad-years	Hosts	Any policy	Severe policy
Explicit provision permitting refoulement	509	34	15.3%	3.9%
Silent or ambiguous	698	29	21.3%	7.0%
Explicit right to non-refoulement	3,044	66	6.8%	2.5%

Note: Restricted to dyad-years in which both measures are coded. Exclusion policy is hand-coded from the archival sources; the statutory position is DWRAP's independent coding of national law.

Table 10: Exclusion policy and the statutory right to non-refoulement

Specification	Explicit right to non-refoulement	Dyad-years
Any, host + year FE	-1.15*** (0.34)	3,321
Any, dyad + year FE	-1.09* (0.48)	2,128
Severe, host + year FE	-1.29* (0.61)	2,268
Severe, dyad + year FE	-1.66* (0.83)	970

Note: Logit of hand-coded exclusion policy on the statutory right to non-refoulement, compared with the two remaining DWRAP positions combined. Standard errors clustered by dyad. The dyad-fixed-effects columns are identified from statutory changes within a host over time.

Consistent with this expectation, Table 10 shows that exclusion policy occurs in 6.8% of dyad-years in which the host grants an explicit right to non-refoulement, compared with 21.3% of dyad-years in which the statute is silent or ambiguous. Across the specifications in Table 10, a statutory right to non-refoulement is negatively associated with both any exclusion policy and severe exclusion policy. Formal protection does not eliminate coercion, however: among dyad-years with an explicit right to non-refoulement, 206 record an exclusion policy and 77 record a severe policy, spanning 30 host countries.

As an additional source-independence check, I am constructing a separately coded measure of exclusion policy from Amnesty International reports; coding and analysis are ongoing.

2.2.2 Conflict and War Endings

2.2.2.1 Alternative Definitions of a War Ending

In the main analysis, war endings are coded using an author-extended version of the Fearon (2017) data, which identify the years in which specific wars begin and end. I define the post-war period as the one to three years after a war ends, provided that another war does not begin during that window. As an alternative, I construct war endings using UCDP data, defining an ending as the last year of a high-intensity conflict spell and keeping the post-war window open unless high-intensity conflict resumes. I prefer the Fearon measure because it more closely captures the end of a war in the sense likely to matter for refugee return. In UCDP, a conflict spell can end simply because battle deaths fall below the high-intensity threshold even when the underlying war continues, which can generate multiple apparent endings within the same war. Refugees may have little reason to interpret such declines in violence as a durable improvement in security. Consistent with these differences, agreement between the two measures is limited: only 21% of UCDP endings have a Fearon ending within two years, and 44% of Fearon endings have a UCDP ending within two years.

I nevertheless examine the robustness of the results to the UCDP measure in Model 2. I also vary the war-ending coding by using only Fearon’s original cases, which cover a shorter period and include only civil wars, removing the 25,000-refugee size threshold, including the war-ending year itself in the post-war window, and adding a lagged peace-agreement indicator. The main results are robust across these specifications. The peace-agreement indicator, however, enters negatively and significantly, which does not align with the positive association between war ending and return in the main specification.

Table 11: Robustness: Alternative Definitions of a War Ending

Model:	Main (1)	UCDP (2)	Total Repatriation		+ Ending yr (5)	+ Peace ag. (6)
			Fearon original (3)	No size floor (4)		
Asylum-side push						
Exclusion policy: severe	1.09*** (0.19)	1.02*** (0.21)	1.10*** (0.19)	1.08*** (0.19)	1.09*** (0.19)	1.07*** (0.18)
Exclusion policy: moderate	0.48* (0.22)	0.50* (0.21)	0.48* (0.22)	0.47* (0.23)	0.49* (0.22)	0.52* (0.21)
Conflict in COA: high	0.59† (0.31)	0.54† (0.30)	0.57† (0.31)	0.59† (0.31)	0.59† (0.31)	0.72* (0.30)
Conflict in COA: low	0.81*** (0.21)	0.82*** (0.21)	0.81*** (0.21)	0.81*** (0.21)	0.81*** (0.21)	0.84*** (0.21)
Existing explanations						
War Ending	1.00*** (0.22)	0.62** (0.20)	1.03*** (0.23)	0.99*** (0.22)	1.00*** (0.21)	0.99*** (0.21)
War Ending (Year Of)					0.05 (0.25)	
Conflict in COO: high	0.11 (0.28)	0.13 (0.35)	0.08 (0.28)	0.11 (0.28)	0.11 (0.28)	0.08 (0.27)
Conflict in COO: low	0.30 (0.22)	0.02 (0.22)	0.28 (0.22)	0.31 (0.22)	0.30 (0.22)	0.24 (0.22)
GDP (COO)	0.39 (0.28)	0.37 (0.29)	0.36 (0.29)	0.38 (0.28)	0.39 (0.28)	0.43 (0.28)
GDP (COA)	-1.09† (0.59)	-1.15† (0.64)	-1.09† (0.59)	-1.12† (0.59)	-1.09† (0.59)	-1.16* (0.56)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.55*** (0.16)	-0.54** (0.17)	-0.53*** (0.16)	-0.51** (0.17)	-0.52** (0.18)
Refugee Rights	-0.02 (0.17)	0.04 (0.16)	-0.02 (0.17)	-0.01 (0.17)	-0.02 (0.17)	0.02 (0.17)
Conflict Duration	-0.03 (0.08)	-0.09 (0.09)	-0.02 (0.08)	-0.03 (0.09)	-0.03 (0.08)	0.01 (0.07)
Peace Agreement (COO, t-1)						-0.44** (0.15)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Ending list	Fearon narrow	UCDP spells	Fearon original	No 25,000 floor	Fearon narrow	Fearon narrow
At-peace test	Fearon spans	UCDP high-int.	Fearon spans	Fearon spans	Fearon spans	Fearon spans
Endings	36	76	33	75	36	36
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,327	2,327	2,327	2,327	2,214
Pseudo R ²	0.80	0.79	0.80	0.80	0.80	0.81

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †: 0.1

PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. All columns are main table col. 1 with the post-war window rebuilt. Col. 1: the paper's window (extended Fearon list plus inter-state wars, endings 1981–2018 with 25,000+ refugees in exile). Col. 2: UCDP high-intensity endings, post-war unless high-intensity war resumes. Cols. 3–6 keep the Fearon at-peace test. Col. 3: Fearon's own cases only. Col. 4: drops the 25,000 filter. Col. 5: yrs 0–3 rather than 1–3. Col. 6: peace agreement at $t - 1$.

2.2.2.2 Alternative Measures of Origin-Country Conflict and Repression

In this section, I vary the way conflict in both the origin and asylum country is coded. Model 1 reproduces the baseline. Model 2 replaces the ordinal conflict indicators with logged battle deaths, a continuous measure of conflict intensity available from 1989. I prefer the ordinal measure for the main analysis because it is available from 1980 and therefore covers the full GRR period. Model 3 dates origin-country conflict at $t - 2$ to allow for a delay between changes in security and refugee return. Model 4 uses an alternative conflict measure from the Political Instability Task Force (PITF), combining its Ethnic War and Revolutionary War datasets (Marshall, 2019). PITF provides an independently coded, episode-based measure of major internal armed conflict. The version used here ends in 2017.

I then vary the way political repression is coded. The main analysis uses the Freedom House civil liberties measure. Model 5 adds two alternative measures. The Political Terror Scale (PTS) measures state-perpetrated physical integrity violations, including torture, political imprisonment, extrajudicial killings, and disappearances, on a 1–5 ordinal scale; I use the mean across its three sources (Amnesty International, Human Rights Watch, and the U.S. State Department), standardized over the estimation sample. The second is a binary indicator for an ongoing episode of genocide or politicide, drawn from the genocide/politicide component of the PITF State Failure Problem Set (Marshall, 2019). The PITF episode list used here ends in 2017, so observations after 2017 are coded as missing and Model 5 is estimated on a shorter panel. The main results are robust across these specifications. Notably, return is also positively associated with both increases in the PITF war and violence measure and ongoing episodes of genocide or politicide.

2.2.3 Repatriation Assistance

The main analysis measures repatriation assistance using UNHCR repatriation expenditure per member of the origin country’s global refugee population. The underlying funding series is hand-coded for 1980–1999 and 2011–2020 and is unavailable for 2000–2010 and 2021–2023. These missing years reflect gaps in the source record rather than years without repatriation assistance and are therefore left missing. Table 13 first separates the consequences of this restricted sample from the association with funding itself. Model 2 re-estimates the baseline specification on the dyad-years for which funding is observed, and Model 3 adds the continuous funding measure. Some coefficients change with the restricted sample, but severe exclusion policy remains strongly associated with return. Funding is positively associated with return, while its inclusion attenuates the post-war coefficient, consistent with the possibility that part of the relationship between war ending and return reflects the assistance that accompanies repatriation after conflict.

I also construct two dyadic measures of UNHCR involvement in repatriation that span the missing years. *UNHCR Repatriation Program* indicates whether a UNHCR-assisted repatriation program was operating in the origin–asylum dyad. The series is coded based on whether there is mention of a UNHCR program in the *World Refugee Survey* reports for 1980–2008. *Tripartite Repatriation Agreement* indicates whether a formal agreement among UNHCR and the origin and asylum governments was in force in the dyad. I construct the measure from an archival register of UNHCR Executive Committee materials covering 40 signings across 36 dyads between 1986 and 2002. Because the register records signing dates but not expiry dates, I code an agreement as active for 3 years following signature. The two additional measures point in the same direction as the main analysis.

2.2.3.1 Distribution of Repatriation Assistance

The main specification enters repatriation funding as a three-level measure rather than imposing a linear relationship between funding and return. Table 14 shows the distribution of the funding variable. Of the 3,015 coded dyad-years, 2,190 (72.6%) record no funding. Among positive observations, the median is \$19.88 per refugee per year. I therefore distinguish between no recorded funding, positive funding at or below the median, and funding above the median. The results suggest a nonlinear relationship: relative to no funding, funding above the median is associated with higher return, while funding below the median is not statistically distinguishable from zero.

A concern is that this result may be sensitive to the choice of cutoff. Table 15 therefore varies the dividing line across six alternatives. The coefficient on the upper category remains positive in all six specifications

Table 12: Robustness: Alternative Measures of Conflict and Repression

Model:	Total Repatriation				
	Main (1)	Battle deaths (2)	COO t-2 (3)	PITF war (4)	+ PTS/Genocide (5)
Asylum-side push					
Exclusion policy: severe	1.09*** (0.19)	0.88*** (0.19)	1.06*** (0.20)	1.10*** (0.20)	1.05*** (0.20)
Exclusion policy: moderate	0.48* (0.22)	0.44† (0.23)	0.54* (0.22)	0.50* (0.23)	0.43† (0.24)
Conflict in COA: high	0.59† (0.31)		0.76* (0.31)	0.58† (0.32)	0.60† (0.32)
Conflict in COA: low	0.81*** (0.21)		0.85*** (0.19)	0.82*** (0.21)	0.79*** (0.22)
Existing explanations					
War Ending	1.00*** (0.22)	0.95*** (0.20)	0.91*** (0.21)	0.94*** (0.21)	0.94*** (0.20)
Conflict in COO: high	0.11 (0.28)			0.06 (0.28)	0.02 (0.29)
Conflict in COO: low	0.30 (0.22)			0.30 (0.22)	0.35 (0.22)
GDP (COO)	0.39 (0.28)	0.39 (0.28)	0.49† (0.28)	0.40 (0.29)	0.29 (0.30)
GDP (COA)	-1.09† (0.59)	-1.17* (0.46)	-1.18† (0.61)	-1.05† (0.59)	-0.67 (0.59)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.47** (0.16)	-0.48** (0.16)	-0.55** (0.17)	-0.56** (0.19)
Refugee Rights	-0.02 (0.17)	0.18 (0.19)	0.07 (0.17)	-0.03 (0.18)	0.01 (0.18)
Conflict Duration	-0.03 (0.08)	0.02 (0.10)	0.07 (0.08)	-0.09 (0.09)	-0.06 (0.09)
Alternative conflict and repression measures					
Battle Deaths (COA, log)		0.13*** (0.04)			
Battle Deaths (COO, log)		0.01 (0.04)			
Conflict in COO: high (t-2)			-0.19 (0.22)		
Conflict in COO: low (t-2)			-0.06 (0.11)		
PITF War/Violence (COO)				1.24* (0.58)	
Political Terror Scale (COO)					-0.01 (0.13)
Genocide/Politicide (COO)					0.81* (0.33)
All controls	Yes	Yes	Yes	Yes	Yes
Years	1980-2022	1990-2022	1982-2022	1980-2022	1980-2022
Fixed-effects					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	2,327	2,134	2,128	2,238	1,994
Pseudo R ²	0.80	0.81	0.81	0.80	0.80

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. All columns carry the full main-table specification and differ only in how conflict and repression are measured. Cols. 2-3 replace the ordinal conflict levels; cols. 4-5 add measures alongside them. Col. 5 adds the Political Terror Scale (standardized) and a genocide/politicide indicator (PITF, ends 2017).

Table 13: Robustness: Measures of Repatriation Assistance

Model:	Total Repatriation				
	Main (1)	Fund.rows (2)	+Funding (3)	Prog.WRS (4)	Trip.02 (5)
Asylum-side push					
Exclusion policy: severe	1.09*** (0.19)	1.24*** (0.32)	1.29*** (0.31)	0.57* (0.28)	0.89** (0.28)
Exclusion policy: moderate	0.48* (0.22)	0.20 (0.27)	0.13 (0.28)	0.19 (0.26)	0.46* (0.20)
Conflict in COA: high	0.59† (0.31)	0.96* (0.43)	0.79† (0.43)	0.86** (0.27)	1.41*** (0.32)
Conflict in COA: low	0.81*** (0.21)	0.93** (0.34)	0.95** (0.36)	0.65** (0.22)	0.95** (0.36)
Existing explanations					
War Ending	1.00*** (0.22)	1.29** (0.40)	1.05** (0.38)	0.82** (0.26)	1.68*** (0.36)
Conflict in COO: high	0.11 (0.28)	0.57† (0.34)	0.38 (0.35)	-0.08 (0.32)	0.28 (0.34)
Conflict in COO: low	0.30 (0.22)	0.29 (0.26)	0.22 (0.25)	0.35 (0.25)	0.23 (0.28)
GDP (COO)	0.39 (0.28)	0.73 (0.49)	0.80 (0.50)	0.18 (0.39)	0.31 (0.39)
GDP (COA)	-1.09† (0.59)	-1.23 (0.83)	-1.40 (0.86)	-0.41 (0.64)	-0.14 (0.67)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.22 (0.24)	-0.10 (0.26)	-0.34† (0.19)	-0.23 (0.22)
Refugee Rights	-0.02 (0.17)	-0.15 (0.22)	-0.19 (0.22)	-0.14 (0.23)	-0.21 (0.26)
Conflict Duration	-0.03 (0.08)	-0.07 (0.13)	-0.06 (0.12)	0.01 (0.11)	0.08 (0.14)
Repatriation assistance					
Repatriation Assistance			0.31** (0.12)		
UNHCR Repatriation Program				0.67** (0.23)	
Tripartite Agreement					0.66* (0.27)
Fixed-effects					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	2,327	1,378	1,378	1,225	873
Pseudo R ²	0.80	0.77	0.78	0.84	0.83

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

PPML; log stock-at-risk offset; dyad and year FE; dyad-clustered SEs. Samples differ by column; only M2 and M3 are comparable. M1: main table col. 1. M2: M1 refitted on M3's rows, no assistance term. M3: main table col. 2; funding coded 1980–1999 and 2011–2020 only, so M2 and M3 drop 949 of M1's 2,327 dyad-years. M4: UNHCR repatriation program, 1980–2008 (*World Refugee Survey* years). M5: tripartite agreement in force, 1980–2002 (ExCom register complete).

(0.30 to 0.84) and is statistically significant at the 5% level in 4 of them, while the lower category is never distinguishable from no funding. The substantive result is therefore that the positive association is concentrated at higher levels of repatriation funding, rather than that \$19.88 represents a sharp threshold.

Table 14: Distribution of UNHCR repatriation funding per refugee and the category cut

Category	Range of funding per refugee	Dyad-years	Share of coded	Median within category
Uncoded (2000–2010, 2021–2023)	–	2,116	–	–
None	exactly \$0	2,190	72.6%	\$0.00
Low	above \$0 to \$19.88	414	13.7%	\$5.80
High	above \$19.88 to \$55.70	411	13.6%	\$50.80

Note: UNHCR repatriation expenditure per member of the origin’s global refugee population, \$/yr, on the 5,131 dyad-year frame. Shares are of the 3,015 coded dyad-years. Cut is the median of the positive coded values. Uncoded yrs are missing, not zero, and are outside the reference category. 171 positive dyad-years sit on the winsorization cap.

Table 15: Sensitivity of the assistance categories to the threshold

Rule	Dyad-years		Coefficient (SE)	
	Low	High	Low	High
Median of the positive values (\$19.88)	414	411	-0.23 (0.34)	0.60* (0.30)
Round threshold at \$10	303	522	-0.15 (0.34)	0.31 (0.32)
Round threshold at \$25	485	340	-0.16 (0.32)	0.76* (0.33)
Top tercile of the positive values (\$35.01)	553	272	-0.01 (0.32)	0.71* (0.35)
Within-year median of the positive values	462	363	0.02 (0.29)	0.30 (0.34)
At the winsorization cap (\$55.70) against below it	654	171	-0.04 (0.33)	0.84* (0.35)

Note: PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. Each row recodes funding per refugee as none / low / high under a different threshold and refits main table col. 2. Zero-funding dyad-years are the omitted baseline, so the counts are the middle and top categories only. Row 1 is the paper’s rule; the last row cuts at the winsorization cap. [†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

2.2.3.2 Robustness of the Repatriation Assistance Result

Table 16: Robustness: the repatriation-assistance coefficient

Departure	Repatriation assistance	Exclusion policy: severe	Dyad-years	Dyads
Baseline (main table col. 2)	0.60* (0.30)	1.31***	1,378	100
Outcome: UNHCR voluntary return	0.55 (0.37)	0.41	1,333	96
Outcome: 3-year moving average	0.44 [†] (0.24)	0.99***	1,309	101
All covariates at $t - 1$	0.05 (0.31)	1.35***	1,132	96
Negative binomial	1.13* (0.44)	2.85***	1,378	100
Clustered by dyad and year	0.60 [†] (0.31)	1.31***	1,378	100
Excl. refugee-warrior dyads	0.58* (0.27)	1.99***	1,318	96
Excl. asylum in Europe & N. America	0.67* (0.31)	1.20***	1,185	82
Top 3 asylum countries per origin	0.51 [†] (0.30)	1.30***	1,221	95
Fled armed conflict only	0.52 (0.32)	1.20***	1,257	93
Core GRR panel (1980–2008, Global South)	0.71* (0.31)	1.15**	580	56
State Department years (2009–)	0.08 (0.53)	2.18***	593	67
Continuous: standardized log funding	0.31** (0.12)	1.29***	1,378	100
Continuous: unstandardized log funding	0.23** (0.08)	1.29***	1,378	100
Extensive margin: any funding recorded	0.14 (0.30)	1.24***	1,378	100

Note: PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs unless the row says otherwise. First group: main table col. 2, one feature changed. Coefficient is funding above the median (\$19.88/refugee/yr) against none; the below-median indicator is in every model, not shown. *Exclusion policy: severe* is a reference point, SEs suppressed. Last three rows change the measure's form, so magnitudes are not comparable.

Table 16 examines whether the funding result survives the same alternative specifications and samples used elsewhere in the appendix. Each model enters the three-level measure as in the main specification, and the table reports the coefficient on funding above the median relative to no recorded funding; the below-median indicator is included in each model but not shown. The coefficient is positive in 15 of the 15 specifications in which it can be estimated and is statistically distinguishable from zero in 7 of them. The final rows substitute continuous versions of the funding measure, so their magnitudes are not directly comparable with the rest of the column. Overall, the positive association with higher repatriation funding is reasonably robust, although less stable than the severe exclusion-policy result. These estimates also rely on smaller samples than the main Model 1 specifications because repatriation funding is not observed in all dyad-years.

2.2.4 Additional Explanatory Variables

This section examines alternative explanatory variables, either by measuring variables from the main analysis differently or by introducing additional measures. Table 17 examines the political status of refugees' ethnic groups in both the origin and asylum country, where shared ethnic ties can be identified. A group that gains political power at home may be more likely to return, while refugees whose co-ethnics hold power in the asylum country may be more likely to remain. I test both possibilities using EPR data.

Model 1 includes the ordinal political-power score of the refugee group in the origin country, ranging from self-exclusion at the bottom to monopoly at the top. Model 2 instead uses a binary indicator for whether the group holds political power. Because both models include dyad fixed effects, the estimates ask whether return changes as the political standing of a refugee group improves relative to its earlier position within the same dyad. I find no evidence that it does. Neither the power score nor the in-power indicator is distinguishable from zero, and both estimates are substantively small. The null is also not driven by the linear coding of political status. Entering the nine EPR categories separately produces no monotonic pattern: the categories that differ most from the omitted category are 'monopoly' and 'self-exclusion,' which lie at opposite ends of the scale. Nor does the result appear to reflect the timing of status changes, which often occur around regime changes at the end of war and are therefore already captured by the post-war window. The political standing of co-ethnics in the asylum country is similarly unrelated to return: neither EPR-TEK indicator is distinguishable from zero for kin in power and for kin excluded.

The ordinal power score is derived from the categorical EPR status of the refugee group, which takes nine values and ranges from ‘self-exclusion’ through ‘powerless,’ ‘discriminated,’ ‘state collapse,’ and the three partner categories to ‘monopoly.’ The binary indicator in Model 2 equals one for groups coded as monopoly, dominant, senior partner, or junior partner, and zero otherwise. Refugee groups are matched to EPR group identifiers using the EPR Refugee Dataset, which links populations of at least 2,000 refugees to EPR groups in neighboring or proximate asylum countries within 950 km of the origin border. For dyads outside this coverage, I code the refugee group manually from *World Refugee Survey* reports.¹ EPR coverage is unavailable for some refugee groups, reducing the estimation sample.

Table 17: Ethnic political status and refugee return

Model:	Total Repatriation		
	Power score (EPR) (1)	In power (EPR) (2)	Kin in host (EPR-TEK) (3)
Asylum-side push			
Exclusion policy: severe	1.24*** (0.21)	1.24*** (0.21)	1.11*** (0.20)
Exclusion policy: moderate	0.50* (0.21)	0.50* (0.21)	0.54* (0.24)
Conflict in COA: high	0.49 (0.35)	0.49 (0.35)	0.63* (0.30)
Conflict in COA: low	1.04*** (0.20)	1.04*** (0.20)	0.84*** (0.22)
Existing explanations			
War Ending	0.69*** (0.19)	0.68*** (0.19)	0.99*** (0.22)
Conflict in COO: high	0.15 (0.28)	0.15 (0.28)	0.09 (0.29)
Conflict in COO: low	0.51* (0.23)	0.50* (0.23)	0.29 (0.23)
GDP (COO)	0.38 (0.26)	0.37 (0.26)	0.38 (0.28)
GDP (COA)	-2.24*** (0.53)	-2.22*** (0.53)	-1.10 [†] (0.59)
Civil Liberties Restrictions (COO)	-0.66*** (0.17)	-0.66*** (0.18)	-0.52** (0.17)
Refugee Rights	0.06 (0.19)	0.06 (0.19)	-0.01 (0.17)
Conflict Duration	-0.18* (0.08)	-0.18* (0.09)	-0.03 (0.09)
Ethnic political status			
Ethnic Power Score (COO)	0.03 (0.07)		
Group In Power (COO)		0.08 (0.25)	
Co-ethnic Kin in Power (COA)			-0.29 (0.45)
Co-ethnic Kin Excluded (COA)			0.21 (0.35)
All controls	Yes	Yes	Yes
Fixed-effects			
Dyad	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Observations	1,718	1,718	2,327
Pseudo R ²	0.81	0.81	0.80

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. All cols. carry the full main-table specification; dyad FE means the EPR terms are identified from groups whose status changes within a dyad. Col. 1: EPR power score of the displaced group, self-exclusion (0) to monopoly (6), entered linearly; the rungs are not evenly spaced, so this is a monotone summary and not a per-step effect. Col. 2: 1 for monopoly, dominant, senior or junior partner. Col. 3: EPR-TEK indicators for co-ethnics in the asylum country holding power or being excluded. EPR coverage is incomplete, hence the smaller samples in cols. 1–2.

¹The same ethnic group is carried forward as long as the refugee population is continuous; the assignment is set to missing when a discontinuity in the population could plausibly indicate a change in ethnic composition. Two case-specific adjustments warrant note: EPR classifies Somalis as ‘irrelevant’ within Somalia on grounds of ethnic homogeneity, and I reclassify this group as high-status; Peuhl/Goula refugees from the Central African Republic appear in the EPR Refugee Dataset but not in the EPR Power Dataset, and are coded as ‘powerless’ based on desk research. The 2021 EPR Refugee values are carried forward from 2020.

Table 18 tests sensitivity to alternative operationalizations and additional covariates. Model 1 reproduces the baseline. Model 2 replaces continuous origin and asylum GDP per capita with quartile indicators (Q1 reference) to relax linearity assumptions. Model 3 lags all independent variables by one year, addressing the possibility that contemporaneous asylum-country conflict partly follows refugee return or refugee mobilization.

Model 4 adds UNHCR funding per refugee in the asylum country. This measure is difficult to interpret, because it is a proxy for two things at once: what UNHCR spends on organizing repatriation, and what it spends on supporting refugees where they are, including camps and health care. The first would be expected to raise return and the second to reduce the pressure to leave, and the funding series does not separate them.

Models 5 and 6 add further conditions in the asylum country. Model 5 includes educational attainment among the resident citizen populations of the origin and asylum countries. This is a national characteristic of each country rather than a measure of refugees themselves, since no comparable cross-national schooling measure exists for refugee populations at this scale. Model 6 adds the asylum country's urban share and an indicator for whether most refugees live in camps. The main results are robust across these checks. The additional measures of economic and social conditions in asylum are not significantly associated with return, while return increases with overall UNHCR funding per refugee. This is consistent with the possibility that the aggregate UNHCR funding measure partly captures increases in repatriation-related spending.

Table 18: Robustness: Alternative Independent Variables

Model:	Main (1)	GDP Quartiles (2)	Total Repatriation Lagged IVs (3)	+ Funding (4)	+ Education (5)	+ Urban/Camp (6)
Asylum-side push						
Exclusion policy: severe	1.09*** (0.19)	1.10*** (0.20)	1.22*** (0.29)	1.23*** (0.22)	1.12*** (0.19)	1.07*** (0.20)
Exclusion policy: moderate	0.48* (0.22)	0.50** (0.19)	0.24 (0.25)	0.55* (0.23)	0.51* (0.22)	0.49* (0.22)
Conflict in COA: high	0.59† (0.31)	0.86** (0.27)	0.49* (0.24)	-0.06 (0.34)	0.51 (0.33)	0.60* (0.28)
Conflict in COA: low	0.81*** (0.21)	0.67*** (0.20)	0.53** (0.18)	0.90*** (0.26)	0.85*** (0.20)	0.79*** (0.21)
Existing explanations						
War Ending	1.00*** (0.22)	0.94*** (0.23)	0.63* (0.28)	0.76*** (0.22)	0.95*** (0.21)	0.99*** (0.23)
Conflict in COO: high	0.11 (0.28)	0.06 (0.28)	-0.43 (0.28)	-0.23 (0.32)	0.14 (0.28)	0.11 (0.29)
Conflict in COO: low	0.30 (0.22)	0.29 (0.22)	-0.14 (0.20)	0.04 (0.25)	0.33 (0.22)	0.31 (0.23)
GDP (COO)	0.39 (0.28)		0.10 (0.29)	0.62† (0.33)	0.44 (0.28)	0.37 (0.28)
GDP (COA)	-1.09† (0.59)		-0.26 (0.66)	-1.07 (0.70)	-1.23* (0.62)	-1.04† (0.59)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.55** (0.19)	-0.27 (0.16)	-0.39* (0.17)	-0.56*** (0.17)	-0.50** (0.16)
Refugee Rights	-0.02 (0.17)	0.05 (0.17)	0.10 (0.16)	0.04 (0.15)	-0.02 (0.18)	0.01 (0.17)
Conflict Duration	-0.03 (0.08)	-0.04 (0.09)	0.11 (0.08)	-0.06 (0.10)	-0.08 (0.09)	-0.06 (0.09)
Alternative and additional measures						
GDP (COA): Q2		-0.68* (0.28)				
GDP (COA): Q3		-0.44 (0.50)				
GDP (COA): Q4		-1.11 (1.36)				
GDP (COO): Q2		-0.72** (0.28)				
GDP (COO): Q3		-0.37 (0.35)				
GDP (COO): Q4		0.02 (0.40)				
UNHCR Funding p.p. (COA)				0.85** (0.30)		
Education (COA)					-0.21 (1.25)	
Education (COO)					-0.23 (0.78)	
Urban						-0.30 (0.25)
Majority in Camps						-0.21 (0.51)
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,327	2,164	1,928	2,202	2,325
Pseudo R ²	0.80	0.80	0.80	0.83	0.80	0.80

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †: 0.1

PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. M1: main table col. 1. M2: GDP (COO/COA) quartiles (Q1 ref.). M3: all covariates at $t-1$, including the war-ending window. M4: + UNHCR funding p.p. (z-log). M5: + educational attainment in both resident citizen populations. M6: + urban share and a majority-in-camps indicator; those series are missing for about 650 dyad-years, so M6 runs on a smaller sample.

2.3 Data Subsets

Table 19 tests whether the results hold across different subsets of the data. Model 1 reproduces the baseline on the extended GRR estimation panel. Models 2 examines the core GRR panel, 1980–2008 with asylum outside Europe and North America, which is the version of the dataset built from the most detailed *World Refugee Survey* reporting; Model 3 restricts to the State Department-report period beginning in 2009.

Model 4 excludes dyads associated in the literature with the “refugee warrior” hypothesis, under which refugee populations, camps, or exile communities may become militarized and contribute to conflict spillover in host states (Zolberg 1989; Salehyan and Gleditsch 2006).² Excluding these cases assesses whether the main results are driven by settings in which refugee presence may be more plausibly endogenous to asylum-country conflict. Model 5 excludes asylum countries in Europe and North America. Model 6 restricts to dyads where refugees originally fled armed conflict rather than non-war forms of persecution. Model 7 restricts to the top three asylum countries per origin by refugee stock, addressing concerns that refugees from some countries are more dispersed across multiple asylum destinations than others; because origins vary in the number of asylum countries they appear in, highly dispersed origins contribute more dyads and receive greater implicit weight when each dyad-year is weighted equally.³

The results are generally robust across these specifications, with one notable exception. In the State Department period (Model 3), high-intensity asylum-country conflict is negatively associated with return, although low-intensity conflict remains positively associated with return and the severe exclusion-policy result is unchanged. This contrasts with the core GRR period (Model 2), where both levels of asylum-country conflict are positively associated with return. The comparison suggests that the relationship between asylum-country conflict and return is less stable in the more recent period.

²The excluded dyads are: RWA–COD, RWA–TZA, AFG–PAK, KHM–THA, PSE–LBN, PSE–JOR, and NIC–HND.

³Origins vary in the number of asylum destinations they appear in because the sample rule adds asylum countries for each origin until the cumulative displaced population reaches 75 per cent of the origin’s total. The median origin appears in 3 asylum countries and the most dispersed in 10, and 33 of 84 origins appear in more than three. The restriction therefore leaves most origins untouched and retains 86 per cent of the estimating sample.

Table 19: Robustness: Alternative samples and periods

Dep. Var.:	Total Repatriation						
	Main	Core.GRR	USHR	Excl.Warr	Excl.ENA	War.Orig	Top.3
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Asylum-side push							
Exclusion policy: severe	1.09*** (0.19)	0.71** (0.24)	1.59*** (0.27)	1.60*** (0.31)	1.04*** (0.19)	1.07*** (0.18)	1.06*** (0.19)
Exclusion policy: moderate	0.48* (0.22)	0.39* (0.18)	0.25 (0.33)	0.36† (0.21)	0.50* (0.22)	0.48* (0.22)	0.51* (0.22)
Conflict in COA: high	0.59† (0.31)	0.95*** (0.29)	-0.94* (0.46)	0.71** (0.23)	0.64* (0.30)	0.55† (0.32)	0.64* (0.31)
Conflict in COA: low	0.81*** (0.21)	0.62** (0.24)	0.64* (0.31)	0.51* (0.22)	0.79*** (0.21)	0.84*** (0.21)	0.81*** (0.22)
Existing explanations							
War Ending	1.00*** (0.22)	1.14*** (0.28)	0.05 (0.52)	0.73** (0.23)	0.97*** (0.23)	1.02*** (0.22)	1.06*** (0.23)
Conflict in COO: high	0.11 (0.28)	0.22 (0.33)	-0.08 (0.50)	-0.08 (0.32)	0.07 (0.29)	0.17 (0.27)	0.12 (0.29)
Conflict in COO: low	0.30 (0.22)	0.41 (0.26)	-0.29 (0.36)	-0.03 (0.22)	0.28 (0.23)	0.32 (0.22)	0.31 (0.23)
GDP (COO)	0.39 (0.28)	0.35 (0.31)	2.58*** (0.53)	0.46† (0.27)	0.30 (0.29)	0.43 (0.28)	0.34 (0.30)
GDP (COA)	-1.09† (0.59)	-0.04 (0.66)	-5.08** (1.74)	-1.59*** (0.44)	-1.12† (0.61)	-1.09† (0.60)	-1.21* (0.61)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.44* (0.19)	-0.33 (0.36)	-0.47** (0.17)	-0.57** (0.18)	-0.54** (0.17)	-0.50** (0.17)
Refugee Rights	-0.02 (0.17)	-0.06 (0.22)	0.01 (0.28)	0.06 (0.15)	0.02 (0.17)	-0.02 (0.19)	0.02 (0.18)
Conflict Duration	-0.03 (0.08)	0.02 (0.11)	0.22 (0.16)	0.02 (0.09)	-0.04 (0.09)	-0.02 (0.09)	-0.03 (0.09)
All controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source	Both	WRS	USHR	Both	Both	Both	Both
Years	1980-2022	1980-2008	2009-2022	1980-2022	1980-2022	1980-2022	1980-2022
Fixed-effects							
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	2,327	1,056	901	2,242	1,877	2,099	2,007
Pseudo R ²	0.80	0.82	0.82	0.79	0.80	0.80	0.80

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. M1: main table col. 1. M2: core GRR panel, 1980–2008 with asylum outside Europe and N. America. M3: State Department yrs. M4: excl. refugee-warrior dyads. M5: excl. asylum in Europe and N. America. M6: war-origin refugees only. M7: top 3 asylum countries per origin-yr. UNHCR-outcome checks are in Table 5.

2.4 Specifications

Table 20 tests whether the results depend on the functional form or fixed-effects structure of the main specification. Model 1 reproduces the baseline Poisson model. Models 2 and 3 instead estimate OLS using $\log(y + 1)$ and $\text{asinh}(y)$, while Model 4 uses a negative binomial model to allow for overdispersion in the return count. Models 5 and 6 impose more demanding fixed effects that absorb time-varying confounders on each side of the dyad. Origin-by-year fixed effects absorb all characteristics shared by refugees from the same origin in a given year, including conflict and the post-war window, so the asylum-side coefficients are identified by differences across their host countries. Asylum-by-year fixed effects absorb all characteristics shared by refugees in the same host-year, including host conflict, income, and refugee-rights policy, so the exclusion-policy coefficients are identified by differences across refugee groups within the same host. The latter remains identifiable because exclusion policy is coded at the dyad rather than host level. The results are robust overall, although the asylum-country conflict coefficients are less stable across specifications.

Table 20: Robustness: Alternative Specifications

Dep. Var.:	Total Repatriation	log(trep+1)	asinh(trep+1)	Total Repatriation		
Model:	Main	OLS log	OLS asinh	Neg.Bin.	Origin-Yr	Asylum-Yr
	(1)	(2)	(3)	(4)	(5)	(6)
	Poisson	OLS	OLS	Neg. Bin.	Poisson	Poisson
Asylum-side push						
Exclusion policy: severe	1.09*** (0.19)	4.58*** (0.42)	4.87*** (0.45)	2.74*** (0.51)	1.34*** (0.34)	1.70* (0.78)
Exclusion policy: moderate	0.48* (0.22)	1.64*** (0.29)	1.79*** (0.32)	1.23* (0.55)	1.36*** (0.16)	0.94* (0.44)
Conflict in COA: high	0.59† (0.31)	0.55 (0.47)	0.57 (0.50)	0.78 (0.51)	0.67 (0.51)	
Conflict in COA: low	0.81*** (0.21)	0.39 (0.30)	0.42 (0.32)	0.23 (0.41)	0.97*** (0.18)	
Existing explanations						
War Ending	1.00*** (0.22)	0.76* (0.35)	0.79* (0.38)	0.89* (0.42)		1.32** (0.45)
Conflict in COO: high	0.11 (0.28)	-0.67 (0.42)	-0.73 (0.46)	0.56 (0.52)	-0.58 (0.78)	0.86† (0.46)
Conflict in COO: low	0.30 (0.22)	-0.24 (0.27)	-0.27 (0.29)	-0.13 (0.31)	0.05 (0.41)	0.77* (0.32)
GDP (COO)	0.39 (0.28)	0.31 (0.26)	0.32 (0.28)	0.23 (0.57)		-0.15 (0.52)
GDP (COA)	-1.09† (0.59)	-0.88 (0.57)	-0.95 (0.62)	-0.61 (1.04)	-2.14† (1.11)	
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.49** (0.18)	-0.53** (0.19)	-0.48 (0.33)		-0.55* (0.26)
Refugee Rights	-0.02 (0.17)	0.21 (0.20)	0.23 (0.22)	0.31 (0.29)	0.10 (0.16)	
Conflict Duration	-0.03 (0.08)	0.28* (0.13)	0.30* (0.14)	0.46* (0.21)	0.09 (0.18)	-0.13 (0.20)
Log Stock at Risk		0.79*** (0.10)	0.84*** (0.11)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	Poisson	OLS	OLS	NegBin	Poisson	Poisson
DV	trep+1	log(y+1)	asinh(y)	trep+1	trep+1	trep+1
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes		
coo_iso-Year					Yes	
coa_iso-Year						Yes
Fit statistics						
Observations	2,327	3,913	3,913	2,327	1,397	1,013
Pseudo R ²	0.80	0.15	0.15	0.04	0.93	0.94

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

Dyad FE and dyad-clustered SEs throughout; alternative clustering reported separately. Origin-Yr and Asylum-Yr are origin-by-year and asylum-by-year FE; y is the return count. M2-M3: OLS on a transformed outcome, log stock at risk as a regressor not an offset, so they run on more dyad-years; magnitudes test sign and significance only. M4: negative binomial. M5: origin-by-year FE. M6: asylum-by-year FE. Absorbed variables are omitted.

2.4.1 Alternative Clustering of the Standard Errors

The baseline clusters by dyad. Alternative groupings account for dependence among dyads that share an origin, a host, or a calendar year. Table 21 reports the standard errors and p -values under each scheme; the point estimates are common to all of them. Clustering by year, or two-way by dyad and year, gives the largest standard errors, and even there all three estimates remain significant: the severe-policy coefficient below the 0.001 level, the war-ending window below the 0.001 level and low-intensity asylum-country conflict below the 0.01 level under every scheme in the table.

Table 21: Alternative clustering of the standard errors

Clustered by	Groups	Exclusion policy: severe		War Ending		Conflict in COA: low	
		SE	p	SE	p	SE	p
Dyad (baseline)	110	0.19***	<0.001	0.22***	<0.001	0.21***	<0.001
Origin country	51	0.16***	<0.001	0.23***	<0.001	0.23***	<0.001
Asylum country	65	0.17***	<0.001	0.28***	<0.001	0.16***	<0.001
Origin + asylum, two-way	116	0.13***	<0.001	0.28***	<0.001	0.19***	<0.001
Year	41	0.26***	<0.001	0.20***	<0.001	0.30**	0.008
Dyad + year, two-way	151	0.22***	<0.001	0.24***	<0.001	0.27**	0.003
Heteroskedastic, no clustering	–	0.24***	<0.001	0.18***	<0.001	0.24**	0.001

Note: Main table col. 1 throughout, so the coefficients do not vary: severe exclusion policy 1.09, war-ending window 1.00, low-intensity asylum conflict 0.81 in every row. Stars are on significance under that variance estimator. ‘Groups’ is the number of clusters, summed across dimensions for the two-way estimators. Read the dyad-plus-year row with care: with 41 yr groups the variance matrix is repaired by fixest, so the year-clustered row is the conservative benchmark.

Table 22: Clustering aligned to the level at which each treatment is assigned

Coefficient	Estimate	Assigned	Dyad p	Aligned SE	Aligned p	Two-way p	Boot p
Exclusion policy: severe	1.09	Dyad	<0.001	0.19***	<0.001	<0.001	0.000
Conflict in COA: high	0.59	Asylum country	0.055	0.28*	0.032	0.094	0.022
Conflict in COA: low	0.81	Asylum country	<0.001	0.16***	<0.001	<0.001	0.000
War Ending	1.00	Origin country	<0.001	0.23***	<0.001	<0.001	0.000
Repatriation funding: high	0.60	Origin country	0.046	0.27*	0.028	0.035	0.014
Repatriation funding: low	-0.23	Origin country	0.502	0.34	0.508	0.572	0.513

Note: Rows 1–4 are main table col. 1; rows 5–6 are col. 2, the only model carrying the funding terms. *Assigned* is the level at which the treatment varies. *Dyad p* is the paper’s clustering; *Aligned* clusters at the assignment level; *Two-way* adds year. *Boot p* is a score wild cluster bootstrap at the assignment level, 5,000 Rademacher draws. Clusters: 110 dyads, 65 asylum countries, 51 origin countries.

2.5 Influence Diagnostics

2.5.1 Largest Cases

The analysis includes 293 dyads. The tables below list the dyads with the most return, the largest single-year movements, and the largest populations with little or no return. They use the full GRR panel rather than the estimation sample, so they also include dyads omitted from the fixed-effects models.

Table 23: The twenty dyads with the most recorded return, 1980–2023

Dyad	Origin → asylum	Years	Total return	Peak stock	Return/peak (%)	% yrs COO conflict	Push ever
AFG-IRN	Afghanistan → Iran	1980–2023	7,507,327	6,352,317	118	100	Both
AFG-PAK	Afghanistan → Pakistan	1980–2023	7,018,397	3,997,133	176	100	Both
IRQ-IRN-KUR	Iraq (Kurds) → Iran	1980–2023	1,426,658	160,000	892	33	Both
RWA-COD	Rwanda → Congo - Kinshasa	1980–2023	1,190,071	1,000,000	119	49	Both
MOZ-MWI	Mozambique → Malawi	1985–2023	1,126,986	1,085,000	104	44	None
BDI-TZA	Burundi → Tanzania	1980–2023	1,049,147	443,700	236	53	Both
ETH-SOM	Ethiopia → Somalia	1980–2023	1,039,754	705,000	147	52	COA conflict
MMR-THA	Myanmar (Burma) → Thailand	1984–2023	735,024	1,653,500	44	72	Both
KHM-THA	Cambodia → Thailand	1981–2023	631,632	365,000	173	45	Both
SDN-SSD-UGA	Sudan → Uganda	1985–2023	623,004	1,037,898	60	97	COA conflict
SSD-SDN	South Sudan → Sudan	2014–2023	572,695	852,080	67	80	COA conflict
SYR-TUR	Syria → Turkey	2009–2023	559,246	3,737,369	15	87	COA conflict
SOM-ETH	Somalia → Ethiopia	1986–2023	532,300	419,000	127	81	COA conflict
UKR-POL	Ukraine → Poland	2002–2023	468,475	958,519	49	45	None
ETH-ERI-SDN	Ethiopia → Sudan	1980–2023	433,357	730,000	59	26	COA conflict
LBR-GIN	Liberia → Guinea	1988–2023	404,223	420,000	96	14	COA conflict
AGO-COD	Angola → Congo - Kinshasa	1980–2023	383,166	312,700	123	40	Both
UGA-SDN	Uganda → Sudan	1980–2010	382,250	250,000	153	95	COA conflict
MOZ-ZAF	Mozambique → South Africa	1984–2023	359,700	300,000	120	50	Both
CIV-LBR	Côte d'Ivoire → Liberia	2002–2023	328,376	128,067	256	18	COA conflict

Note: Return over peak stock can exceed 100 per cent where a population arrived and left more than once over the period.

Table 24: The twenty largest single-year return episodes

Dyad	Year	Origin → asylum	Return	Stock at risk	Rate (%)	COO	COA	Forced
AFG-PAK	2002	Afghanistan → Pakistan	1,700,000	2,100,000	81.0	Low	Low	Moderate
IRQ-IRN-KUR	1991	Iraq (Kurds) → Iran	1,295,000	1,366,400 [†]	94.8	Low	Low	None
AFG-IRN	2020	Afghanistan → Iran	1,151,789	2,701,142	42.6	High	Low	Severe
AFG-PAK	1992	Afghanistan → Pakistan	910,000	3,997,133	22.8	High	Low	None
RWA-COD	1996	Rwanda → Congo - Kinshasa	700,000	900,000	77.8	Low	High	None
AFG-IRN	2019	Afghanistan → Iran	668,209	2,701,142	24.7	High	Low	Severe
MOZ-MWI	1994	Mozambique → Malawi	600,000	700,000	85.7	None	None	None
AFG-IRN	2005	Afghanistan → Iran	489,600	1,702,800	28.8	High	Low	Severe
AFG-PAK	2005	Afghanistan → Pakistan	475,100	967,200	49.1	High	None	Severe
AFG-IRN	1993	Afghanistan → Iran	471,500	2,862,500	16.5	High	Low	Severe
AFG-IRN	2004	Afghanistan → Iran	419,500	1,600,000	26.2	Low	None	Severe
AFG-PAK	2004	Afghanistan → Pakistan	409,100	1,200,000	34.1	Low	Low	None
AFG-IRN	2008	Afghanistan → Iran	409,000	2,414,700	16.9	High	Low	Severe
UKR-POL	2022	Ukraine → Poland	403,000	1,361,519 [†]	29.6	High	None	None
MOZ-MWI	1993	Mozambique → Malawi	400,000	1,085,000	36.9	None	None	None
SSD-SDN	2023	South Sudan → Sudan	386,793	796,831	48.5	None	High	Moderate
AFG-PAK	2016	Afghanistan → Pakistan	381,275	2,366,950	16.1	High	Low	None
AFG-IRN	2007	Afghanistan → Iran	379,650	2,190,400	17.3	High	Low	Severe
AFG-IRN	2002	Afghanistan → Iran	376,300	2,855,000	13.2	Low	None	Severe
AFG-PAK	1993	Afghanistan → Pakistan	358,800	2,075,000	17.3	High	None	None

Note: [†] marks dyad-years in which recorded return exceeds the lagged stock, where the denominator is the year-end stock plus return rather than the lagged stock, so the rate is an upper bound. COO and COA give conflict intensity in the origin and asylum country that year; Forced gives the asylum country's exclusion policy.

Table 25: The non-return cases

Dyad	Origin → asylum	Years	Refugee-years (m)	Peak stock	Total return	% yrs COO conflict	Push ever
PSE-JOR	Palestinian Territories → Jordan	1981–2008	44.62	2,439,718	233	89	None
VNM-CHN	Vietnam → China	1980–2021	11.97	321,502	0	15	COA conflict
VEN-USA	Venezuela → Colombia	2009–2023	11.93	2,875,743	0	0	COA conflict
PSE-SYR	Palestinian Territories → Syria	1980–2016	10.62	592,200	233	83	COA conflict
PSE-LBN	Palestinian Territories → Lebanon	1981–2023	9.24	500,000	233	81	COA conflict
VEN-PER	Venezuela → Peru	2003–2023	6.98	1,542,004	0	0	COA conflict
ESH-DZA	Western Sahara → Algeria	1980–2023	5.89	165,000	0	21	COA conflict
COL-ECU	Colombia → Ecuador	1990–2023	4.71	674,500	5,102	85	COA conflict
CHN-IND	China → India	1984–2023	4.01	123,000	0	0	COA conflict
UKR-RUS	Ukraine → Russia	1996–2023	3.56	1,275,925	0	42	COA conflict
PHL-MYS	Philippines → Malaysia	1981–2023	2.70	136,075	0	91	COA conflict
BTN-NPL	Bhutan → Nepal	1991–2023	2.33	133,700	0	0	Both
RUS-USA	Russia → United States	1988–2023	2.22	245,238	0	0	None
ERI-ETH	Eritrea → Ethiopia	1993–2023	2.03	181,271	0	8	COA conflict
TUR-DEU	Turkey → Germany	2000–2023	1.98	181,344	0	96	None
IRN-TUR	Iran → Turkey	1982–2023	1.98	257,000	2,208	53	Both
CHN-USA	China → United States	1988–2023	1.96	173,873	0	0	Moderate only
VEN-BRA	Venezuela → Brazil	2011–2023	1.93	510,499	0	0	None
GTM-USA	Guatemala → United States	1988–2023	1.76	192,596	0	20	Moderate only
ARM-AZE	Armenia → Azerbaijan	1992–2013	1.56	229,000	0	38	COA conflict

Note: The twenty largest dyads whose cumulative return never reaches 1 per cent of peak stock. Ranked by refugee-years, the sum of the annual stock over the dyad's observed history, which measures how much displacement the dyad represents rather than how long it is observed. The three Palestinian dyads each record the same 233 returns in 1986, which is one reported figure carried across all three rather than three independent observations.

2.5.2 Dropping the Largest Cases

Recorded return is heavily concentrated, and how concentrated depends on which sample the share is taken over. Across the full GRR panel, the top five percent of dyad-years account for 93.8 percent of recorded return. Within the estimation frame of 5,131 dyad-years the figure is 88.5 percent, and within the Model 1 estimation sample of 2,327 dyad-years it is 75.1 percent; each successive restriction removes dyad-years with little or no recorded return and so spreads the total more evenly. Table 26, which is estimated on the Model 1 sample, tests whether this concentration drives the estimates by dropping the largest dyads or dyad-years and, separately, restricting the sample to the ten largest dyads. The severe-policy coefficient is larger after each exclusion. The post-war estimate is less stable. Low-intensity asylum-country conflict also becomes imprecise after excluding the largest cases, but is strongly associated with return among the ten largest dyads.

Table 26: Robustness: Concentration of Returns

Dependent Variable:	Total Repatriation				
Model:	Full (1)	Drop top 5 dyads (2)	Drop top 1% yrs (3)	Drop top 5% yrs (4)	Top 10 dyads (5)
Asylum-side push					
Exclusion policy: severe	1.09*** (0.19)	1.44*** (0.25)	1.54*** (0.18)	1.45*** (0.34)	0.86* (0.42)
Exclusion policy: moderate	0.48* (0.22)	0.16 (0.19)	0.45** (0.17)	0.58** (0.22)	0.68 (0.43)
Conflict in COA: high	0.59† (0.31)	0.53* (0.25)	0.51* (0.24)	0.35 (0.25)	0.49 (0.65)
Conflict in COA: low	0.81*** (0.21)	0.27 (0.26)	0.26 (0.25)	0.01 (0.17)	1.24*** (0.37)
Existing explanations					
War Ending	1.00*** (0.22)	0.84** (0.28)	0.95*** (0.25)	0.41* (0.20)	0.87* (0.40)
Conflict in COO: high	0.11 (0.28)	0.07 (0.31)	0.06 (0.29)	-0.53* (0.27)	-0.71*** (0.18)
Conflict in COO: low	0.30 (0.22)	0.16 (0.24)	0.29 (0.21)	-0.25 (0.17)	-0.34 (0.27)
GDP (COO)	0.39 (0.28)	0.58† (0.30)	0.36 (0.27)	0.60** (0.22)	1.63† (0.86)
GDP (COA)	-1.09† (0.59)	-0.50 (0.58)	-0.59 (0.48)	0.12 (0.57)	-3.49* (1.39)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.39* (0.19)	-0.47* (0.20)	-0.29 (0.19)	-0.89** (0.28)
Refugee Rights	-0.02 (0.17)	0.06 (0.15)	0.12 (0.15)	0.02 (0.13)	0.39* (0.18)
Conflict Duration	-0.03 (0.08)	0.04 (0.10)	0.01 (0.09)	0.07 (0.09)	0.36 (0.31)
Fixed-effects					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	2,327	2,199	2,289	2,081	275
Squared Correlation	0.76	0.62	0.59	0.33	0.89
Pseudo R ²	0.80	0.67	0.65	0.44	0.87

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

2.5.3 Leave-One-Out Analyses

Figure 4 re-estimates the main model after excluding each origin country, asylum country, origin region, and asylum region in turn. Table 27 reports exclusions that change a coefficient's sign or move it substantially. No single origin, host, region, or dyad accounts for the severe-policy result.

The less precisely estimated asylum-side coefficients are more sensitive to individual exclusions. High-intensity asylum-country conflict, significant only at the ten percent level in the baseline, changes appreciably when Rwanda–DR Congo and the two Afghan dyads are excluded. Low-intensity asylum-country conflict is also sensitive to the Afghan dyads, while the moderate-policy coefficient changes sign when Afghanistan–Iran is excluded. At the regional level, high-intensity asylum-country conflict becomes negative when Sub-Saharan Africa is excluded as either an origin or asylum region, although neither estimate is statistically significant. Origin-country conflict also changes sign under some exclusions, but is not distinguishable from zero in the baseline. By contrast, the severe exclusion-policy result is highly stable across the leave-one-out exercises.

Figure 4: Coefficient stability across leave-one-out analyses

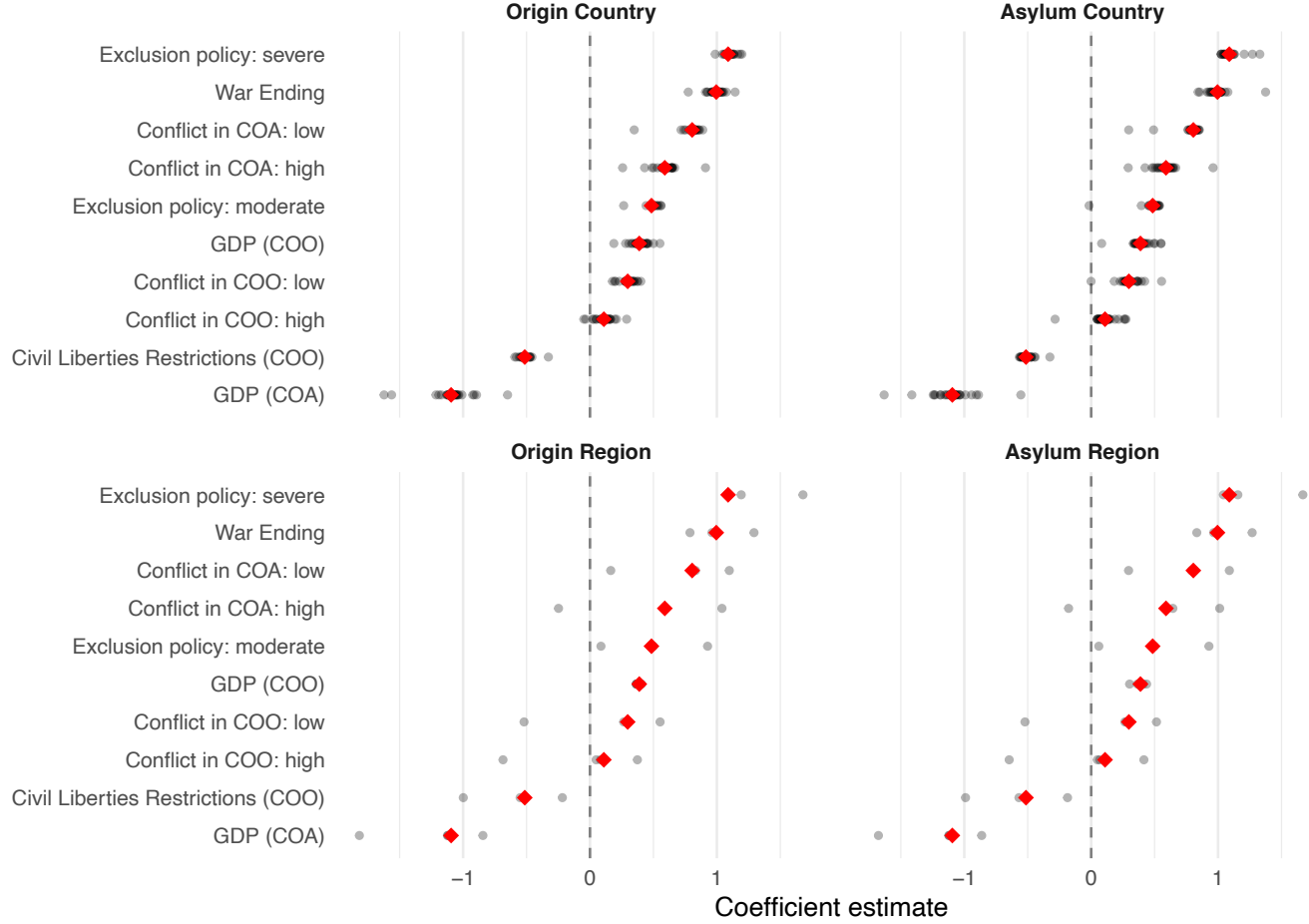


Table 27: Influential regions

Variable	Region Dropped	Drop Type	Main Est.	New Est.	Change
Conflict in COA: high	Sub-Saharan Africa	Origin Region	0.59+	-0.25	0.84
Conflict in COA: high	Sub-Saharan Africa	Asylum Region	0.59+	-0.18	0.77
Conflict in COO: high	Sub-Saharan Africa	Origin Region	0.11	-0.69	0.79
Conflict in COO: high	Sub-Saharan Africa	Asylum Region	0.11	-0.65	0.76
Conflict in COO: low	Sub-Saharan Africa	Asylum Region	0.30	-0.52+	0.82
Conflict in COO: low	Sub-Saharan Africa	Origin Region	0.30	-0.52+	0.82

Note: Regions whose exclusion causes a sign change or an absolute coefficient shift above 1. Stars indicate significance of each coefficient: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Bold rows indicate sign reversals where the new coefficient remains statistically significant at $p < 0.05$; sign flips on coefficients that fall back to non-significance after exclusion are shown but not bolded.

Table 28: Most influential dyads

Variable	Dyad	Main Est.	New Est.	Change
Conflict in COA: high	AFG-PAK	0.59+	0.96***	0.37
Conflict in COA: high	RWA-COD	0.59+	0.31	0.28
Conflict in COA: high	AFG-IRN	0.59+	0.34	0.25
Conflict in COA: high	SOM-ETH	0.59+	0.50	0.09
Conflict in COA: high	MOZ-MWI	0.59+	0.52+	0.07
Conflict in COA: low	AFG-IRN	0.81***	0.27	0.54
Conflict in COA: low	AFG-PAK	0.81***	0.49*	0.31
Conflict in COA: low	KHM-THA	0.81***	0.89***	0.08
Conflict in COA: low	MMR-THA	0.81***	0.73***	0.08
Conflict in COA: low	RWA-COD	0.81***	0.76***	0.05
Exclusion policy: severe	AFG-PAK	1.09***	1.33***	0.24
Exclusion policy: severe	RWA-COD	1.09***	1.29***	0.20
Exclusion policy: severe	AFG-IRN	1.09***	0.97***	0.12
Exclusion policy: severe	BDI-TZA	1.09***	1.19***	0.11
Exclusion policy: severe	MMR-BGD	1.09***	1.03***	0.06
Exclusion policy: moderate	AFG-IRN	0.48*	-0.01	0.49
Exclusion policy: moderate	AFG-PAK	0.48*	0.40*	0.09
Exclusion policy: moderate	BDI-TZA	0.48*	0.54*	0.06
Exclusion policy: moderate	ETH-SOM	0.48*	0.53*	0.05
Exclusion policy: moderate	RWA-COD	0.48*	0.53*	0.04
GDP (COA)	AFG-IRN	-1.09+	-0.54	0.56
GDP (COA)	KHM-THA	-1.09+	-1.63**	0.53
GDP (COA)	UGA-COD	-1.09+	-1.36*	0.26
GDP (COA)	UGA-SDN	-1.09+	-1.31*	0.21
GDP (COA)	MMR-BGD	-1.09+	-0.91	0.19
War Ending	AFG-IRN	1.00***	1.23***	0.24
War Ending	MOZ-MWI	1.00***	0.84***	0.15
War Ending	AFG-PAK	1.00***	0.86***	0.14
War Ending	LBR-GIN	1.00***	0.92***	0.08
War Ending	IRQ-IRN-IRN	1.00***	1.07***	0.07
Civil Liberties Restrictions (COO)	AFG-IRN	-0.51**	-0.39*	0.13
Civil Liberties Restrictions (COO)	ETH-ERI-SDN	-0.51**	-0.60***	0.09
Civil Liberties Restrictions (COO)	AFG-PAK	-0.51**	-0.44**	0.07
Civil Liberties Restrictions (COO)	MMR-THA	-0.51**	-0.45**	0.06
Civil Liberties Restrictions (COO)	IRQ-IRN-IRN	-0.51**	-0.45**	0.06
Conflict in COO: high	RWA-COD	0.11	-0.13	0.24
Conflict in COO: high	ETH-SOM	0.11	0.27	0.16
Conflict in COO: high	AFG-PAK	0.11	0.26	0.15
Conflict in COO: high	AGO-ZMB	0.11	0.20	0.09
Conflict in COO: high	SDN-SSD-UGA	0.11	0.20	0.09
Conflict in COO: low	RWA-COD	0.30	0.10	0.20
Conflict in COO: low	AFG-IRN	0.30	0.50*	0.20
Conflict in COO: low	ETH-SOM	0.30	0.42+	0.12
Conflict in COO: low	AFG-PAK	0.30	0.18	0.11
Conflict in COO: low	MOZ-MWI	0.30	0.37	0.07
GDP (COO)	KHM-THA	0.39	0.19	0.20
GDP (COO)	AFG-PAK	0.39	0.55+	0.16
GDP (COO)	MMR-BGD	0.39	0.55*	0.16
GDP (COO)	SOM-ETH	0.39	0.49+	0.11
GDP (COO)	MMR-THA	0.39	0.29	0.10

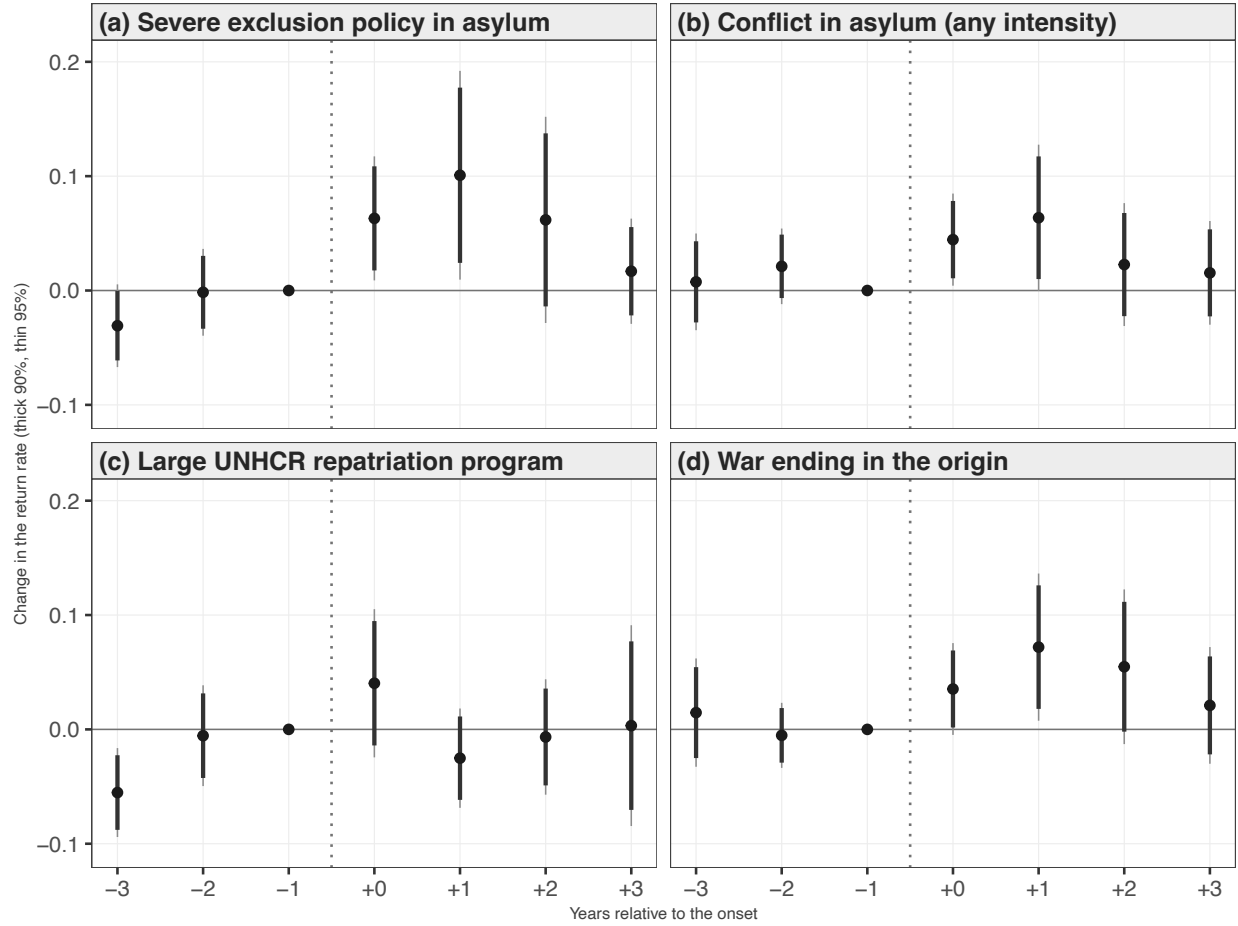
Note: For each key variable, the five dyads whose exclusion produces the largest absolute change in the coefficient. Stars indicate significance of each coefficient: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Bold rows indicate sign reversals where the new coefficient remains statistically significant at $p < 0.05$.

2.6 Event Study

The main text reports stacked event studies around the onset of each of the four conditions, addressing the concern that a condition is adopted or recorded only after return has already begun to rise. Those models estimate PPML on the return count. Figure 5 reports the same four event studies using the linear analogue, with return measured as a share of the stock at risk. The linear model retains dyads with no within-dyad variation in return and therefore uses two to four times as many stacked dyad-years as the PPML models; its coefficients represent changes in the return rate and are not directly comparable in magnitude with the log-scale PPML coefficients in the main text. The results nevertheless show the same substantive patterns.

For large UNHCR repatriation programs, return is lower at $t - 3$ relative to the omitted period. This pattern runs counter to the main endogeneity concern, which would predict rising return before program onset if UNHCR programs were simply initiated in response to returns already underway.

Figure 5: Event studies around condition onsets, linear model on the return rate



3 Extensions and Supporting Analyses

3.1 Simulations of Aggregate Importance

Table 29: Exposure and model-adjusted aggregate contrasts

Condition	Dyad-years present	Stock-years exposed	Reduction in fitted return
Severe exclusion policy	132 (9.6%)	99.9m	22.1%
Post-war window (1–3 years)	189 (13.7%)	21.6m	9.9%
High-intensity asylum conflict	280 (20.3%)	55.0m	12.1%
High repatriation assistance	271 (19.7%)	29.4m	15.1%

Note: Main-text Model 2 throughout: PPML with dyad and yr FE, funding none/low/high, on the 1,378 dyad-years and 100 dyads for which funding is coded; high-assistance coefficient 0.60 (SE 0.30). Final column: percentage reduction in aggregate fitted return with the listed condition set to its reference value, other covariates fixed; the assistance row switches off the high category only. Dyad FE hold each dyad’s average exposure fixed, so these are within-dyad contrasts. Conditions overlap, so rows are not additive. Descriptive contrasts, not causal effects.

Table 29 provides the details behind the aggregate simulations reported in the main text. For each condition, I predict aggregate return under the observed data and then again after setting that condition to its reference value while leaving the remaining covariates unchanged. The final column reports the percentage difference between these two quantities. The model is Model 2 from the main analysis, including dyad and year fixed effects and the categorical measure of UNHCR repatriation assistance. It is therefore estimated only on the 1,378 dyad-years and 100 dyads for which the funding measure is coded and the fixed-effects estimator retains the dyad. Within this sample, removing severe exclusion policy produces the largest reduction in fitted return (22.1%), followed by large repatriation assistance (15.1%), high-intensity asylum conflict (12.1%), and the post-war window (9.9%). These contrasts combine the estimated association with exposure: a condition can matter substantially in aggregate either because its coefficient is large, because many refugees are exposed to it, or both. Because the model carries dyad and year fixed effects, each contrast holds the dyad’s average exposure fixed and therefore reports the within-dyad contribution of the condition rather than its share of return globally.

3.2 Interaction between Asylum-Side Shock and Origin-Country Conflict

Table 30 examines whether asylum-side shocks are associated with return differently when conflict remains underway in the origin country. I collapse origin-country conflict into a single indicator and interact it separately with severe exclusion policy and low- and high-intensity conflict in the asylum country. None of the interaction terms is statistically distinguishable from zero. The associations between these asylum-side shocks and return are therefore similar whether the origin country is at peace or still experiencing conflict.

Table 30: Asylum-side shocks interacted with ongoing conflict at the origin

Term	Origin at peace	Interaction	p (interaction)	Origin at war
Severe exclusion policy	1.19** (0.36)	-0.15 (0.32)	0.64	1.04 (0.18)
High-intensity asylum conflict	0.57 (0.36)	-0.05 (0.49)	0.92	0.52 (0.38)
Low-intensity asylum conflict	0.38 (0.31)	0.49 (0.30)	0.10	0.87 (0.19)

Note: PPML; log stock-at-risk offset; dyad and yr FE; dyad-clustered SEs. Two models, both main table col. 1 with origin conflict collapsed to a single indicator. Row 1 interacts it with severe exclusion policy; rows 2–3 with both asylum-conflict intensities at once, so they are mutually adjusted. *Origin at peace* is the asylum-side coefficient at peace, *Interaction* what conflict adds, *Origin at war* their sum. No stars on the final column: it is a linear combination.

3.3 Cross-Dyad Correlates

The main analysis identifies relationships from changes within origin-asylum dyads over time. Because dyad fixed-effects PPML drops dyads with no observed return over the estimation period, it excludes a substantial set of refugee populations for which return never occurs. This section instead examines variation across dyads, retaining those zero-return dyads and allowing the analysis to incorporate characteristics that do not vary over time. Table 31 reports the seven additional dyad-level measures and their sources.

Table 32 reports the cross-dyad specification alongside the main fixed-effects model. The central asylum-side patterns are broadly similar: severe exclusion policies and low-intensity conflict in the asylum country remain positively associated with return, and the positive association with the post-war period is also unchanged. Some other relationships differ once the comparison is made across dyads rather than within them. In particular, refugees from poorer origin countries and those hosted in poorer asylum countries tend to have higher return rates, while greater civil-liberties restrictions in the origin are associated with lower return. Among the time-invariant dyadic characteristics added to this specification, most show little systematic relationship with return. Religious similarity is positively associated with return, while common spoken language is weakly negatively associated; distance, contiguity, Refugee Convention membership, and cross-border kin ties are not distinguishable from zero. These estimates are descriptive cross-dyad associations rather than robustness checks on the within-dyad analysis, but they show that the paper’s principal asylum-side relationships are not confined to the subset of dyads that experience return.

Table 31: Provenance of the cross-dyad measures

Term	Measure	Source
Refugee Convention	Host is party to the 1967 Protocol and is not barred from this origin by an article 1B(1)(a) declaration limiting the Convention to events in Europe	UN Treaty Collection ch. V-2 and V-5.
Distance (log)	Log population-weighted great-circle distance between the two countries, km	CEPII GeoDist (Mayer and Zignago 2011); seven dyads GeoDist does not cover filled with great-circle distances between hand-entered centroids
Contiguous	Shared land border	CEPII GeoDist, same merge as distance
Common spoken language	A language spoken by at least 9 per cent of the population of both countries (CEPII <code>comlang_ethno</code>)	CEPII GeoDist (Mayer and Zignago 2011)
Religious similarity (z)	Probability that one person drawn at random from each country adheres to the same broad tradition, over sixteen traditions, averaged across the waves from 1980; standardized	Correlates of War World Religion Project v1.1 (Maoz and Henderson 2013)
Kin share of origin population (z)	Share of the origin population belonging to a transnational ethnic kin group that is also present in the asylum country, averaged over the dyad’s years; standardized	EPR-Core and EPR-TEK 2021 (Vogt et al. 2015)
Log Stock at Risk	The model’s own exposure offset, entered again as a regressor	GRR, same construction as the offset

Note: The seven terms entering only the cross-dyad column. (z) measures are standardized on the estimation frame, so a coefficient is the association with a one-SD difference between dyads. Religious similarity is the inner product of the two countries’ adherence-share vectors; kin share comes from the EPR group panel intersected with EPR-TEK links.

To complement the regression analysis, Figure 6 compares the share of recorded return occurring under each condition with the share of the global population at risk exposed to it. The conditions are not mutually exclusive, and these comparisons are descriptive, without controls or fixed effects.

Figure 7 plots the twenty origin-asylum dyads with the most recorded return over time, showing origin-country conflict intensity alongside the asylum-side conditions summarized in the descriptive figure. Asylum-side shocks are common across these major displacement contexts rather than concentrated in a small number

Table 32: Cross-dyad correlates

Model:	Total Repatriation	
	Dyad & year FE (1)	Cross-dyad (2)
Asylum-side push		
Exclusion policy: severe	1.09*** (0.19)	0.98*** (0.23)
Exclusion policy: moderate	0.48* (0.22)	0.27 (0.17)
Conflict in COA: high	0.59 [†] (0.31)	0.50 (0.31)
Conflict in COA: low	0.81*** (0.21)	0.59** (0.18)
Existing explanations		
War Ending	1.00*** (0.22)	1.16*** (0.25)
Conflict in COO: high	0.11 (0.28)	0.14 (0.26)
Conflict in COO: low	0.30 (0.22)	0.30 (0.28)
GDP (COO)	0.39 (0.28)	-0.46*** (0.13)
GDP (COA)	-1.09 [†] (0.59)	-0.80*** (0.17)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.50*** (0.10)
Refugee Rights	-0.02 (0.17)	0.00 (0.08)
Conflict Duration	-0.03 (0.08)	0.04 (0.10)
Cross-dyad measures (absorbed by dyad fixed effects)		
Refugee Convention		0.25 (0.22)
Distance (log)		0.04 (0.15)
Contiguous		0.54 (0.43)
Log Stock at Risk		-0.04 (0.06)
Common spoken language		-0.32 [†] (0.18)
Religious similarity (z)		0.24** (0.08)
Kin share of origin population (z)		-0.06 (0.07)
Constant		-4.87** (1.61)
Fixed-effects		
Dyad	Yes	
Year	Yes	
Fit statistics		
Observations	2,327	3,810
Pseudo R ²	0.80	0.72

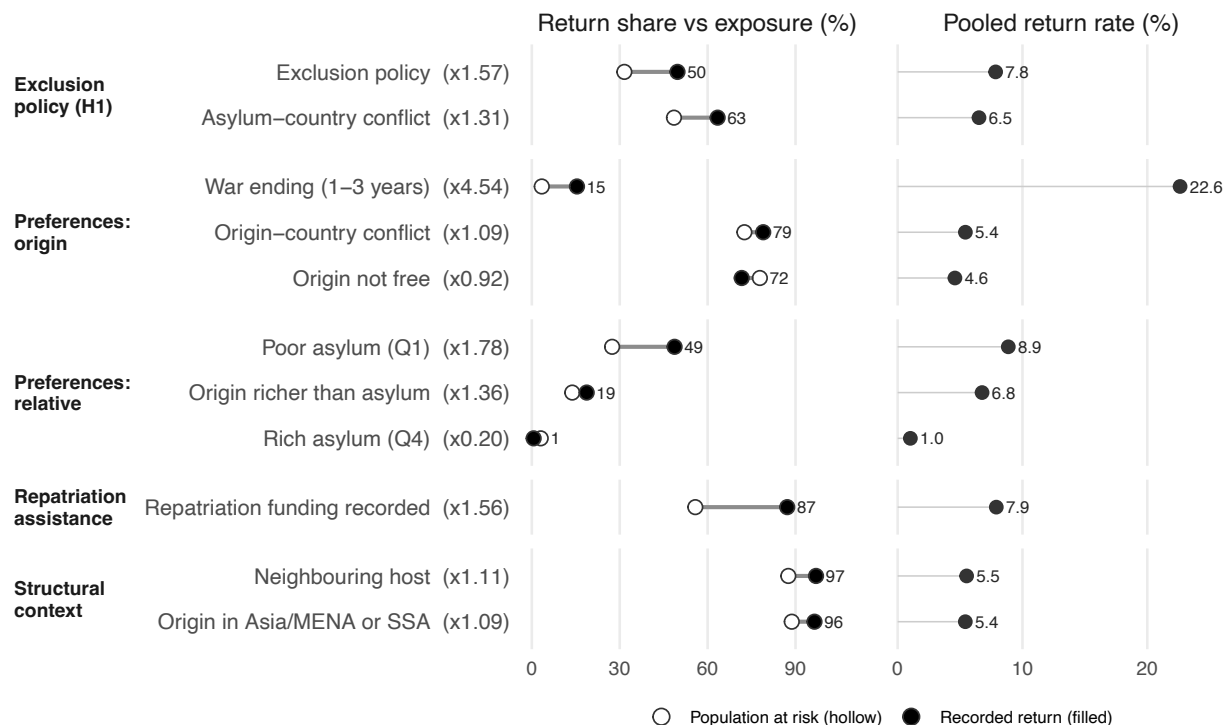
Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

PPML; log stock-at-risk offset; dyad-clustered SEs. Col. 1: main table col. 1. Col. 2: dyad and yr FE removed and the absorbed dyad characteristics entered directly, so the log stock at risk is both offset and regressor. Col. 2 retains 249 of 285 dyads and 66.5% of stock-years, against 110 and 58.7% under FE. Sources in Table 31.

of cases: nine of the twenty dyads record a severe exclusion policy at some point and fifteen record one at any level, against 35% of dyads panel-wide that ever record a policy of any kind. They also frequently occur while conflict in the origin country is still ongoing, visible as symbols appearing on shaded rather than unshaded portions of each row.

Figure 6: Correlates of global refugee return



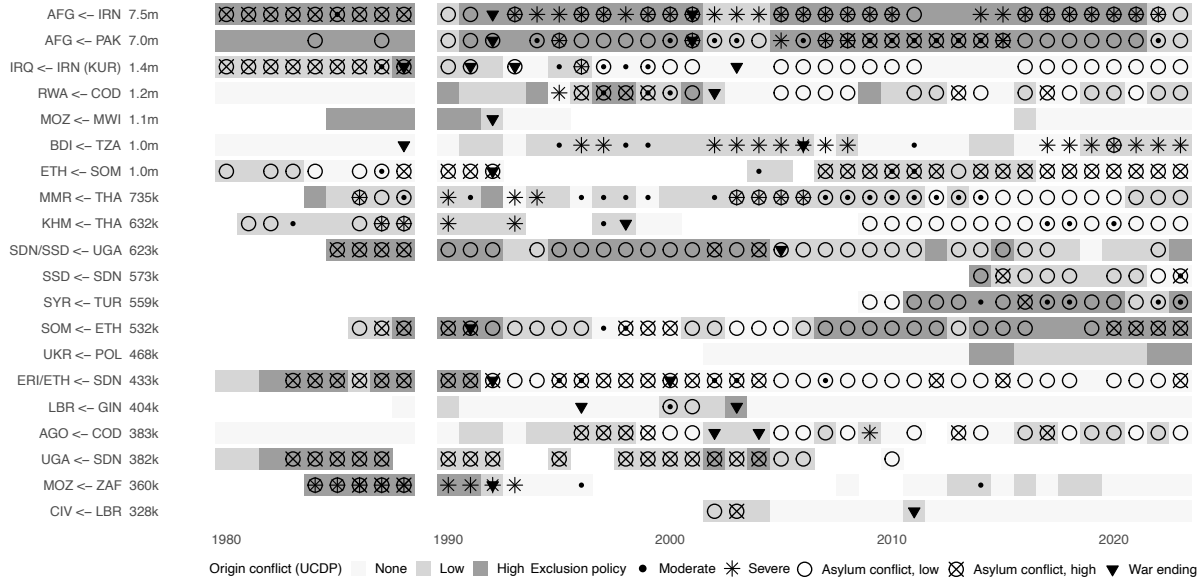
Note: Descriptive; no covariates, FE or model. Rows are conditions, grouped by the account they belong to. Left panel: the share of all recorded return occurring under the condition (filled) against the condition's share of stock-ys at risk (hollow); the multiplier in each row label is the ratio of the two. Right panel: the pooled return rate under the condition, total return over total stock-ys at risk. Denominators are the full rate sample except the repatriation-funding row, which is taken over the yrs in which the funding series is coded, since 2000–2010 and 2021–2023 are missing rather than zero. Conditions overlap, so rows are not additive.

3.4 Conditional Effects of the Post-War Window

A stronger version of the preference-based account implies that the association between peace at home and return should depend on how attractive remaining in asylum is. The main text reports the substantive results; Table 33 reports the full interaction models. Each column interacts the post-war window with a binary condition divided at the sample median, while retaining the full main specification and the continuous measure underlying the split.

Overall, there is little evidence that the association between war ending and return systematically varies with these conditions. Asylum-country income is the exception: the post-war association is substantially weaker in wealthier asylum countries and is not distinguishable from zero there. The interactions with origin-country conditions and repatriation funding are not precisely estimated. These interaction results are distinct from the main associations of the moderating variables themselves: outside the post-war window, high repatriation funding is associated with more return, while greater freedom in the origin is associated with less.

Figure 7: Conflict, war endings, and exclusion policy in the 20 dyads with the most return



Note: The 20 origin-asylum dyads with the most recorded return, 1980–2023, together 78% of all recorded return; rows ordered by that total, labels giving the direction of return and the dyad’s total. Shading: origin conflict intensity (UCDP). Symbols: exclusion policy, asylum conflict and war endings, per the legend; policy symbols are drawn over conflict symbols, so a dyad-year with both reads as a star or dot inside a circle. War endings here are the extended civil-war list merged with the coded interstate wars, a wider set than the UCDP high-intensity spells used elsewhere. The gap in every row is 1989, for which GRR has no observations.

Table 33: Interactions between the post-war window and permissive conditions

Term	(1) Poor asylum	(2) Rich origin	(3) Large funding	(4) Free origin
Window and condition				
Condition absent means	Rich asylum	Poor origin	No large funding	Repressive origin
War ending, condition absent	0.03 (0.24)	1.16*** (0.26)	0.57 (0.42)	0.84*** (0.21)
Condition alone, no war ending	0.21 (0.34)	0.79** (0.30)	0.69** (0.24)	-0.69† (0.40)
War ending x condition	1.42*** (0.25)	-0.59 (0.42)	0.66 (0.56)	0.63 (0.40)
War ending, condition present	1.44*** (0.22)	0.57† (0.32)	1.23** (0.48)	1.47*** (0.44)
Support and fit				
Post-war dyad-years, condition absent	62	87	53	90
Post-war dyad-years, condition present	73	48	42	45
All main-specification variables	Yes	Yes	Yes	Yes
Dyad and year fixed effects	Yes	Yes	Yes	Yes
Dyad-years	2,327	2,327	1,378	2,327
Pseudo R2	0.808	0.802	0.783	0.801

3.5 The Implementation of Exclusion Policies

In the main analysis, exclusion policies are a key explanatory variable and predict higher rates of return. This section instead treats severe exclusion policy as the dependent variable and examines what predicts its occurrence across the global data. Possible explanations include refugee population size, host institutions, conflict and militarization, and social ties.

Table 34 reports three comparisons: across dyads with year fixed effects, within the same dyad over time, and across refugee groups from the same origin in the same year. The clearest pattern is population size. A ten-fold increase in the lagged refugee stock is associated with 8.10 times the odds of severe exclusion in the cross-dyad model, and the association remains positive within dyads and within origin-years. By contrast, the host's total refugee burden, conflict, militarization, and social ties do not predict severe policy consistently across specifications. Host political measures are less uniform: the V-Dem polyarchy measure is not robust, while the DWRAP refugee-rights index is positively associated with severe policy in all three comparisons. The additional measures in the final block are entered separately rather than jointly. Overall, severe exclusion is most consistently associated with the size of the particular refugee population a government is hosting, rather than with a general measure of host burden or a single alternative security explanation.

Table 35 examines whether this pattern is concentrated early in a displacement episode. Severe-policy onset is most common two to three years after a dyad enters the panel and declines thereafter overall. Among large refugee populations, however, onset risk remains elevated even many years after displacement, suggesting that severe policy is not simply a short-run response to a new influx.

Table 34: Predictors of severe exclusion policy

Variable	(1) Cross-dyad	(2) Within dyad	(3) Same origin-year
Scale of the refugee population			
Refugee stock in this dyad (log, lagged)	1.71*** (0.32)	1.19** (0.40)	2.20 [†] (1.32)
Total refugees hosted by the asylum country (log)	-0.05 (0.24)	-0.29 (0.46)	-0.74 (1.19)
Camp residence (ordinal)	0.34 (0.24)	0.67 (0.69)	-0.86 (0.68)
Host government and host pressure			
GDP p.c. in asylum (log)	0.18 (0.34)	-1.26 (1.63)	3.52* (1.42)
V-Dem polyarchy (host)	0.94** (0.36)	0.55 (0.77)	-1.03 (1.26)
DWRAP refugee-rights index (host)	0.40* (0.19)	0.64* (0.32)	1.04* (0.48)
Conflict in asylum: high	-0.84 (0.56)	-0.18 (0.83)	-1.39 (1.94)
Conflict in asylum: low	0.48 (0.35)	0.53 (0.51)	-0.57 (1.08)
Origin-country conditions			
Conflict in origin: high	0.04 (0.45)	0.03 (0.86)	—
Conflict in origin: low	0.03 (0.48)	0.30 (0.46)	—
GDP p.c. in origin (log)	-0.85** (0.30)	-0.91 (0.99)	—
Civil-liberties restrictions in origin	-0.17 (0.24)	0.56 (0.39)	—
Cumulative years of origin conflict (log)	0.10 (0.24)	0.10 (0.37)	—
Dyadic relationship and duration			
Distance origin-asylum (log)	-0.09 (0.26)	—	-0.43 (0.57)
Origin and asylum contiguous	-0.01 (0.58)	—	6.94* (2.88)
Composite cultural-tie index	0.19 (0.21)	—	-1.84 (1.36)
Years since the dyad entered the panel	0.15 (0.36)	—	-3.71* (1.58)
Added one at a time			
Origin battle deaths (log, per SD)	0.45 (0.51)	-0.42 (0.55)	-1.00 (1.40)
Refugees reported militarized	0.81 [†] (0.47)	-0.78 (0.64)	0.12 (1.63)
Common spoken language	0.09 (0.57)	—	0.65 (2.07)
Religious similarity	0.19 (0.22)	—	-1.51 (1.21)
Kin share of origin population	-0.16 (0.15)	—	0.80 (0.68)
Specification			
Fixed effects	Year	Dyad, year	Origin × year
Dyad-years	3,622	917	285

Note: Logit on severe exclusion policy; dyad-clustered SEs. Col. 1 pools dyads with yr FE. Col. 2 is the paper's within-dyad comparison. Col. 3 compares refugee groups from the same origin in the same year across asylum countries. A dash means the design absorbs the variable; *sep.* marks a cell with no counter-examples rather than an estimate. Final-block rows are each added singly to the column's own specification; battle deaths are UCDP counts from 1989. All models converged.

Table 35: Onset risk of severe policy by displacement age and population size

Years since the dyad entered the panel	All sizes	Under 100,000 refugees	100,000 refugees or more
0–1	0.00% (0/20)	0.00% (0/11)	0.00% (0/9)
2–3	3.98% (9/226)	3.64% (6/165)	4.92% (3/61)
4–5	1.53% (4/262)	1.04% (2/193)	2.90% (2/69)
6–10	1.36% (9/663)	0.83% (4/481)	2.75% (5/182)
11–15	1.16% (8/687)	0.60% (3/497)	2.63% (5/190)
16–20	1.16% (7/604)	0.00% (0/464)	5.00% (7/140)
21+	0.79% (11/1400)	0.18% (2/1106)	3.06% (9/294)

Note: Each cell is the probability that a new severe exclusion policy begins, with onsets over dyad-years at risk in parentheses. The risk set is dyad-years whose previous CALENDAR year is observed in the sample and carried no severe policy, so 1988 and 1990 are never treated as consecutive. Age is measured from the dyad's first year of observed refugee presence in the full panel.

4 Literature Review

Figure 1 is based on a systematic search of nineteen journals from 2006 through 2026:

- **Political science** (twelve): *American Political Science Review*, *American Journal of Political Science*, *Journal of Politics*, *Comparative Political Studies*, *British Journal of Political Science*, *World Politics*, *International Organization*, *International Studies Quarterly*, *Journal of Peace Research*, *Journal of Conflict Resolution*, *Political Science Research and Methods*, and *PNAS*.
- **Migration and refugee studies** (five): *Journal of Refugee Studies*, *International Migration Review*, *Journal of Ethnic and Migration Studies*, *International Migration*, and *Comparative Migration Studies*.
- **Development economics** (two): *World Development* and the *Journal of Development Economics*.

I supplemented the journal search by tracing citations to and from core studies, reviewing the 2024 World Bank–UNHCR Joint Data Center review of refugee return, and searching for combinations of *refugee return*, *repatriation*, *return intentions*, *IDP return*, and *internally displaced return*.

I conducted an additional search focused on qualitative research. It combined return and repatriation terms with *interview*, *ethnography*, *focus group*, *life history*, *decision-making*, *home*, *belonging*, *waiting*, *refusal*, and *circular*, together with publisher-site searches and citation tracing from key qualitative studies. The qualitative search was targeted rather than a second journal-by-journal census.

A study is included if it (a) reports original empirical evidence, (b) examines refugees or internally displaced people displaced by conflict, and (c) analyzes qualitative return decisions, stated return intentions or preferences, or realized return. I exclude reviews, conceptual and normative work, studies of state policy or pressure to leave, and studies of post-return reintegration unless return itself is an outcome. Working papers and online-first articles are included.

I code three things. *Outcome* distinguishes qualitative evidence on return decisions, stated intentions or preferences, and realized return. *Scope* distinguishes studies of one displacement context from multi-context or cross-national studies. The first category includes studies of several origin populations in one host and of one origin population across several hosts, provided that the study concerns a single displacement context. Only studies spanning several distinct displacement contexts are coded as cross-national. *Population* differentiates internally displaced people, Ukrainian refugees, Syrian refugees, and other refugee populations.

Table 36: Studies plotted in Figure 1 of the paper

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Peace Research 61(6)	Costalli, Stefano (2024)	2024	Do UN Peace Operations Help Forcibly Displaced People?	Multiple	Multi-context	Realized return	Other refugees
J Refugee Studies	Zakirova, Komila (2021)	2021	The Road Back Home is Never Long: Refugee Return Migration	Multiple	Multi-context	Realized return	Other refugees
Peace Economics, Peace Science and Public Policy Development in Practice 30	Arias, María Alejandra (2014)	2014	The Desire to Return during Civil War: Evidence for Internally Displaced Populations in Colombia	Colombian	One context	Intentions	IDPs
	Collado, Zenaida C. (2020)	2020	Living in Displacement Contexts: Return Intentions among Internally Displaced Persons in Marawi, Philippines	Marawi IDPs	One context	Intentions	IDPs
International Migration 64(3)	Malashkhia, Mariam (2026)	2026	Integration and Return Intentions among Displaced Individuals	Georgian	One context	Intentions	IDPs
European Journal of Psychotraumatology 14(2)	Thompson, Peter Onah (2023)	2023	Posttraumatic Stress Moderates Return Intentions: A Factorial Survey Experiment with Internally Displaced Persons in Nigeria	Conflict-displaced Nigerians	One context	Intentions	IDPs
JEMS	Di Saint Pierre, Francesca (2015)	2015	Return Wishes of Refugees in the Netherlands: The Role of Integration, Host National Identification and Perceived Discrimination	Multiple (Afghan, Iranian, Iraqi, Somali)	One context	Intentions	Other refugees
Comparative Migration Studies Population, Space and Place 18(1)	Tefera, Tadele Muche (2026)	2026	Conflict, Drought, and Displacement: Determinants of Return Intentions among South Sudanese Refugees	South Sudanese	One context	Intentions	Other refugees
International Migration	Zimmermann (2012)	2012	Understanding Repatriation: Refugee Perspectives on the Importance of Safety, Reintegration, and Hope	Somali and Afghan	One context	Intentions	Other refugees
	Özer, Yeşim (2024)	2024	Back for Good? Return Aspirations of Immigrants in the Netherlands	Multiple immigrant groups	One context	Intentions	Other refugees
IMR	Al Husein, Nawras (2023)	2023	Determinants of Intended Return Migration among Refugees: A Comparison of Syrian Refugees in Germany and Turkey	Syrian	One context	Intentions	Syrian
BJPS	Alrababah, Ala (2023)	2023	The Dynamics of Refugee Return: Syrian Refugees and Their Migration Intentions	Syrian	One context	Intentions	Syrian
Political Studies	Gerver, Mollie (2025)	2025	Refugee Resettlement and Preferences	Syrian	One context	Intentions	Syrian
APSR	Ghosn, Faten (2021)	2021	The Journey Home: Violence, Anchoring, and Refugee Decisions to Return	Syrian	One context	Intentions	Syrian
Journal of Population Economics 36	Hannafi, Cyrine (2023)	2023	Social Integration of Syrian Refugees and Their Intention to Stay in Germany	Syrian	One context	Intentions	Syrian
Journal of Immigrant and Refugee Studies	Kaya, Serdar (2020)	2020	Prospects of Return: The Case of Syrian Refugees in Germany	Syrian	One context	Intentions	Syrian
Journal of Immigrant and Refugee Studies	Kayaoglu, Ayselin (2021)	2021	Return Aspirations of Syrian Refugees in Turkey	Syrian	One context	Intentions	Syrian
Journal of International Migration and Integration 24(4)	Özkan, Zafer (2023)	2023	Intentions to Return and Migrate to Third Countries: A Socio-Demographic Investigation among Syrians in Turkey	Syrian	One context	Intentions	Syrian
CESifo Working Paper 12118	Adema, Joop (2025)	2025	What Drives Refugees' Return after Conflict?	Ukrainian	One context	Intentions	Ukrainian
Journal of Ethnic and Migration Studies	Hernes, Vilde (2024)	2024	Where Does the Future Lie? Initial Aspirations for Return among Newly Arrived Ukrainian Refugees in Norway	Ukrainian	One context	Intentions	Ukrainian
International Migration 64(3)	Kochaniak, Katarzyna (2026)	2026	Return Intentions under Temporary Protection: Theoretical Perspectives on Migrant Households	Ukrainian households	One context	Intentions	Ukrainian
PLOS ONE 18	Kohlenberger, Judith (2023)	2023	High Self-Selection of Ukrainian Refugees into Europe: Evidence from Krakow and Vienna	Ukrainian	One context	Intentions	Ukrainian
IBS Working Paper 01/2025	Lewandowski, Piotr (2025)	2025	The Role of Job Task Routinization in Shaping Return Intentions: Evidence from Ukrainian War Refugees in Poland	Ukrainian	One context	Intentions	Ukrainian
Population, Space and Place	Novotný, Josef (2026)	2026	Drivers of Return Plans among Ukrainian War Refugees in Czechia: Challenging Common Assumptions and Refining the Theory	Ukrainian	One context	Intentions	Ukrainian

Table 36: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Ethnic and Migration Studies, online first	Olivo Rumpf, Karelis (2026)	2026	The Interplay between Origin- and Receiving-Country Conditions: Experimental Evidence on Ukrainian Refugees' Return Intentions	Ukrainian	One context	Intentions	Ukrainian
International Migration Review, online first	Perelli-Harris, Brienna (2025)	2025	The Uncertainty of Forced Displacement: How Language and Violence Shaped Displacement Trajectories during Russia's Invasion of Ukraine	Ukrainian	One context	Intentions	Ukrainian
International Migration 63(5)	Saijo, Harunobu (2025)	2025	Which Policy Attributes Affect Assisted Voluntary Return and Reintegration Uptake among Ukrainian Evacuees in Japan?	Ukrainian	One context	Intentions	Ukrainian
Journal of Behavioral and Experimental Economics	Vakhitov, Volodymyr (2025)	2025	Return Intentions of Ukrainian Refugees: The Role of National Identity and Pride	Ukrainian	One context	Intentions	Ukrainian
International Migration 62	van Tubergen, Frank (2024)	2024	Return Intentions among Ukrainian Refugees in Europe: A Cross-National Study	Ukrainian women	One context	Intentions	Ukrainian
J Refugee Studies	Žila, Ondřej (2026)	2026	'Partial Returns': Displacement, Mobility, and Translocal Connections in Post-war Bosnia and Herzegovina	Bosnian	One context	Qualitative	DPs
Journal of Refugee Studies 34(1)	Antwi-Boateng (2021)	2021	The Dilemma of Ghana-Based Liberian Refugees and the Challenges of Repatriation and Integration	Liberian	One context	Qualitative	Other refugees
Journal of Ethnic and Migration Studies	Durodola (2026)	2026	Return to What? Reimagining Home and Belonging after Conflict and Refugee Status Cessation among Liberians in Nigeria	Liberian	One context	Qualitative	Other refugees
International Migration	Eastmond, Marita (2006)	2006	Transnational Returns and Reconstruction in Post-war Bosnia and Herzegovina	Bosnian	One context	Qualitative	Other refugees
J Refugee Studies	Hardgrove, Abby (2009)	2009	Liberian Refugee Families in Ghana: The Implications of Family Demands and Capabilities for Return to Liberia	Liberian	One context	Qualitative	Other refugees
International Migration 52(6)	Harpviken, Kristian Berg (2014)	2014	Split Return: Transnational Household Strategies in Afghan Repatriation	Afghan	One context	Qualitative	Other refugees
Journal of Eastern African Studies 4(1)	Kaiser (2010)	2010	Dispersal, Division and Diversification: Durable Solutions and Sudanese Refugees in Uganda	Sudanese	One context	Qualitative	Other refugees
African Human Mobility Review 6(2)	Khan (2020)	2020	Is Voluntary Repatriation the Preferred Durable Solution? The View of Refugees in South Africa	Multiple refugee origins	One context	Qualitative	Other refugees
Journal of Refugee Studies	Krupalija, Mustafa (2026)	2026	Rebuilding Home on Hot Ground: Forced Migration and Return to Srebrenica	Bosnian	One context	Qualitative	Other refugees
Journal of Refugee Studies 19(4)	Muggeridge (2006)	2006	Back Home? Refugees' Experiences of their First Visit back to their Country of Origin	Multiple refugee origins	One context	Qualitative	Other refugees
J Refugee Studies	Nandi, Sarah (2025)	2025	"We were together and we had our own family in each other": Refusing Repatriation and Forging Gendered Belonging as Hijra Refugees in Kolkata	Bangladeshi (Hijra)	One context	Qualitative	Other refugees
Transnational Social Review 6(1-2)	Omata (2016)	2016	Home-making during Protracted Exile: Diverse Responses of Refugee Families in the Face of Remigration	Liberian	One context	Qualitative	Other refugees
JEMS	Omata, Naohiko (2013)	2013	The Complexity of Refugees' Return Decision-Making in a Protracted Exile: Beyond the Home-Coming Model and Durable Solutions	Liberian	One context	Qualitative	Other refugees
International Migration 55(5)	Porobić, Selma (2017)	2017	Daring Life-Return Projects to Post-Dayton Bosnia and Herzegovina	Bosnian returnees	One context	Qualitative	Other refugees
International Migration	Stefansson, Anders H. (2006)	2006	Homes in the Making: Property Restitution, Refugee Return, and Senses of Belonging in a Post-war Bosnian Town	Bosnian	One context	Qualitative	Other refugees
Geoforum 43(1)	Tete (2012)	2012	'Any Place Could Be Home': Embedding Refugees' Voices into Displacement Resolution and State Refugee Policy	Liberian	One context	Qualitative	Other refugees
Comparative Migration Studies	Van Houte, Marieke (2016)	2016	Deconstructing the Meanings of and Motivations for Return: An Afghan Case Study	Afghan	One context	Qualitative	Other refugees
Refugee Survey Quarterly 45(1)	Diab (2026)	2026	Prolonging the Temporary: Female-Headed Syrian Refugee Households on Displacement, Return and Waiting post-Assad	Syrian women heading households	One context	Qualitative	Syrian

Table 36: Studies plotted in Figure 1 of the paper (*continued*)

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
International Migration 64(3)	Dinçşahin (2026)	2026	From Protracted Liminality to Fractured Homecoming: Intrafamilial Bargaining and the Limits of Voluntary Repatriation for Syrians in Turkey	Syrian	One context	Qualitative	Syrian
Journal of Refugee Studies 39(1)	Enna (2026)	2026	'The Threat of Forced Return Is the Government's Last Resort': Structural, Cultural, and Direct Violence towards Syrian Refugees in Lebanon	Syrian	One context	Qualitative	Syrian
International Migration Review, OnlineFirst	Masterson (2026)	2026	Circular Refugee Migration: Understanding Protracted Displacement Beyond a Refugee-Returnee Binary	Syrian	One context	Qualitative	Syrian
International Migration Review 57(4)	Müller-Funk, Lea (2023)	2023	I Will Return Strong: The Role of Life Aspirations in Refugees' Return Aspirations	Syrian	One context	Qualitative	Syrian
Comparative Migration Studies 14:32	Kokonowskyj, Larissa (2026)	2026	The Role of Family Dynamics in (Return) Migration Aspirations among Ukrainian Protection Holders in Germany	Ukrainian	One context	Qualitative	Ukrainian
Comparative Migration Studies 13:66	Ash, Konstantin (2025)	2025	Who Leaves and Who Returns? IDPs and Returnees after the Russian Invasion of Ukraine	Ukrainian	One context	Realized return	IDPs
JCR	Bove, Vincenzo (2025)	2025	What it Takes to Return: UN Peacekeeping and the Safe Return of Displaced People	South Sudanese	One context	Realized return	IDPs
AJPS	Camarena, Kara Ross (2020)	2020	When Do Displaced Persons Return? Postwar Migration among Christians in Mount Lebanon	Lebanese (Christians)	One context	Realized return	IDPs
International Migration 55(5)	Joireman, Sandra F. (2017)	2017	Ethnic Violence, Local Security and Return Migration: Enclave Communities in Kosovo	Kosovar displaced	One context	Realized return	IDPs
J Refugee Studies	Stefanovic, Djordje (2015)	2015	Home is Where the Heart Is? Forced Migration and Voluntary Return in Turkey's Kurdish Regions	Kurdish	One context	Realized return	IDPs
Journal of Conflict Resolution 69(1)	Tellez, Juan F. (2025)	2025	Social Cohesion, Economic Security, and Forced Displacement in the Long Run: Evidence from Rural Colombia	Colombian IDPs	One context	Realized return	IDPs
Working Paper	Weber, Sigrid (2022)	2022	Property Rights and Post-conflict Recovery: Theory and Evidence from IDP Return Movements in Iraq	Iraqi (Yazidi and Sunni Muslim)	One context	Realized return	IDPs
Journal of Refugee Studies 23(1)	Huttunen (2010)	2010	Sedentary Policies and Transnational Relations: A 'Non-sustainable' Case of Return to Bosnia	Bosnian	One context	Realized return	Other refugees
J Refugee Studies	Klinthäll, Martin (2007)	2007	Refugee Return Migration: Return Migration from Sweden to Chile, Iran and Poland 1973-1996	Chilean, Iranian, Polish	One context	Realized return	Other refugees
International Migration 55(5)	Kutlay, Mustafa (2017)	2017	The Turks of Bulgaria: An Outlier Case of Forced Migration and Voluntary Return	Turks of Bulgaria	One context	Realized return	Other refugees
Journal of Ethnic and Migration Studies 51(1)	Omata (2025)	2025	Returning to Fund Refugeehood: Dispersal and Survival between Uganda and South Sudan	South Sudanese	One context	Realized return	Other refugees
International Migration 55(5)	Stefanovic, Djordje (2017)	2017	Peaceful Returns: Reversing Ethnic Cleansing after the Bosnian War	Bosnian	One context	Realized return	Other refugees
Middle Black Sea Journal of Communication Studies 11	Altınışık (2026)	2026	From Forced Migration to Voluntary Return: Psychosocial Factors Influencing Syrian Refugees' Decisions to Return to Their Homeland After the War	Syrian	One context	Realized return	Syrian
J Development Economics	Beaman, Lori (2022)	2022	When Do Refugees Return Home? Evidence from Syrian Displacement in Mashreq	Syrian	One context	Realized return	Syrian
International Review of Sociology	Bimay (2026)	2026	Between Hope and Hesitation: Perceptions and Experiences of Refugee Women Returning from Turkey to Syria	Syrian women	One context	Realized return	Syrian
Working Paper (EBRD)	Adema, Joop (2026)	2026	Refugee Return	Ukrainian	One context	Realized return	Ukrainian

Note: Drawn from the literature audit of 23 August 2026. Scope and Outcome are the two axes of that figure; Marker is the population class the figure shows by shape. Inclusion follows the Figure decision and Strict figure columns of the audit's two inventory tabs; two studies logged there as out of scope are excluded.

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